

Discrete-Time Fourier Transform

6.1 INTRODUCTION

In the previous chapter, we introduced the continuous-time Fourier transform and developed the many characteristics of that transform which makes the methods of Fourier analysis of such great value in analysing and understanding the properties of continuous-time signals and systems. In this chapter, we complete our development of the basic tools of Fourier analysis by introducing and examining the discrete-time Fourier transform.

6.2 FOURIER TRANSFORM REPRESENTATION OF APERIODIC DISCRETE-TIME SIGNALS

This derivation proceeds along a path analogous to the derivation of the CTFT from the CTFS. The DTFS representation of a periodic signal $x(n)$ (with period N and frequency $\omega_0 = 2\pi/N$) can be written as

$$x(n) = \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n}$$

Substituting $X_k = 1/N \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n}$ in the DTFS definition yields

$$x(n) = \sum_{k=\langle N \rangle} \left(\frac{1}{N} \sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n} \right) e^{jk\omega_0 n}$$

or

$$x(n) = \sum_{k=\langle N \rangle} \left(\sum_{n=\langle N \rangle} x(n) e^{-jk\omega_0 n} \right) e^{jk\omega_0 n} \frac{\omega_0}{2\pi}$$

Since the inner summation is over any arbitrary range of n of width N , let the range be $-(N/2) \leq n \leq (N/2) - 1$ for N even or $-(N - 1/2) \leq n \leq (N - 1/2)$ for N odd. The outer summation is over any arbitrary range of k of width N , so let its range be

$k_0 \leq k \leq (k_0 + N - 1)$. Then

$$x(n) = \sum_{k=k_0}^{k_0+N-1} \left(\sum_{n=-\frac{N}{2}}^{\frac{N}{2}-1} x(n) e^{-jk\omega_0 n} \right) e^{jk\omega_0 n \frac{\omega_0}{2\pi}}, \quad \text{for } N \text{ even} \quad (6.1)$$

or

$$x(n) = \sum_{k=k_0}^{k_0+N-1} \left(\sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} x(n) e^{-jk\omega_0 n} \right) e^{jk\omega_0 n \frac{\omega_0}{2\pi}}, \quad \text{for } N \text{ odd} \quad (6.2)$$

Now let the fundamental period $N \rightarrow \infty$. In that limit $\omega_0/2\pi \rightarrow d\omega/2\pi$ and $k\omega_0 \rightarrow \omega$. The inner summation covers an infinite range. The outer summation approaches an integral in $\omega = k\omega_0$ covering a range of

$$k_0 \leq k \leq (k_0 + N - 1)$$

$$k_0 \leq k < (k_0 + N)$$

$$k_0 \leq \frac{\omega}{\omega_0} < (k_0 + N)$$

$$k_0\omega_0 < \omega < (k_0\omega_0 + N\omega_0)$$

$$k_0\omega_0 < \omega < (k_0\omega_0 + 2\pi)$$

Therefore, Eqs (6.1) and (6.2) both become

$$x(n) = \frac{1}{2\pi} \int_{2\pi} \left(\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right) e^{j\omega n} d\omega \quad (6.3)$$

where $\int_{2\pi}$ indicates integration over any continuous interval of 2π . The inner summation of the above equation is a function of only ω and not n . Denoting the summation by $X(e^{j\omega})$, we can write Eq. (6.3) as

$$x(n) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \quad (6.4)$$

where

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \quad (6.5)$$

Equation (6.4) is usually referred to as the *synthesis equation* because it synthesizes an arbitrary signal from its complex exponential components. On the other hand, Eq. (6.5) is referred to as the *analysis equation* because it analyses how much of each complex exponential signal is present in the original signal. We call $X(e^{j\omega})$ the discrete-time Fourier transform (DTFT) of $x(n)$, and $x(n)$ the inverse discrete-time Fourier transform (IDTFT) of $X(e^{j\omega})$. This nomenclature can be represented as

$$X(e^{j\omega}) = \text{DTFT}[x(n)] = \mathcal{F}[x(n)] \quad \text{and} \quad x(n) = \text{IDTFT}[X(e^{j\omega})] = \mathcal{F}^{-1}[X(e^{j\omega})]$$

The same information is conveyed by the statement that $x(n)$ and $X(e^{j\omega})$ are a (discrete-time) Fourier transform pair. Symbolically, this is expressed as

$$x(n) \longleftrightarrow X(e^{j\omega})$$

In general the Fourier transform $X(e^{j\omega})$ is a complex function of the real variable ω and can be written in the rectangular form as

$$X(e^{j\omega}) = X_R(e^{j\omega}) + jX_I(e^{j\omega}) \quad (6.6)$$

where $X_R(e^{j\omega})$ and $X_I(e^{j\omega})$ are, respectively, the real and imaginary parts of $X(e^{j\omega})$. From Eq. (6.6), it follows that

$$X_R(e^{j\omega}) = \frac{1}{2}[X(e^{j\omega}) + X^*(e^{j\omega})] \quad (6.7)$$

$$X_I(e^{j\omega}) = \frac{1}{2j}[X(e^{j\omega}) - X^*(e^{j\omega})] \quad (6.8)$$

where $X^*(e^{j\omega})$ denotes the complex conjugate of $X(e^{j\omega})$.

The Fourier transform $X(e^{j\omega})$ can alternatively be expressed in the polar form as

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\theta(\omega)} \quad (6.9)$$

where

$$\theta(\omega) = \angle X(e^{j\omega}) \quad (6.10)$$

The quantity $|X(e^{j\omega})|$ is called the *magnitude spectrum* and the quantity $\theta(\omega) = \angle X(e^{j\omega})$ is called the *phase spectrum*.

The relations between the rectangular and polar forms of $X(e^{j\omega})$ follow from Eqs (6.6) and (6.9) and are given by

$$\begin{aligned} X_R(e^{j\omega}) &= |X(e^{j\omega})| \cos(\theta(\omega)) \\ X_I(e^{j\omega}) &= |X(e^{j\omega})| \sin(\theta(\omega)) \\ |X(e^{j\omega})| &= \sqrt{X_R^2(e^{j\omega}) + X_I^2(e^{j\omega})} \end{aligned} \quad (6.11)$$

$$\theta(\omega) = \angle X(e^{j\omega}) = \tan^{-1} \left[\frac{X_I(e^{j\omega})}{X_R(e^{j\omega})} \right] \quad (6.12)$$

Thus, for a real signal, it follows from the Eq. (6.11) that

$$\begin{aligned} |X(e^{-j\omega})| &= \sqrt{X_R^2(e^{-j\omega}) + X_I^2(e^{-j\omega})} \\ |X(e^{-j\omega})| &= \sqrt{X_R^2(e^{j\omega}) + X_I^2(e^{j\omega})} \\ |X(e^{-j\omega})| &= |X(e^{j\omega})| \end{aligned} \quad (6.13)$$

The magnitude spectrum $|X(e^{j\omega})|$ is an even function of ω . Likewise, for a real signal, we note from Eq. (6.12) that

$$\angle X(e^{-j\omega}) = \tan^{-1} \left[\frac{X_I(e^{-j\omega})}{X_R(e^{-j\omega})} \right] = \tan^{-1} \left[-\frac{X_I(e^{j\omega})}{X_R(e^{j\omega})} \right] = -\tan^{-1} \left[\frac{X_I(e^{j\omega})}{X_R(e^{j\omega})} \right] = -\angle X(e^{j\omega})$$

$$\angle X(e^{-j\omega}) = -\angle X(e^{j\omega}) \quad (6.14)$$

The above equation implies that the phase spectrum $\angle X(e^{j\omega})$ is an odd function of ω .

6.3 PERIODICITY OF THE DTFT

The DTFT is a periodic function in ω with a period 2π , i.e.,

$$X(e^{j(\omega+2\pi)}) = X(e^{j\omega}) \quad (6.15)$$

Proof We know that

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

For any integer k , we have

$$\begin{aligned} X(e^{j(\omega+2\pi k)}) &= \sum_{n=-\infty}^{\infty} x(n) e^{-j(\omega+2\pi k)n} \\ &= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} e^{-j2\pi kn} \\ X(e^{j(\omega+2\pi k)}) &= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \\ X(e^{j(\omega+2\pi k)}) &= X(e^{j\omega}) \end{aligned}$$

where we have used the fact that $e^{-j2\pi kn} = 1$. Hence $X(e^{j\omega})$ is periodic with period 2π . But this property is just a consequence of the fact that frequency range for any discrete-time signal is unique over the frequency interval of $(-\pi, \pi)$ or, equivalently, $(0, 2\pi)$, and any frequency outside this interval is equivalent to a frequency within this interval. ■

6.4 CONVERGENCE OF DTFT

An infinite series of Eq. (6.5) may or may not converge. The Fourier transform $X(e^{j\omega})$ of $x(n)$ is said to exist if the series in Eq. (6.5) converges in some sense. Let

$$X_K(e^{j\omega}) = \sum_{n=-K}^K x(n) e^{-j\omega n} \quad (6.16)$$

denote the partial sum of the weighted complex exponentials in Eq. (6.5). Then for uniform convergence of $X(e^{j\omega})$,

$$\lim_{K \rightarrow \infty} [X(e^{j\omega}) - X_K(e^{j\omega})] = 0$$

$$\lim_{K \rightarrow \infty} X_K(e^{j\omega}) = X(e^{j\omega})$$

Uniform convergence is guaranteed if $x(n)$ is absolutely summable. Indeed, if

$$\sum_{n=-\infty}^{\infty} |x(n)| < \infty \quad (6.17)$$

then

$$|X(e^{j\omega})| = \left| \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right|$$

$$\leq \sum_{n=-\infty}^{\infty} |x(n)| |e^{-j\omega n}|$$

$$|X(e^{j\omega})| \leq \sum_{n=-\infty}^{\infty} |x(n)| < \infty$$

for all values of ω guaranteeing the existence of $X(e^{j\omega})$. Thus, Eq. (6.17) is a sufficient condition for the existence of the Fourier transform $X(e^{j\omega})$ of the sequence $x(n)$. We note that this is the discrete-time counterpart of the first Dirichlet condition for the Fourier transform of continuous-time signals. The last two conditions do not apply because of the discrete-time nature of $x(n)$.

A large class of sequences encountered in practice are of finite length with finite sample values. These sequences are absolutely summable, and hence, their Fourier transform converge uniformly. Some sequences are not absolutely summable, but they are square summable. Since

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 \leq \left(\sum_{n=-\infty}^{\infty} |x(n)| \right)^2$$

an absolutely summable sequence always has a finite energy. However, a finite-energy sequence is not necessarily absolutely summable. We would like to define the Fourier transform of finite-energy sequences, but we may relax the condition of uniform convergence. For such sequences we can impose a mean-square convergence condition:

$$\lim_{K \rightarrow \infty} \int_{-\pi}^{\pi} |X(e^{j\omega}) - X_K(e^{j\omega})|^2 d\omega = 0 \quad (6.18)$$

Thus, the energy in the error $X(e^{j\omega}) - X_K(e^{j\omega})$ tends towards zero, but the error $|X(e^{j\omega}) - X_K(e^{j\omega})|$ does not necessarily tend to zero. In this way we can include finite-energy signals in the class of signals for which the Fourier transform exists. These can also be defined for a certain class of sequences that are neither absolutely summable nor square-summable. Example of such sequences are the unit step sequence, the sinusoidal sequence, and the complex exponential sequence, which are neither absolutely summable nor square-summable.

6.4.1 Gibbs Phenomenon

Let us consider an example from the class of finite-energy signals. Suppose that

$$X(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases}$$

The inverse DTFT of $X(e^{j\omega})$ is given by

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega$$

$$x(n) = \frac{1}{2\pi} \frac{e^{j\omega n}}{jn} \Big|_{-\omega_c}^{\omega_c}$$

$$x(n) = \frac{\sin(\omega_c n)}{n\pi}, \quad n \neq 0$$

For $n = 0$, the inverse Fourier transform expression reduces to

$$x(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} d\omega = \frac{\omega_c}{\pi}$$

Hence

$$x(n) = \begin{cases} \frac{\omega_c}{\pi}, & n = 0 \\ \frac{\sin(\omega_c n)}{n\pi}, & n \neq 0 \end{cases} \quad (6.19)$$

It should be noted that often the above sequence is expressed in a compact form as

$$x(n) = \frac{\sin(\omega_c n)}{n\pi}, \quad -\infty < n < \infty \quad (6.20)$$

with the understanding that at $n = 0$, $x(n) = \omega_c/\pi$. Now let us consider the determination of the Fourier transform of the sequence given by Eq. (6.20). The sequence $x(n)$ is not absolutely summable. Hence the infinite series

$$\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} = \sum_{n=-\infty}^{\infty} \frac{\sin(\omega_c n)}{n\pi} e^{-j\omega n} \quad (6.21)$$

does not converge uniformly for all ω . We shall show later that the energy of the above sequence is given by ω_c/π , and therefore, $x(n)$ is a finite-energy sequence. Hence, the sum in the Eq. (6.21) is guaranteed to converge to the $X(e^{j\omega})$ in the mean-square sense.

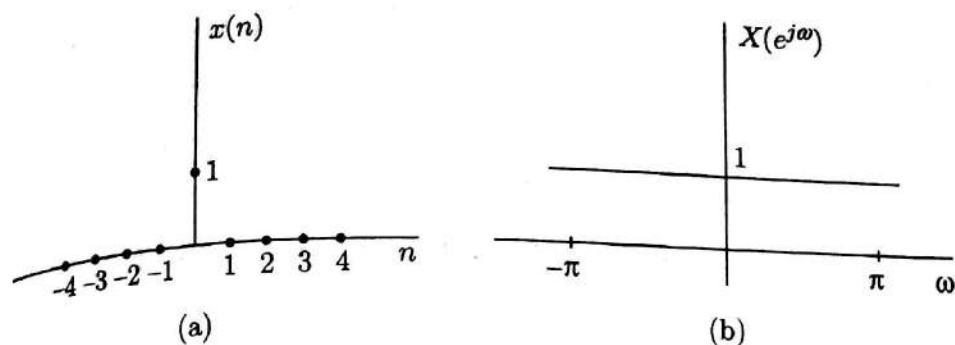


Fig. 6.2 (a) Unit impulse sequence and (b) its Fourier transform

The unit sample sequence has a Fourier transform representation consisting of equal contributions at all frequencies. The unit sample sequence and its Fourier transform are shown in Figs 6.2(a) and (b), respectively.

Example 6.2 Find the Fourier transform of the causal sequence [shown in Fig. 6.3(a)]

$$x(n) = a^n u(n), \quad |a| < 1.$$

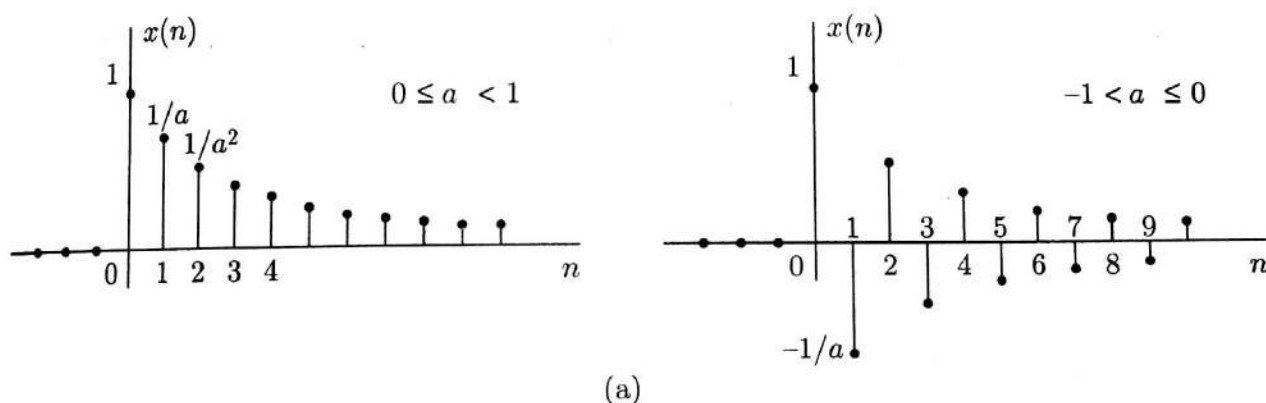


Fig. 6.3(a) Signal $x(n]$

Solution

From Eq.(6.5), the Fourier transform $X(e^{j\omega})$ of the sequence $x(n]$ is given by

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} = \sum_{n=-\infty}^{\infty} a^n u(n) e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} a^n e^{-j\omega n} \\ X(e^{j\omega}) &= \sum_{n=0}^{\infty} (a e^{-j\omega})^n \\ X(e^{j\omega}) &= \frac{1}{1 - a e^{-j\omega}} \end{aligned}$$

$$a^n u(n) \longleftrightarrow \frac{1}{1 - a e^{-j\omega}}, \quad |a| < 1 \quad (6.24)$$

By multiplying both the numerator and denominator of Eq. (6.24) by the complex conjugate of the denominator, we obtain

$$X(e^{j\omega}) = \frac{1 - ae^{j\omega}}{(1 - ae^{-j\omega})(1 - ae^{j\omega})} = \frac{1 - a \cos(\omega) - ja \sin(\omega)}{1 - 2a \cos(\omega) + a^2}$$

This expression can be subdivided into real and imaginary parts. Thus, we obtain

$$X_R(e^{j\omega}) = \frac{1 - a \cos(\omega)}{1 - 2a \cos(\omega) + a^2}$$

$$X_I(e^{j\omega}) = -\frac{a \sin(\omega)}{1 - 2a \cos(\omega) + a^2}$$

Using $X_R(e^{j\omega})$ and $X_I(e^{j\omega})$, the magnitude spectrum is given by

$$|X(e^{j\omega})| = \sqrt{X_R^2(e^{j\omega}) + X_I^2(e^{j\omega})} = \frac{1}{\sqrt{1 - 2a \cos(\omega) + a^2}}$$

and the phase spectrum is given by

$$\angle X(e^{j\omega}) = \tan^{-1} \left[\frac{X_I(e^{j\omega})}{X_R(e^{j\omega})} \right] = \tan^{-1} \left[\frac{a \sin(\omega)}{1 - a \cos(\omega)} \right]$$

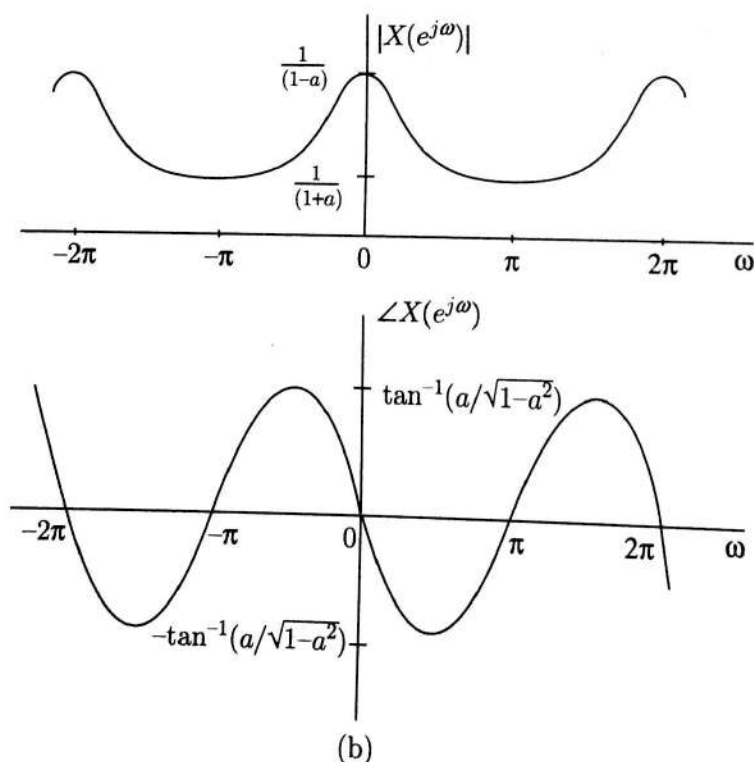


Fig. 6.3(b) Magnitude and phase of the Fourier transform of Example 6.2 for $0 < a < 1$

The magnitude and phase spectrum are shown in Fig. 6.3(b) for $0 < a < 1$ and in Fig. 6.3(c) for $-1 < a < 0$. Note that the magnitude and phase spectrum are periodic in ω with period 2π .

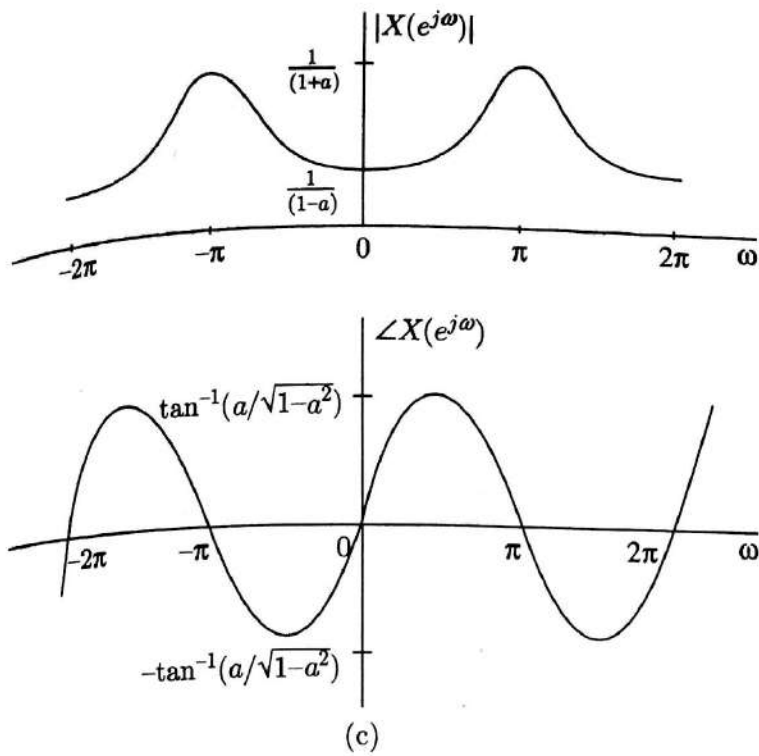


Fig. 6.3(c) Magnitude and phase of the Fourier transform for $-1 < a < 0$

Example 6.3 Find the Fourier transform of the sequence [shown in Fig. 6.4]

$$x(n) = a^n u(-n-1), \quad |a| > 1.$$

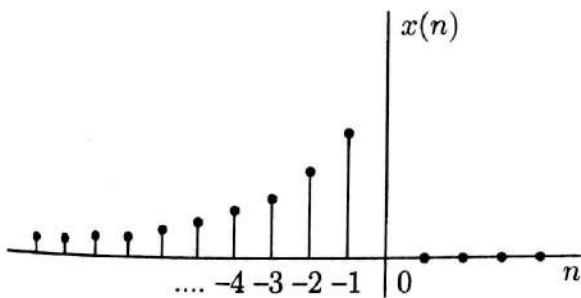


Fig. 6.4

Solution

From Eq. (6.5), the Fourier transform $X(e^{j\omega})$ of the sequence $x(n)$ is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} = \sum_{n=-\infty}^{\infty} a^n u(-n-1) e^{-j\omega n}$$

Since

$$u(-n-1) = \begin{cases} 1, & (-n-1) \geq 0 \rightarrow n \leq -1 \\ 0, & (-n-1) < 0 \rightarrow n > -1 \end{cases}$$

We have

$$X(e^{j\omega}) = \sum_{n=-\infty}^{-1} a^n e^{-j\omega n} = \sum_{n=-\infty}^{-1} \left(\frac{1}{a} e^{j\omega}\right)^{-n} = \sum_{n=1}^{\infty} \left(\frac{1}{a} e^{j\omega}\right)^n$$

A change of variables is performed by letting $m = (n - 1)$, which also yields $n = (m + 1)$, $m = 0$ as $n = 1$ and $m = \infty$ as $n = \infty$. Therefore,

$$\begin{aligned} X(e^{j\omega}) &= \sum_{m=0}^{\infty} \left(\frac{1}{a} e^{j\omega}\right)^{m+1} \\ &= \frac{1}{a} e^{j\omega} \sum_{m=0}^{\infty} \left(\frac{1}{a} e^{j\omega}\right)^m \end{aligned}$$

$$X(e^{j\omega}) = \frac{1/a e^{j\omega}}{1 - 1/a e^{j\omega}}$$

$$X(e^{j\omega}) = \frac{1}{a e^{-j\omega} - 1}, \quad |a| > 1$$

$$a^n u(-n - 1) \longleftrightarrow \frac{1}{a e^{-j\omega} - 1}, \quad |a| > 1 \quad (6.25)$$

Except for the change of sign, this Fourier transform is identical to that of $x(n) = a^n u(n)$, $|a| < 1$. Yet there is no ambiguity in determining the IDTFT of $X(e^{j\omega}) = \frac{1}{a e^{-j\omega} - 1}$ because of the restrictions on the value of a in each case. If $|a| < 1$, then its IDTFT is $x(n) = -a^n u(n)$. If $|a| > 1$, its IDTFT is $x(n) = a^n u(-n - 1)$.

By multiplying both the numerator and denominator of Eq. (6.25) by the complex conjugate of the denominator, we obtain

$$X(e^{j\omega}) = \frac{a e^{j\omega} - 1}{(a e^{-j\omega} - 1)(a e^{j\omega} - 1)} = \frac{a \cos(\omega) - 1 + ja \sin(\omega)}{1 - 2a \cos(\omega) + a^2}$$

This expression can be subdivided into real and imaginary parts. Thus, we obtain

$$X_R(e^{j\omega}) = \frac{a \cos(\omega) - 1}{1 - 2a \cos(\omega) + a^2}$$

$$X_I(e^{j\omega}) = \frac{a \sin(\omega)}{1 - 2a \cos(\omega) + a^2}$$

Using $X_R(e^{j\omega})$ and $X_I(e^{j\omega})$, the magnitude spectrum is given by

$$|X(e^{j\omega})| = \sqrt{X_R^2(e^{j\omega}) + X_I^2(e^{j\omega})} = \frac{1}{\sqrt{1 - 2a \cos(\omega) + a^2}}$$

and the phase spectrum is given by

$$\angle X(e^{j\omega}) = \tan^{-1} \left[\frac{X_I(e^{j\omega})}{X_R(e^{j\omega})} \right] = \tan^{-1} \left[\frac{a \sin(\omega)}{a \cos(\omega) - 1} \right]$$

Example 6.4 Find the Fourier transform of the anticausal sequence

$$x(n) = a^{-n}u(-n-1), \quad |a| < 1$$

Solution

From Eq. (6.5), the Fourier transform $X(e^{j\omega})$ of the sequence $x(n)$ is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} = \sum_{n=-\infty}^{\infty} a^{-n} u(-n-1) e^{-j\omega n}$$

Since

$$u(-n-1) = \begin{cases} 1, & (-n-1) \geq 0 \rightarrow n \leq -1 \\ 0, & (-n-1) < 0 \rightarrow n > -1 \end{cases}$$

We have

$$X(e^{j\omega}) = \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n} = \sum_{n=-\infty}^{-1} (a e^{j\omega})^{-n} = \sum_{n=1}^{\infty} (a e^{j\omega})^n$$

A change of variables is performed by letting $m = (n - 1)$, which also yields $n = (m + 1)$, $m = 0$ as $n = 1$ and $m = \infty$ as $n = \infty$. Therefore,

$$X(e^{j\omega}) = \sum_{m=0}^{\infty} (a e^{j\omega})^{m+1} = a e^{j\omega} \sum_{m=0}^{\infty} (a e^{j\omega})^m = \frac{a e^{j\omega}}{1 - a e^{j\omega}}$$

$$a^{-n}u(-n-1) \longleftrightarrow \frac{a e^{j\omega}}{1 - a e^{j\omega}}, \quad |a| < 1 \quad (6.26)$$

Example 6.5 Find the Fourier transform of the noncausal sequence

$$x(n) = a^{|n|}, \quad |a| < 1$$

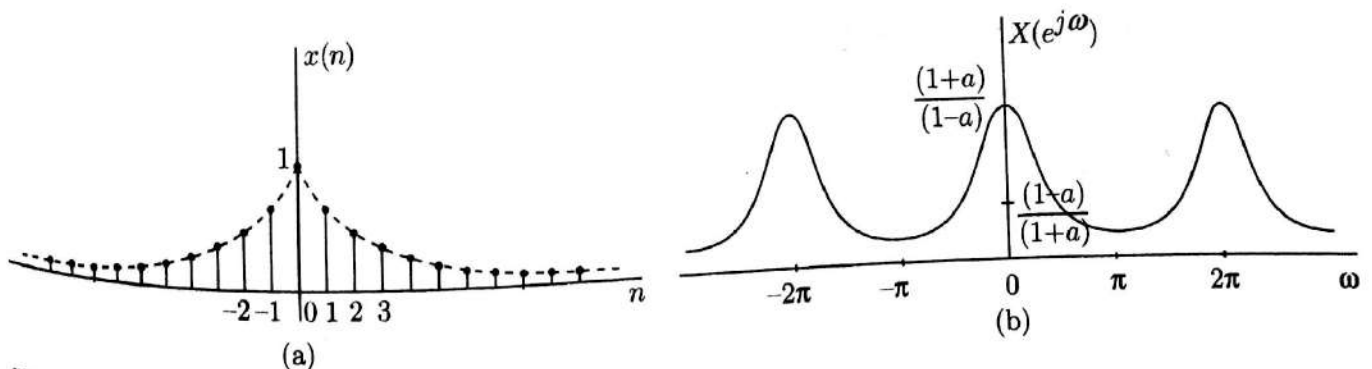


Fig. 6.5 (a) Signal $x(n) = a^{|n|}$ and (b) its Fourier transform

Solution

Consider the given sequence

$$x(n) = a^{|n|}, \quad |a| < 1$$

It can be written as

$$x(n) = \begin{cases} a^n, & n \geq 0 \\ a^{-n}, & n \leq -1 \end{cases}$$

$$x(n) = a^n u(n) + a^{-n} u(-n - 1)$$

The signal $x(n)$ is plotted for $0 < a < 1$ in Fig. 6.5(a). Taking the Fourier transform of the above equation, we have

$$\mathcal{F}[x(n)] = \mathcal{F}[a^n u(n)] + \mathcal{F}[a^{-n} u(-n - 1)]$$

Using Eqs (6.24) and (6.26), we have

$$X(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}} + \frac{a e^{j\omega}}{1 - a e^{j\omega}}$$

$$X(e^{j\omega}) = \frac{1 - a^2}{1 - 2a \cos(\omega) + a^2}$$

$$a^{|n|} \longleftrightarrow \frac{1 - a^2}{1 - 2a \cos(\omega) + a^2}, \quad |a| < 1 \quad (6.27)$$

In this case $X(e^{j\omega})$ is real and is illustrated in Fig. 6.5(b).

Example 6.6 Find the Fourier transform of the rectangular pulse

$$x(n) = u(n + N_1) - u(n - N_1 - 1) = \begin{cases} 1, & |n| \leq N_1 \\ 0, & |n| > N_1 \end{cases}$$

which is illustrated in Fig. 6.6(a) for $N_1 = 2$.

Solution

From Eq. (6.5), the Fourier transform $X(e^{j\omega})$ of the sequence $x(n)$ is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} = \sum_{n=-N_1}^{N_1} e^{-j\omega n}$$

A change of variables is performed by letting $m = (n + N_1)$, which also yields $n = (m - N_1)$, $m = 0$ as $n = -N_1$ and $m = 2N_1$ as $n = N_1$. Therefore,

$$\begin{aligned}
 X(e^{j\omega}) &= \sum_{m=0}^{2N_1} e^{-j\omega(m-N_1)} \\
 X(e^{j\omega}) &= e^{j\omega N_1} \sum_{m=0}^{2N_1} e^{-j\omega m} \\
 &= e^{j\omega N_1} \left(\frac{1 - e^{-j\omega(2N_1+1)}}{1 - e^{-j\omega}} \right) \\
 &= e^{j\omega N_1} \left(\frac{e^{-j\omega/2} (e^{j\omega/2} - e^{-j\omega(2N_1+1/2)})}{e^{-j\omega/2} (e^{j\omega/2} - e^{-j\omega/2})} \right) \\
 &= \frac{(e^{j\omega(N_1+1/2)} - e^{-j\omega(N_1+1/2)})}{(e^{j\omega/2} - e^{-j\omega/2})} \\
 X(e^{j\omega}) &= \frac{\sin(\omega(N_1 + 1/2))}{\sin(\omega/2)}
 \end{aligned}$$

$$x(n) = \begin{cases} 1, & |n| \leq N_1 \\ 0, & |n| > N_1 \end{cases} \longleftrightarrow X(e^{j\omega}) = \frac{\sin(\omega(N_1 + 1/2))}{\sin(\omega/2)} \quad (6.28)$$

The Fourier transform is sketched in Fig. 6.6(b) for $N_1 = 2$. Since $X(e^{j\omega})$ is real, the magnitude and phase spectrum are given by

$$|X(e^{j\omega})| = \left| \frac{\sin(\omega(N_1 + 1/2))}{\sin(\omega/2)} \right|$$

and

$$\angle X(e^{j\omega}) = \begin{cases} 0, & X(e^{j\omega}) > 0 \\ \pi, & X(e^{j\omega}) < 0 \end{cases}$$

The magnitude and phase spectrum are plotted, respectively, in Figs 6.6(c) and (d) for $N_1 = 2$.

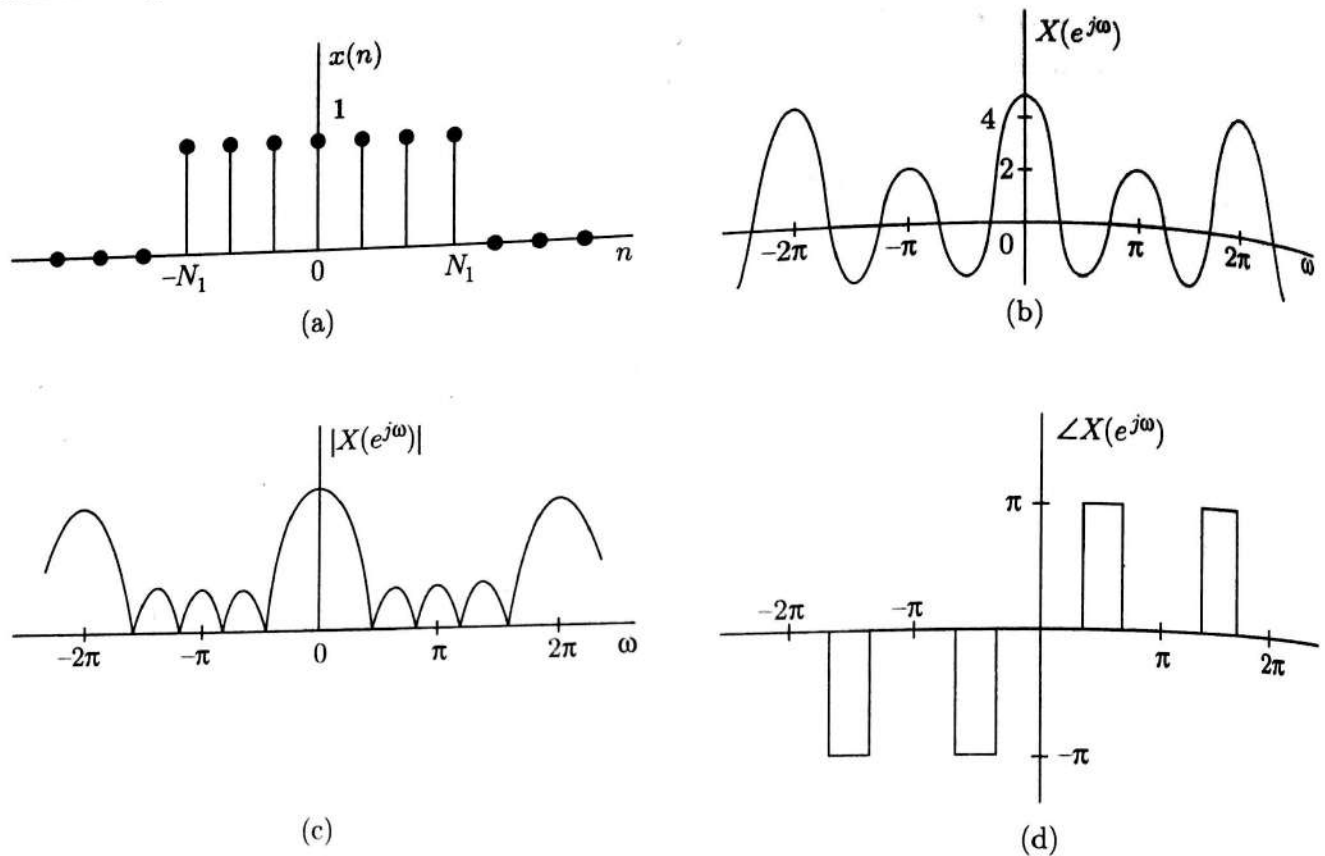


Fig. 6.6 (a) Rectangular pulse signal, (b) its Fourier transform, (c) magnitude spectrum and (d) its phase spectrum

Example 6.7 Find the inverse Fourier transform of

$$X(e^{j\omega}) = \sum_{m=-\infty}^{\infty} 2\pi\delta(\omega - \omega_0 - 2\pi m)$$

which is illustrated in Fig. 6.7.

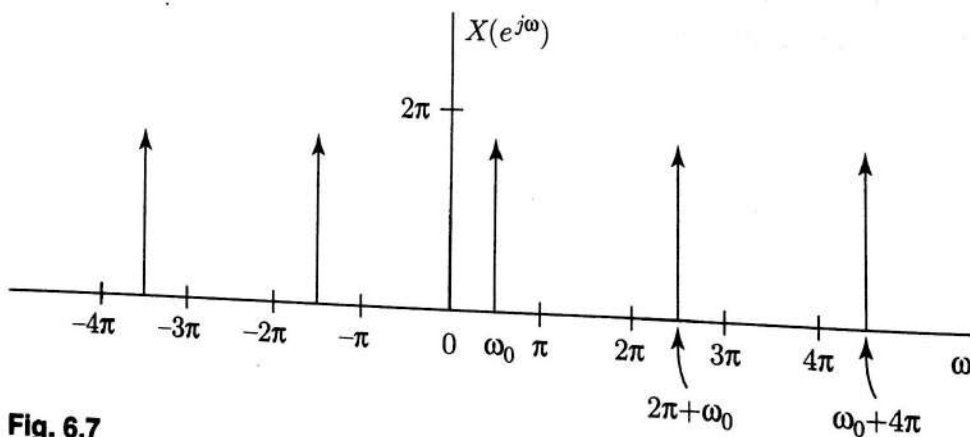


Fig. 6.7

Solution

From Fig. 6.7, we can observe that

$$X(e^{j\omega}) = 2\pi\delta(\omega - \omega_0), \quad 0 \leq \omega < 2\pi$$

Now, from Eq. (6.4), we have

$$x(n) = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

where $\int_{2\pi}$ indicates integration over any continuous interval of 2π . Therefore,

$$\begin{aligned} x(n) &= \frac{1}{2\pi} \int_0^{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_0^{2\pi} 2\pi \delta(\omega - \omega_0) e^{j\omega n} d\omega \\ x(n) &= e^{j\omega_0 n} \Big|_{\omega=\omega_0} = e^{j\omega_0 n} \end{aligned}$$

$$e^{j\omega_0 n} \longleftrightarrow \sum_{m=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 - 2\pi m) \quad (6.29)$$

Example 6.8 Find the DTFT of the unit step function $u(n)$.

Solution

From Eq. (6.5), the Fourier transform $X(e^{j\omega})$ of $x(n)$ is defined by

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} = \sum_{n=0}^{\infty} e^{-j\omega n} \\ &= \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} e^{-j\omega n} \\ X(e^{j\omega}) &= \lim_{N \rightarrow \infty} \frac{1 - e^{-j\omega N}}{1 - e^{-j\omega}} \\ X(e^{j\omega}) &= \frac{1}{1 - e^{-j\omega}} - \lim_{N \rightarrow \infty} \frac{e^{-j\omega N}}{1 - e^{-j\omega}} \end{aligned} \quad (6.30)$$

Consider the second term $\lim_{N \rightarrow \infty} \frac{e^{-j\omega N}}{1 - e^{-j\omega}}$ on the RHS of the above equation. For any test function $f(\omega)$, we have

$$\int_{-\infty}^{\infty} f(\omega) \left(\lim_{N \rightarrow \infty} \frac{e^{-j\omega N}}{1 - e^{-j\omega}} \right) d\omega = \lim_{N \rightarrow \infty} \int_{-\infty}^{\infty} f(\omega) \frac{e^{-j\omega N}}{1 - e^{-j\omega}} d\omega$$

A change of variables is performed by letting $\omega N = y$, which also yields $\omega = y/N$, $y = -\infty$ as $\omega = -\infty$ and $y = \infty$ as $\omega = \infty$. Therefore,

$$\begin{aligned}
 \lim_{N \rightarrow \infty} \int_{-\infty}^{\infty} f(\omega) \frac{e^{-j\omega N}}{1 - e^{-j\omega}} d\omega &= \lim_{N \rightarrow \infty} \int_{-\infty}^{\infty} f(y/N) \frac{e^{-jy}}{1 - e^{-jy/N}} \frac{dy}{N} \\
 &= f(0) \int_{-\infty}^{\infty} \left(\lim_{N \rightarrow \infty} \frac{e^{-jy}}{1 - e^{-jy/N}} \frac{dy}{N} \right) \\
 &= f(0) \int_{-\infty}^{\infty} \left(\lim_{N \rightarrow \infty} \frac{e^{-jy}}{N(1 - e^{-jy/N})} \right) dy \\
 &= f(0) \int_{-\infty}^{\infty} \left(\lim_{N \rightarrow \infty} \frac{e^{-jy}}{N(1 - (1 - jy/N + (-jy/N)^2/2! - \dots))} \right) dy \\
 &= f(0) \int_{-\infty}^{\infty} \frac{e^{-jy}}{jy} dy
 \end{aligned}$$

$$\begin{aligned}
 \lim_{N \rightarrow \infty} \int_{-\infty}^{\infty} f(\omega) \frac{e^{-j\omega N}}{1 - e^{-j\omega}} d\omega &= f(0) \int_{-\infty}^{\infty} \frac{\cos(y) - j \sin(y)}{jy} dy \\
 &= f(0) \left[\int_{-\infty}^{\infty} \frac{\cos(y)}{jy} dy - \int_{-\infty}^{\infty} \frac{j \sin(y)}{jy} dy \right] \\
 &= f(0) \left[\int_{-\infty}^{\infty} \frac{\cos(y)}{jy} dy - \int_{-\infty}^{\infty} \frac{\sin(y)}{y} dy \right]
 \end{aligned}$$

The first term on the RHS is odd and so its integration is zero:

$$\lim_{N \rightarrow \infty} \int_{-\infty}^{\infty} f(\omega) \frac{e^{-j\omega N}}{1 - e^{-j\omega}} d\omega = f(0) \left[0 - \int_{-\infty}^{\infty} \frac{\sin(y)}{y} dy \right] = -f(0) \int_{-\infty}^{\infty} \frac{\sin(y)}{y} dy = -f(0)\pi$$

$$\int_{-\infty}^{\infty} \left[\lim_{N \rightarrow \infty} \frac{e^{-j\omega N}}{1 - e^{-j\omega}} \right] f(\omega) d\omega = -f(0)\pi = \int_{-\infty}^{\infty} [-\pi\delta(\omega)] f(\omega) d\omega$$

Therefore,

$$\lim_{N \rightarrow \infty} \frac{e^{-j\omega N}}{1 - e^{-j\omega}} = -\pi\delta(\omega)$$

Substituting $\lim_{N \rightarrow \infty} (e^{-j\omega N}) / (1 - e^{-j\omega}) = -\pi\delta(\omega)$ into Eq. (6.30), we obtain

$$X(e^{j\omega}) = \frac{1}{1 - e^{-j\omega}} + \pi\delta(\omega)$$

However, the DTFT must be periodic in ω with period 2π . Therefore,

$$X(e^{j\omega}) = U(e^{j\omega}) = \frac{1}{1 - e^{-j\omega}} + \pi \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m)$$

$$u(n) \longleftrightarrow \frac{1}{1 - e^{-j\omega}} + \pi \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m) \quad (6.31)$$

Example 6.9 Find the inverse DTFT of the rectangular pulse spectrum defined only for $-\pi \leq \omega \leq \pi$

$$X(e^{j\omega}) = \text{rect}\left(\frac{\omega}{2\omega_c}\right) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases}$$

which is depicted in Fig. 6.8(a). Alternatively, we can define $X(e^{j\omega})$ over all ω by writing it as an infinite sum of rectangle functions shifted by integer multiple of 2π

$$X(e^{j\omega}) = \sum_{m=-\infty}^{\infty} \text{rect}\left(\frac{\omega - 2\pi m}{2\omega_c}\right)$$

Solution

The inverse DTFT of $X(e^{j\omega})$ is given by

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega$$

$$x(n) = \frac{1}{2\pi} \frac{e^{j\omega n}}{jn} \Big|_{-\omega_c}^{\omega_c}$$

$$x(n) = \frac{\sin(\omega_c n)}{n\pi}, \quad n \neq 0$$

6.5.1 Linearity

If

$$x_1(n) \longleftrightarrow X_1(e^{j\omega}) \quad \text{and} \quad x_2(n) \longleftrightarrow X_2(e^{j\omega})$$

then

$$ax_1(n) + bx_2(n) \longleftrightarrow aX_1(e^{j\omega}) + bX_2(e^{j\omega}) \quad (6.37)$$

Proof The Fourier transform of $ax_1(n) + bx_2(n)$ is

$$\begin{aligned} \mathcal{F}[ax_1(n) + bx_2(n)] &= \sum_{n=-\infty}^{\infty} [ax_1(n) + bx_2(n)] e^{-j\omega n} \\ &= a \underbrace{\sum_{n=-\infty}^{\infty} x_1(n) e^{-j\omega n}}_{X_1(e^{j\omega})} + b \underbrace{\sum_{n=-\infty}^{\infty} x_2(n) e^{-j\omega n}}_{X_2(e^{j\omega})} \\ &= aX_1(e^{j\omega}) + bX_2(e^{j\omega}) \end{aligned}$$

6.5.2 Time Shifting

If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$y(n) = x(n - n_0) \longleftrightarrow Y(e^{j\omega}) = X(e^{j\omega})e^{-j\omega n_0} \quad (6.38)$$

Proof The Fourier transform of $x(n - n_0)$ is

$$\mathcal{F}[x(n - n_0)] = \sum_{n=-\infty}^{\infty} x(n - n_0) e^{-j\omega n}$$

A change of variables is performed by letting $m = (n - n_0)$, which also yields $n = (m + n_0)$, $m = -\infty$ as $n = -\infty$ and $m = \infty$ as $n = \infty$. Therefore,

$$\begin{aligned} \mathcal{F}[x(n - n_0)] &= \sum_{m=-\infty}^{\infty} x(m) e^{-j\omega(m+n_0)} \\ &= e^{-j\omega n_0} \sum_{m=-\infty}^{\infty} x(m) e^{-j\omega m} \end{aligned}$$

$$\mathcal{F}[x(n - n_0)] = X(e^{j\omega}) e^{-j\omega n_0}$$

One consequence of this property is that, when a signal is shifted in time, the magnitudes of its Fourier transform remain unaltered. That is, if we express $X(e^{j\omega})$

in polar form as

$$\mathcal{F}[x(n)] = X(e^{j\omega}) = |X(e^{j\omega})| e^{j\angle X(e^{j\omega})}$$

then

$$\mathcal{F}[x(n - n_0)] = X(e^{j\omega}) e^{-j\omega n_0} = |X(e^{j\omega})| e^{j(\angle X(e^{j\omega}) - \omega n_0)}$$

Thus, the effect of a time shift on a signal is to introduce into its Fourier transform a phase shift, ωn_0 , which is a linear function of ω . ■

Example 6.11 Find the DTFT of the sequence

$$x(n) = \frac{1}{4} \text{sinc} \left(\frac{1}{4}(n - 2) \right)$$

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shown in Fig. 6.10 (a).

Solution

From Eq. (6.34), we have

$$\frac{\omega_c}{\pi} \text{sinc} \left(\frac{\omega_c n}{\pi} \right) \longleftrightarrow \sum_{m=-\infty}^{\infty} \text{rect} \left(\frac{\omega - 2\pi m}{2\omega_c} \right)$$

Substituting $\omega_c = \pi/4$ into the above equation gives

$$\frac{1}{4} \text{sinc} \left(\frac{1}{4}n \right) \longleftrightarrow \sum_{m=-\infty}^{\infty} \text{rect} \left(\frac{\omega - 2\pi m}{\pi/2} \right)$$

The time-shifting property [Eq. (6.38)] yields

$$\frac{1}{4} \text{sinc} \left(\frac{1}{4}(n - 2) \right) \longleftrightarrow \sum_{m=-\infty}^{\infty} \text{rect} \left(\frac{\omega - 2\pi m}{\pi/2} \right) e^{-j2\omega}$$

The magnitude spectrum of the shifted signal is shown in Fig. 6.10(b).

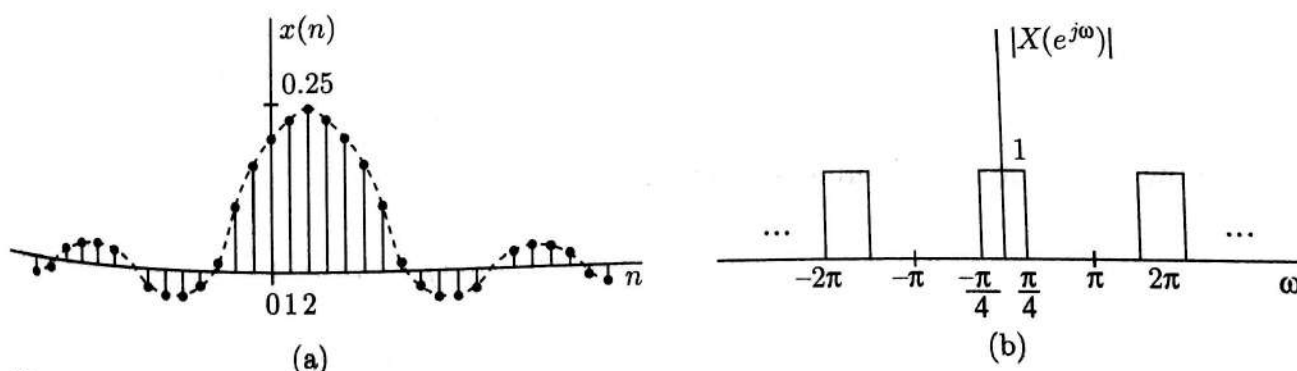


Fig. 6.10 Signal $x(n)$ of Example 6.11 and (b) its DTFT

6.5.3 Frequency Shifting

If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$y(n) = x(n) e^{j\omega_0 n} \longleftrightarrow Y(e^{j\omega}) = X(e^{j(\omega - \omega_0)}) \quad (6.39)$$

Proof The Fourier transform of $x(n) e^{j\omega_0 n}$ is given by

$$\begin{aligned} \mathcal{F}[x(n) e^{j\omega_0 n}] &= \sum_{n=-\infty}^{\infty} [x(n) e^{j\omega_0 n}] e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} x(n) e^{-j(\omega - \omega_0)n} \\ \mathcal{F}[x(n) e^{j\omega_0 n}] &= X(e^{j(\omega - \omega_0)}) \end{aligned}$$

Hence, a frequency shift corresponds to multiplication in time domain by a complex sinusoid whose frequency is equal to the frequency shift. ■

Example 6.12 Find the Fourier transform of an exponential sequence with alternating sign

$$y(n) = (-1)^n a^n u(n), \quad |a| < 1$$

Solution

Define a sequence

$$x(n) = a^n u(n), \quad |a| < 1$$

From Eq. (6.24), its Fourier transform is given by

$$X(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}}$$

Now consider the given sequence

$$y(n) = (-1)^n a^n u(n), \quad |a| < 1$$

Substituting $(-1)^n = e^{j\pi n}$ into the above equation yields

$$y(n) = e^{j\pi n} a^n u(n) = e^{j\pi n} x(n)$$

Now apply the frequency-shifting property [Eq. (6.39)] to get the Fourier transform of $y(n)$:

$$\begin{aligned} Y(e^{j\omega}) &= X(e^{j(\omega-\pi)}) \\ &= \frac{1}{1 - a e^{-j(\omega-\pi)}} \end{aligned}$$

$$Y(e^{j\omega}) = \frac{1}{1 - a e^{j\pi} e^{-j\omega}}$$

$$Y(e^{j\omega}) = \frac{1}{1 + a e^{-j\omega}}$$

$$(-1)^n a^n u(n) \longleftrightarrow \frac{1}{1 + a e^{-j\omega}}, \quad |a| < 1 \quad (6.40)$$

6.5.4 Time Reversal

If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$y(n) = x(-n) \longleftrightarrow Y(e^{j\omega}) = X(e^{-j\omega}) \quad (6.41)$$

Proof The Fourier transform of $y(n)$ is given by

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} y(n) e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} x(-n) e^{-j\omega n} \end{aligned}$$

Substituting $m = -n$ into the above equation gives

$$\begin{aligned} Y(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(m) e^{j\omega m} \\ &= \sum_{n=-\infty}^{\infty} x(m) e^{-j(-\omega)m} \\ Y(e^{j\omega}) &= X(e^{-j\omega}) \end{aligned}$$

An interesting consequence of the time-reversal property is that if $x(n)$ is even, then its Fourier transform is also even, i.e.,

$$\text{if } x(-n) = x(n), \text{ then } X(e^{-j\omega}) = X(e^{j\omega}) \quad (6.42)$$

Similarly, if $x(n)$ is odd, then so is its Fourier transform, i.e.,

$$\text{if } x(-n) = -x(n), \text{ then } X(e^{-j\omega}) = -X(e^{j\omega}) \quad (6.43)$$

6.5.5 Time Expansion

Because of the discrete-time nature of the time index for discrete-time signals, the relation between time and frequency scaling in discrete time takes on a somewhat different form from its continuous-time counterpart. If we try to define the signal $x(an)$, we run into difficulties if a is not an integer. Therefore, we cannot slow down the signal by choosing $a < 1$. On the other hand, if we let a be an integer other than ± 1 , we do not merely speed up the original signal. For example, since n can take on only integer values, the signal $x(2n)$ consists of the even samples of $x(n)$ alone.

Let m be a positive integer, and define the signal

$$x_{(m)}(n) = \begin{cases} x\left(\frac{n}{m}\right), & \text{if } n \text{ is a multiple of } m \\ 0, & \text{if } n \text{ is not a multiple of } m. \end{cases} \quad (6.44)$$

$x_m(n)$ can be obtained from $x(n)$ by placing $(m-1)$ zeros between successive values of the original signal. Intuitively, we can think of $x_{(m)}(n)$ as a slowed-down version of $x(n)$. Now, if

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$y(n) = x_{(m)}(n) \longleftrightarrow Y(e^{j\omega}) = X(e^{jm\omega}) \quad (6.45)$$

Proof The Fourier transform of $y(n) = x_{(m)}(n)$ is given by

$$Y(e^{j\omega}) = \sum_{n=-\infty}^{\infty} y(n) e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x_{(m)}(n) e^{-j\omega n} = \sum_{n=-\infty}^{\infty} x\left(\frac{n}{m}\right) e^{-j\omega n}$$

A change of variables is performed by letting $r = n/m$, which also yields $r = -\infty$ as $n = -\infty$ and $r = \infty$ as $n = \infty$. Therefore,

$$Y(e^{j\omega}) = \sum_{r=-\infty}^{\infty} x(r) e^{-j\omega mr}$$

$$Y(e^{j\omega}) = X(e^{jm\omega})$$

Note that as the signal spread out and slowed down in time by taking $m > 1$, its Fourier transform is compressed.

6.5.6 Differencing in Time Domain

The discrete-time parallel to the differentiation property of the continuous-time Fourier transform involves the use of the first-difference operation, which is defined as

$y(n) = x(n) - x(n-1)$. Now, if

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$y(n) = x(n) - x(n-1) \longleftrightarrow Y(e^{j\omega}) = (1 - e^{-j\omega})X(e^{j\omega}) \quad (6.46)$$

A common use of this property is in situations where evaluation of the Fourier transform is easier for the first difference than for the original sequence.

Proof Given that

$$x(n) \longleftrightarrow X(e^{j\omega})$$

Using the time-shifting property, we get

$$x(n-1) \longleftrightarrow X(e^{j\omega})e^{-j\omega}$$

Now, using the linearity property, we get

$$x(n) - x(n-1) \longleftrightarrow X(e^{j\omega}) - X(e^{j\omega})e^{-j\omega}$$

$$x(n) - x(n-1) \longleftrightarrow (1 - e^{-j\omega})X(e^{j\omega})$$

■

6.5.7 Differentiation in Frequency Domain

If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$-jnx(n) \longleftrightarrow \frac{dX(e^{j\omega})}{d\omega} \quad (6.47)$$

or

$$nx(n) \longleftrightarrow j \frac{dX(e^{j\omega})}{d\omega} \quad (6.48)$$

Proof From Eq. (6.5), we have

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

By differentiating both sides w.r.t. ω , we obtain

$$\frac{dX(e^{j\omega})}{d\omega} = \sum_{n=-\infty}^{\infty} [-jnx(n)] e^{-j\omega n}$$

$$\frac{dX(e^{j\omega})}{d\omega} = \mathcal{F}[-jnx(n)]$$

Therefore,

$$-jnx(n) \longleftrightarrow \frac{dX(e^{j\omega})}{d\omega}$$

Multiplying both sides by j yields

$$nx(n) \longleftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$$

Example 6.13 Determine the Fourier transform of the sequence

$$y(n) = na^n u(n), \quad |a| < 1$$

Solution

Define a sequence

$$x(n) = a^n u(n), \quad |a| < 1$$

From Eq. (6.24), its Fourier transform is given by

$$X(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}}$$

Now consider the given sequence

$$y(n) = na^n u(n) = nx(n)$$

Now apply the differentiation in frequency-domain property [Eq. (6.48)] to get the Fourier transform of $y(n)$:

$$Y(e^{j\omega}) = j \frac{dX(e^{j\omega})}{d\omega} = j \frac{d}{d\omega} \left(\frac{1}{1 - a e^{-j\omega}} \right) = \frac{a e^{-j\omega}}{(1 - a e^{-j\omega})^2}$$

$$na^n u(n) \longleftrightarrow \frac{a e^{-j\omega}}{(1 - a e^{-j\omega})^2}, \quad |a| < 1 \quad (6.49)$$

Example 6.14 Determine the inverse DTFT of

$$X(e^{j\omega}) = \frac{1}{(1 - a e^{-j\omega})^2}, \quad |a| < 1$$

Solution

From Eq. (6.49), we have

$$na^n u(n) \longleftrightarrow \frac{a e^{-j\omega}}{(1 - a e^{-j\omega})^2}, \quad |a| < 1$$

Using the time-shifting property, we obtain

$$\begin{aligned}(n+1)a^{n+1}u(n+1) &\longleftrightarrow \left[\frac{ae^{-j\omega}}{(1-ae^{-j\omega})^2} \right] e^{j\omega} \\(n+1)a^{n+1}u(n+1) &\longleftrightarrow \frac{a}{(1-ae^{-j\omega})^2} \\(n+1)a^n u(n+1) &\longleftrightarrow \frac{1}{(1-ae^{-j\omega})^2}\end{aligned}$$

Therefore,

$$x(n) = (n+1)a^n u(n+1)$$

It is worth noting that although the RHS is multiplied by a step function that begins at $n = -1$, the sequence $(n+1)a^n u(n+1)$ is still zero prior to $n = 0$, since the factor $n+1$ is zero at $n = -1$. Thus, we can alternatively express $x(n)$ as

$$x(n) = (n+1)a^n u(n)$$

and hence,

$$(n+1)a^n u(n) \longleftrightarrow \frac{1}{(1-ae^{-j\omega})^2} \quad (6.50)$$

6.5.8 Convolution Property

If

$$x_1(n) \longleftrightarrow X_1(e^{j\omega}) \quad \text{and} \quad x_2(n) \longleftrightarrow X_2(e^{j\omega})$$

then

$$x_1(n) * x_2(n) \longleftrightarrow X_1(e^{j\omega})X_2(e^{j\omega}) \quad (6.51)$$

The Fourier transform maps the convolution of two signals into the product of their Fourier transforms.

Proof The Fourier transform of $x_1(n) * x_2(n)$ is

$$\begin{aligned}\mathcal{F}[x_1(n) * x_2(n)] &= \sum_{n=-\infty}^{\infty} [x_1(n) * x_2(n)] e^{-j\omega n} \\&= \sum_{n=-\infty}^{\infty} \left(\sum_{m=-\infty}^{\infty} x_1(m)x_2(n-m) \right) e^{-j\omega n}\end{aligned}$$

By interchanging the order of summation, we have

$$\mathcal{F}[x_1(n) * x_2(n)] = \sum_{m=-\infty}^{\infty} x_1(m) \left(\sum_{n=-\infty}^{\infty} x_2(n-m) e^{-j\omega n} \right)$$

By applying the time-shifting property [Eq. (6.38)], the bracketed term is $X_2(e^{j\omega})e^{-j\omega m}$. Substituting this into the above equation yields

$$\begin{aligned}\mathcal{F}[x_1(n) * x_2(n)] &= \sum_{m=-\infty}^{\infty} x_1(m) \left(X_2(e^{j\omega}) e^{-j\omega m} \right) \\ &= X_2(e^{j\omega}) \sum_{m=-\infty}^{\infty} x_1(m) e^{-j\omega m} \\ \mathcal{F}[x_1(n) * x_2(n)] &= X_2(e^{j\omega}) X_1(e^{j\omega}) = X_1(e^{j\omega}) X_2(e^{j\omega})\end{aligned}$$

Therefore,

$$x_1(n) * x_2(n) \longleftrightarrow X_1(e^{j\omega}) X_2(e^{j\omega})$$

The convolution property states that convolution in the time domain corresponds to multiplication in the frequency domain. ■

Example 6.15 Using the convolution property determine the convolution $x(n) = x_1(n) * x_2(n)$ of the sequences

$$x_1(n) = x_2(n) = \{1, \underset{\uparrow}{1}, 1\} = \delta(n+1) + \delta(n) + \delta(n-1) = \begin{cases} 1, & -1 \leq n \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

Solution

The Fourier transform of $x_1(n) = x_2(n)$ is given by

$$\begin{aligned}X_1(e^{j\omega}) = X_2(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_1(n) e^{-j\omega n} = \sum_{n=-1}^1 e^{-j\omega n} \\ &= e^{j\omega} + 1 + e^{-j\omega} \\ &= 1 + 2 \left(\frac{e^{j\omega} + e^{-j\omega}}{2} \right) \\ &= 1 + 2 \cos(\omega)\end{aligned}$$

Now using the convolution property, we have

$$\begin{aligned}\mathcal{F}[x_1(n) * x_2(n)] &= X(e^{j\omega}) = X_1(e^{j\omega}) X_2(e^{j\omega}) \\ &= [1 + 2 \cos(\omega)]^2 \\ X(e^{j\omega}) &= 3 + 4 \cos(\omega) + 2 \cos(2\omega) \\ &= 3 + 2(e^{j\omega} + e^{-j\omega}) + (e^{j2\omega} + e^{-j2\omega}) \\ X(e^{j\omega}) &= e^{j2\omega} + 2e^{j\omega} + 3 + 2e^{-j\omega} + e^{-j2\omega}\end{aligned}$$

Taking the inverse Fourier transform of the above equation, we obtain

$$x(n) = \delta(n+2) + 2\delta(n+1) + 3\delta(n) + 2\delta(n-1) + \delta(n-2) = \{1, 2, \underset{\uparrow}{3}, 2, 1\}$$

6.5.9 Accumulation Property

If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$\sum_{k=-\infty}^n x(k) \longleftrightarrow \frac{1}{1 - e^{-j\omega}} X(e^{j\omega}) + \pi X(e^{j0}) \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m) \quad (6.52)$$

Proof Convolution of a signal $x(n)$ with a unit step function $u(n)$ yields

$$x(n) * u(n) = \sum_{k=-\infty}^{\infty} x(k) u(n-k)$$

Since

$$u(n-k) = \begin{cases} 1, & (n-k) \geq 0 \longrightarrow k \leq n \\ 0, & (n-k) < 0 \longrightarrow k > n \end{cases}$$

we have

$$x(n) * u(n) = \sum_{k=-\infty}^n x(k)$$

Now we can prove the accumulation property of the Fourier transform:

$$\begin{aligned} \sum_{k=-\infty}^n x(k) &= x(n) * u(n) \\ \mathcal{F} \left[\sum_{k=-\infty}^n x(k) \right] &= \mathcal{F}[x(n) * u(n)] \end{aligned}$$

Using the convolution property [Eq. (6.51)] and Eq. (6.31), we obtain

$$\begin{aligned}
 \mathcal{F}\left[\sum_{k=-\infty}^n x(k)\right] &= X(e^{j\omega})U(e^{j\omega}) \\
 &= X(e^{j\omega})\left(\frac{1}{1-e^{-j\omega}} + \pi \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m)\right) \\
 &= \frac{1}{1-e^{-j\omega}}X(e^{j\omega}) + \pi X(e^{j\omega}) \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m) \\
 &= \frac{1}{1-e^{-j\omega}}X(e^{j\omega}) + \pi X(e^{j0}) \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m)
 \end{aligned}$$

therefore,

$$\sum_{k=-\infty}^n x(k) \longleftrightarrow \frac{1}{1-e^{-j\omega}}X(e^{j\omega}) + \pi X(e^{j0}) \sum_{m=-\infty}^{\infty} \delta(\omega - 2\pi m)$$

6.5.10 Multiplication (or Modulation or Windowing) Property

If

$$x_1(n) \longleftrightarrow X_1(e^{j\omega}) \quad \text{and} \quad x_2(n) \longleftrightarrow X_2(e^{j\omega})$$

then

$$x_1(n)x_2(n) \longleftrightarrow \frac{1}{2\pi}[X_1(e^{j\omega}) \circledast X_2(e^{j\omega})] = \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta})X_2(e^{j(\omega-\theta)})d\theta \quad (6.53)$$

where $\circledast$ denotes the periodic convolution. The Fourier transform maps the multiplication of two signals into the convolution of their Fourier transforms.

Proof The Fourier transform of $x_1(n)x_2(n)$ is given by

$$\begin{aligned}
 \mathcal{F}[x_1(n)x_2(n)] &= \sum_{n=-\infty}^{\infty} [x_1(n)x_2(n)] e^{-j\omega n} \\
 &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta}) e^{j\theta n} d\theta \right) x_2(n) e^{-j\omega n}
 \end{aligned}$$

Interchanging the order of integration and summation and noting that $X_1(e^{j\theta})$ does not depend on n , we obtain

$$\mathcal{F}[x_1(n)x_2(n)] = \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta}) \left(\sum_{n=-\infty}^{\infty} [x_2(n) e^{j\theta n}] e^{-j\omega n} \right) d\theta$$

By applying the frequency-shifting property [Eq. (6.39)], the bracketed term is $X_2(e^{j(\omega-\theta)})$. Substituting this into the above equation yields

$$\begin{aligned}\mathcal{F}[x_1(n)x_2(n)] &= \frac{1}{2\pi} \int_{2\pi} X_1(e^{j\theta})X_2(e^{j(\omega-\theta)})d\theta \\ &= \frac{1}{2\pi} [X_1(e^{j\omega}) \circledast X_2(e^{j\omega})]\end{aligned}$$

Therefore,

$$x_1(n)x_2(n) \longleftrightarrow \frac{1}{2\pi} [X_1(e^{j\omega}) \circledast X_2(e^{j\omega})]$$

The convolution integral in Eq. (6.53) is known as the periodic convolution of $X_1(e^{j\omega})$ and $X_2(e^{j\omega})$ because it is the convolution of two periodic functions having the same period. We note that the limits of integration extend over a single period. Furthermore, we note that due to the periodicity of the Fourier transform for discrete-time signals, there is no 'perfect' duality between the time and frequency domains w.r.t. the convolution operation, as in the case of continuous-time signals. Indeed, convolution in the time domain (aperiodic summation) is equivalent to multiplication of continuous periodic Fourier transforms. However, multiplication of aperiodic sequences is equivalent to periodic convolution of their Fourier transforms. ■

6.5.11 Conjugation and Conjugate Symmetry

If

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$x^*(n) \longleftrightarrow X^*(e^{-j\omega}) \quad (6.54)$$

Proof From Eq. (6.5) the Fourier transform of $x^*(n)$ is

$$\begin{aligned}\mathcal{F}[x^*(n)] &= \sum_{n=-\infty}^{\infty} x^*(n)e^{-j\omega n} = \left[\sum_{n=-\infty}^{\infty} x(n)e^{j\omega n} \right]^* \\ &= \left[\sum_{n=-\infty}^{\infty} x(n)e^{-j(-\omega)n} \right]^* \\ &= [X(e^{-j\omega})]^* \\ \mathcal{F}[x^*(n)] &= X^*(e^{-j\omega})\end{aligned}$$

Case I If $x(n)$ is real, i.e., if

$$x^*(n) = x(n)$$

then

$$\begin{aligned}\mathcal{F}[x^*(n)] &= \mathcal{F}[x(n)] \\ X^*(e^{-j\omega}) &= X(e^{j\omega}) \\ X(e^{-j\omega}) &= X^*(e^{j\omega})\end{aligned}\tag{6.55}$$

So, if $x(n)$ is real, the Fourier transform will be *conjugate symmetric*. Also if $x(n)$ is real and even, then from Eqs (6.42) and (6.55), we have

$$X(e^{-j\omega}) = X^*(e^{j\omega}) = X(e^{j\omega})\tag{6.56}$$

That is, if $x(n)$ is real and even, then so is its Fourier transform.

Case II If $x(n)$ is real and odd, then its Fourier transform is purely imaginary and odd. Using Eqs (6.43) and (6.55), we get

$$X(e^{-j\omega}) = X^*(e^{j\omega}) = -X(e^{j\omega})\tag{6.57}$$

Case III Even and odd decomposition of real signals: If $x(n)$ is real and

$$x(n) \longleftrightarrow X(e^{j\omega})$$

then

$$1. \quad x_e(n) = \mathcal{E}\{x(n)\} \longleftrightarrow \operatorname{Re}\{X(e^{j\omega})\} = X_R(e^{j\omega})\tag{6.58}$$

Proof The even part of a signal $x(n)$ is defined as

$$x_e(n) = \frac{1}{2}[x(n) + x(-n)]$$

Taking the Fourier transform, we obtain

$$\begin{aligned}\mathcal{F}[x_e(n)] &= \mathcal{F}\left[\frac{1}{2}[x(n) + x(-n)]\right] \\ &= \frac{1}{2}(\mathcal{F}[x(n)] + \mathcal{F}[x(-n)]) \\ &= \frac{1}{2}(X(e^{j\omega}) + X^*(e^{j\omega})) \\ &= \frac{1}{2}2X_R(e^{j\omega})\end{aligned}$$

$$\mathcal{F}[x_e(n)] = X_R(e^{j\omega})$$

Therefore,

$$x_e(n) = \mathcal{E}\{x(n)\} \longleftrightarrow X_R(e^{j\omega})$$

and

$$2. \quad x_o(n) = \mathcal{O}\{x(n)\} \longleftrightarrow j\operatorname{Im}\{X(e^{j\omega})\} = jX_I(e^{j\omega})\tag{6.59}$$

Proof The odd part of a signal $x(n)$ is defined as

$$x_o(n) = \frac{1}{2}[x(n) - x(-n)]$$

Taking the Fourier transform, we obtain

$$\begin{aligned}
 \mathcal{F}[x_o(n)] &= \mathcal{F}\left[\frac{1}{2}[x(n) - x(-n)]\right] \\
 &= \frac{1}{2}(\mathcal{F}[x(n)] - \mathcal{F}[x(-n)]) \\
 &= \frac{1}{2}(X(e^{j\omega}) - X^*(e^{j\omega})) \\
 &= \frac{1}{2}2jX_I(e^{j\omega}) \\
 \mathcal{F}[x_o(n)] &= jX_I(e^{j\omega})
 \end{aligned}$$

Therefore,

$$x_o(n) = \mathcal{O}\{x(n)\} \longleftrightarrow jX_I(e^{j\omega})$$

6.5.12 Parseval's Relation

Let $x(n)$ be an energy signal, and if

$$x(n) \longleftrightarrow X(e^{j\omega})$$

@sTc

then

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega \quad (6.60)$$

Proof Consider the LHS of Eq. (6.60), we have

$$\begin{aligned}
 E_x &= \sum_{n=-\infty}^{\infty} |x(n)|^2 = \sum_{n=-\infty}^{\infty} x(n)x^*(n) \\
 &= \sum_{n=-\infty}^{\infty} x(n) \left(\frac{1}{2\pi} \int_{2\pi} X(e^{j\omega}) e^{j\omega n} d\omega \right)^* \\
 &= \sum_{n=-\infty}^{\infty} x(n) \left(\frac{1}{2\pi} \int_{2\pi} X^*(e^{j\omega}) e^{-j\omega n} d\omega \right) \\
 &= \frac{1}{2\pi} \int_{2\pi} X^*(e^{j\omega}) \left(\sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \right) d\omega \\
 &= \frac{1}{2\pi} \int_{2\pi} X^*(e^{j\omega}) X(e^{j\omega}) d\omega \\
 \sum_{n=-\infty}^{\infty} |x(n)|^2 &= \frac{1}{2\pi} \int_{2\pi} |X(e^{j\omega})|^2 d\omega
 \end{aligned}$$

Example 6.16 Find the energy of the sequence

$$x(n) = \text{sinc}\left(\frac{\omega_c n}{\pi}\right)$$

assuming $\omega_c < \pi$.

Solution

From Eq. (6.34), we have

$$\begin{aligned} \frac{\omega_c}{\pi} \text{sinc}\left(\frac{\omega_c n}{\pi}\right) &\longleftrightarrow \sum_{m=-\infty}^{\infty} \text{rect}\left(\frac{\omega - 2\pi m}{2\omega_c}\right) \\ x(n) = \text{sinc}\left(\frac{\omega_c n}{\pi}\right) &\longleftrightarrow X(e^{j\omega}) = \frac{\pi}{\omega_c} \sum_{m=-\infty}^{\infty} \text{rect}\left(\frac{\omega - 2\pi m}{2\omega_c}\right) \end{aligned}$$

Hence, from the Parseval's relation in Eq. (6.60), we have

$$\begin{aligned} E_x &= \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega \\ E_x &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{\pi^2}{\omega_c^2} \left| \text{rect}\left(\frac{\omega}{2\omega_c}\right) \right|^2 d\omega \end{aligned} \quad (6.61)$$

We know that

$$\text{rect}\left(\frac{\omega}{2\omega_c}\right) = \begin{cases} 1, & |\omega| < \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases}$$

Substituting the value of $\text{rect}(\omega/2\omega_c)$ into Eq. (6.61) yields

$$E_x = \frac{1}{2\pi} \frac{\pi^2}{\omega_c^2} \int_{-\omega_c}^{\omega_c} 1 d\omega = \frac{\pi}{2\omega_c^2} 2\omega_c = \frac{\pi}{\omega_c}$$

6.6 SOME IMPORTANT RESULTS

Let $X(e^{j\omega})$ denote the DTFT of $x(n)$.

1.

$$X(e^{j0}) = \sum_{n=-\infty}^{\infty} x(n) \quad (6.62)$$

Proof From Eq. (6.5), we have

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

Substituting $\omega = 0$ into the above equation gives

$$X(e^{j\omega})\Big|_{\omega=0} = X(e^{j0}) = \sum_{n=-\infty}^{\infty} x(n)$$

2.

$$X(e^{j\pi}) = \sum_{n=-\infty}^{\infty} (-1)^n x(n) \quad (6.63)$$

Proof By definition

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

Substituting $\omega = \pi$ into the above equation yields

$$\begin{aligned} X(e^{j\omega})\Big|_{\omega=\pi} &= X(e^{j\pi}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\pi n} = \sum_{n=-\infty}^{\infty} x(n) (e^{-j\pi})^n \\ &= \sum_{n=-\infty}^{\infty} x(n) (\cos(\pi) - j \sin(\pi))^n \\ X(e^{j\pi}) &= \sum_{n=-\infty}^{\infty} x(n) (-1 - j0)^n \\ X(e^{j\pi}) &= \sum_{n=-\infty}^{\infty} (-1)^n x(n) \end{aligned}$$

3.

$$\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega = 2\pi x(0) \quad (6.64)$$

Proof From Eq. (6.4), we have

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Substituting $n = 0$ into the above equation yields

$$x(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$$

$$2\pi x(0) = \int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$$

4.

$$\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega = 2\pi \sum_{n=-\infty}^{\infty} |x(n)|^2 \quad (6.65)$$

Proof From the Parseval's relation in Eq. (6.60), we have

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

$$2\pi \sum_{n=-\infty}^{\infty} |x(n)|^2 = \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

5.

$$\int_{-\pi}^{\pi} \left| \frac{dX(e^{j\omega})}{d\omega} \right|^2 d\omega = 2\pi \sum_{n=-\infty}^{\infty} |nx(n)|^2 \quad (6.66)$$

Proof From the differentiation in frequency-domain property [Eq. (6.48)], we have

$$y(n) = nx(n) \longleftrightarrow Y(e^{j\omega}) = j \frac{dX(e^{j\omega})}{d\omega}$$

Now, from the Parseval's relation in Eq. (6.60), we have

$$\sum_{n=-\infty}^{\infty} |y(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |Y(e^{j\omega})|^2 d\omega$$

$$\sum_{n=-\infty}^{\infty} |nx(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left| j \frac{dX(e^{j\omega})}{d\omega} \right|^2 d\omega$$

$$2\pi \sum_{n=-\infty}^{\infty} |nx(n)|^2 = \int_{-\pi}^{\pi} \left| \frac{dX(e^{j\omega})}{d\omega} \right|^2 d\omega$$

Example 6.17 Let $X(e^{j\omega})$ denote the Fourier transform of the signal $x(n]$ depicted in Fig. 6.11. Perform the following calculations without explicitly evaluating $X(e^{j\omega})$:

- Evaluate $X(e^{j0})$.
- Find $\angle X(e^{j\omega})$.
- Evaluate $\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$.
- Find $X(e^{j\pi})$.
- Determine and sketch the signal whose Fourier transform is $\Re\{X(e^{j\omega})\}$.
- Evaluate $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$.
- Evaluate $\int_{-\pi}^{\pi} \left| \frac{dX(e^{j\omega})}{d\omega} \right|^2 d\omega$.

Solution

From Fig. 6.11, we can observe that

$$x(n) = \{-1 \quad 0 \quad 1 \quad \underset{\uparrow}{2} \quad 1 \quad 0 \quad 1 \quad 2 \quad 1 \quad 0 \quad -1\}, \quad -3 \leq n \leq 7$$

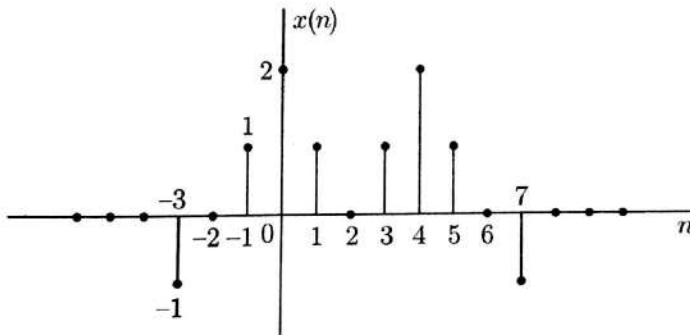


Fig. 6.11 Signal $x(n)$ for Example 6.17

- (a) From Eq. (6.62), we have

$$X(e^{j0}) = \sum_{n=-\infty}^{\infty} x(n) = \sum_{n=-3}^7 x(n) = 6$$

- (b) We can observe that $y(n) = x(n+2)$ is real and even. Therefore (using the properties of the DTFT) the DTFT $Y(e^{j\omega}) = \mathcal{F}[y(n)]$ is real and even. Hence, $Y(e^{j\omega})$ has zero phase. We also know that

$$y(n) = x(n+2)$$

$$\mathcal{F}[y(n)] = \mathcal{F}[x(n+2)]$$

$$Y(e^{j\omega}) = X(e^{j\omega}) e^{j2\omega}$$

or, equivalently,

$$X(e^{j\omega}) = Y(e^{j\omega}) e^{-j2\omega}$$

Since $Y(e^{j\omega})$ has zero phase, we have

$$\angle X(e^{j\omega}) = -2\omega$$

(c) From Eq. (6.64), we have

$$\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega = 2\pi x(0) = 4\pi$$

(d) From Eq. (6.63), we have

$$\begin{aligned} X(e^{j\pi}) &= \sum_{n=-\infty}^{\infty} (-1)^n x(n) \\ &= \sum_{n=-3}^7 (-1)^n x(n) \\ &= -x(-3) + x(-2) - x(-1) + x(0) - x(1) + x(2) - x(3) + x(4) - x(5) \\ &\quad + x(6) - x(7) \\ &= 1 + 0 - 1 + 2 - 1 + 0 - 1 + 2 - 1 + 0 + 1 = 2 \end{aligned}$$

(e) From Eq. (6.58), we have

$$x_e(n) \longleftrightarrow \Re\{X(e^{j\omega})\}$$

Therefore, the desired signal is

$$x_e(n) = \frac{x(n) + x(-n)}{2}$$

n	-7	-6	-5	-4	-3	-2	-1	0	1	2	3	4	5	6	7
$x(n)$	0	0	0	0	-1	0	1	2	1	0	1	2	1	0	-1
$x(-n)$	-1	0	1	2	1	0	1	2	1	0	-1	0	0	0	0
$\mathcal{E}\{x(n)\} = x_e(n)$	-1/2	0	1/2	1	0	0	1	2	1	0	0	1	1/2	0	-1/2

The signal is as shown in Fig. 6.12.

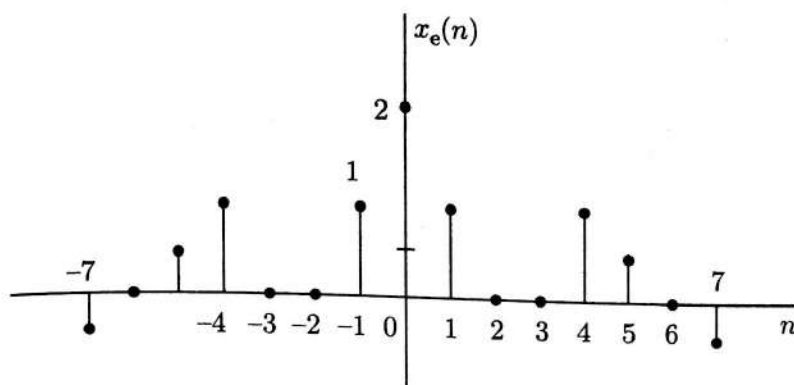


Fig. 6.12

(f) From Eq. (6.65), we have

$$\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega = 2\pi \sum_{n=-\infty}^{\infty} |x(n)|^2 = 2\pi \sum_{n=-3}^7 |x(n)|^2 = 28\pi$$

(g) From Eq. (6.66), we have

$$\int_{-\pi}^{\pi} \left| \frac{dX(e^{j\omega})}{d\omega} \right|^2 d\omega = 2\pi \sum_{n=-\infty}^{\infty} |nx(n)|^2 = 2\pi \sum_{n=-\infty}^{\infty} |n|^2 |x(n)|^2 = 2\pi \sum_{n=-3}^7 |n|^2 |x(n)|^2 = 316\pi$$

6.7 FOURIER TRANSFORM OF PERIODIC SIGNALS

We can construct the Fourier transform of a periodic signal directly from its Fourier series representation. The resulting transform consists of a train of impulses in the frequency domain, with the areas of the impulses proportional to the Fourier series coefficients.

Consider the periodic signal $x(n)$ with period N and fundamental frequency $\omega_0 = 2\pi/N$. The signal $x(n)$ has the discrete-time Fourier series representation

$$x(n) = \sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n}$$

Taking the Fourier transform on both sides yields

$$\begin{aligned} \mathcal{F}[x(n)] &= \mathcal{F} \left[\sum_{k=\langle N \rangle} X_k e^{jk\omega_0 n} \right] \\ X(e^{j\omega}) &= \sum_{k=\langle N \rangle} X_k \mathcal{F}[e^{jk\omega_0 n}] \end{aligned}$$

Using Eq. (6.29), we have

$$\mathcal{F}[e^{jk\omega_0 n}] = 2\pi\delta(\omega - k\omega_0) \quad 0 \leq \omega < 2\pi$$

Therefore,

$$X(e^{j\omega}) = \frac{1}{2} \mathcal{F}[e^{j\omega_0 n}] + \frac{1}{2} \mathcal{F}[e^{-j\omega_0 n}]$$

$$X(e^{j\omega}) = \frac{1}{2} \sum_{m=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 - 2\pi m) + \frac{1}{2} \sum_{m=-\infty}^{\infty} 2\pi \delta(\omega + \omega_0 - 2\pi m)$$

$$X(e^{j\omega}) = \frac{1}{2} \sum_{m=-\infty}^{\infty} 2\pi \delta\left(\omega - \frac{2\pi}{5} - 2\pi m\right) + \frac{1}{2} \sum_{m=-\infty}^{\infty} 2\pi \delta\left(\omega + \frac{2\pi}{5} - 2\pi m\right)$$

The DTFT $X(e^{j\omega})$ is illustrated in Fig. 6.15.

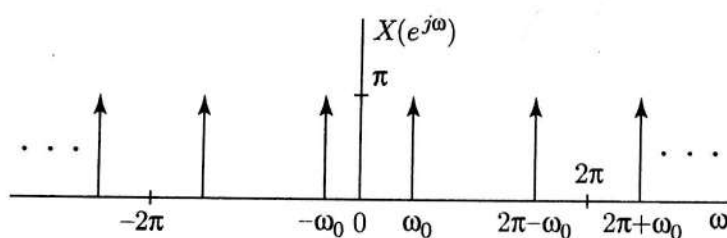


Fig. 6.15 DTFT of $x(n) = \cos(\omega_0 n)$

Example 6.20 Find the Fourier transform of the periodic signal

$$x(n) = \sin(\omega_0 n)$$

with $\omega_0 = 2\pi/5$.

Solution

Consider the given signal

$$x(n) = \sin(\omega_0 n) = \frac{1}{2j} e^{j\omega_0 n} - \frac{1}{2j} e^{-j\omega_0 n}$$

We know that [Eq. (6.29)]

$$e^{j\omega_0 n} \longleftrightarrow \sum_{m=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 - 2\pi m)$$

Therefore,

$$X(e^{j\omega}) = \frac{1}{2j} \mathcal{F}[e^{j\omega_0 n}] - \frac{1}{2j} \mathcal{F}[e^{-j\omega_0 n}]$$

$$X(e^{j\omega}) = \frac{1}{2j} \sum_{m=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 - 2\pi m) - \frac{1}{2j} \sum_{m=-\infty}^{\infty} 2\pi \delta(\omega + \omega_0 - 2\pi m)$$

$$X(e^{j\omega}) = \frac{1}{2j} \sum_{m=-\infty}^{\infty} 2\pi \delta\left(\omega - \frac{2\pi}{5} - 2\pi m\right) - \frac{1}{2j} \sum_{m=-\infty}^{\infty} 2\pi \delta\left(\omega + \frac{2\pi}{5} - 2\pi m\right)$$

The DTFT $X(e^{j\omega})$ is illustrated in Fig. 6.16.

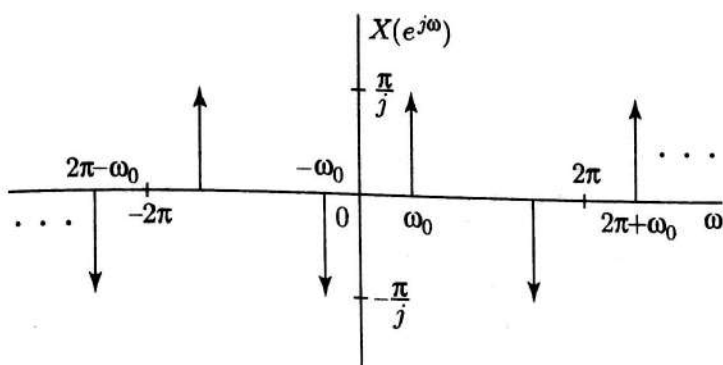


Fig. 6.16 DTFT of $x(n) = \sin(\omega_0 n)$

6.8 SIGNAL TRANSMISSION THROUGH LTI SYSTEMS

If $x(n)$ and $y(n)$ are the input and output of an LTI system with impulse response $h(n)$, then

$$y(n) = x(n) * h(n)$$

Application of the time convolution property to the above equation yields [assuming that both $x(n)$ and $h(n)$ are Fourier transformable]

$$\mathcal{F}[y(n)] = \mathcal{F}[x(n)] * \mathcal{F}[h(n)]$$

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega}) \quad (6.68)$$

or

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} \quad (6.69)$$

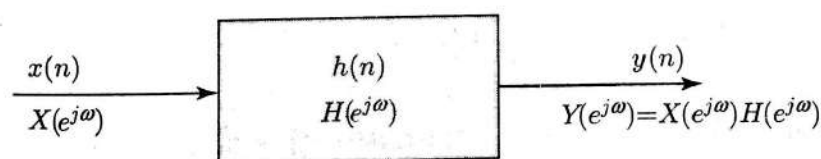


Fig. 6.17

The use of the convolution property for LTI systems is demonstrated in Fig. 6.17. Equation (6.68) can be expressed in the polar form as

$$|Y(e^{j\omega})| e^{j\angle Y(e^{j\omega})} = \left[|X(e^{j\omega})| e^{j\angle X(e^{j\omega})} \right] \left[|H(e^{j\omega})| e^{j\angle H(e^{j\omega})} \right]$$

$$|Y(e^{j\omega})| e^{j\angle Y(e^{j\omega})} = |X(e^{j\omega})| |H(e^{j\omega})| e^{j[\angle X(e^{j\omega}) + \angle H(e^{j\omega})]}$$

Therefore,

$$|Y(e^{j\omega})| = |X(e^{j\omega})| |H(e^{j\omega})| \quad (6.70)$$

and

$$\angle Y(e^{j\omega}) = \angle X(e^{j\omega}) + \angle H(e^{j\omega}) \quad (6.71)$$

During transmission, the input signal magnitude spectrum $|X(e^{j\omega})|$ is changed to $|X(e^{j\omega})||H(e^{j\omega})|$. Similarly, the input signal phase spectrum $\angle X(e^{j\omega})$ is changed to $\angle X(e^{j\omega}) + \angle H(e^{j\omega})$. An input signal spectral component of frequency ω is

attenuates a given sinusoidal component of the input and how much of a phase shift the system adds to that particular component.

We say that the complex sinusoid $x(n) = e^{j\omega_0 n}$ is an *eigenfunction* of the system associated with the *eigenvalue* $H(e^{j\omega_0})$. An eigenfunction of a system is an input signal that produces an output that differs from the input by a constant multiplicative factor.

6.8.2 Steady-State and Transient Responses

The output $y(n)$ of an LTI discrete-time system characterized by a constant coefficient difference equation is a sum of two parts: the complementary solution $y_c(n)$ (zero-input response) resulting with the input $x(n) = 0$ and a particular solution $y_p(n)$ (forced response) resulting from the specified input $x(n)$. Consider an LTI discrete-time system characterized by a linear constant coefficient difference equation of the form

$$\sum_{k=0}^N d_k y(n-k) = \sum_{k=0}^M x(n-k) \quad (6.73)$$

where $x(n)$ and $y(n)$ are, respectively, the input and output of the system, and d_k and p_k are constants. The *order* of the discrete-time system is given by $\max(N, M)$, which is the order of the difference equation characterizing the system. The complementary solution $y_c(n)$ is the solution of Eq. (6.73) with the input $x(n) = 0$, i.e., it is the solution of the homogeneous difference equation:

$$\sum_{k=0}^N d_k y(n-k) = 0 \quad (6.74)$$

We assume that the complementary solution $y_c(n)$ is of the form

$$y_c(n) = \lambda^n$$

Substituting the above in Eq. (6.74), we arrive at

$$\begin{aligned} \sum_{k=0}^N d_k y(n-k) &= \sum_{k=0}^N d_k \lambda^{n-k} = 0 \\ \lambda^{n-N} (d_0 \lambda^N + d_1 \lambda^{N-1} + \cdots + d_{N-1} \lambda + d_N) &= 0 \end{aligned}$$

The polynomial $\sum_{k=0}^N d_k \lambda^{n-k}$ is called the characteristic polynomial of the discrete-time system of Eq. (6.73). Let $\lambda_1, \lambda_2, \dots, \lambda_N$ denote its N roots. If these roots are all distinct, then the general form of the complementary solution is given by

$$y_c(n) = \alpha_1 \lambda_1^n + \alpha_2 \lambda_2^n + \cdots + \alpha_N \lambda_N^n \quad (6.75)$$

where $\alpha_1, \alpha_2, \dots, \alpha_N$ are constants determined from the specified initial conditions of the discrete-time system and λ_i are the roots of the characteristic polynomial. Moreover, for a stable system, $|\lambda_i| < 1$, and as a result, the complementary solution $y_c(n)$ decays to zero for large values of n . The complementary solution part $y_c(n)$ of the output is also referred to as the *transient response*. We now examine the behaviour of the particular solution $y_p(n)$ for two specific types of input.

1. Consider first an input sequence with a constant amplitude starting at some time instant, say n_0 , and then continuing forever afterwards. The output of a causal stable LTI discrete-time system will then be composed of a *steady-state response* (the particular solution), which is also a constant amplitude sequence, and a transient response with zero-valued samples after some time instant $n_1 > n_0$, resulting in a constant amplitude output after the time instant n_1 .
2. Consider a constant amplitude sinusoidal input sequence. The output of a causal stable LTI system with a causal input, i.e., a constant amplitude sinusoidal sequence, will have after some time instant a steady-state output that is also a constant amplitude sinusoidal sequence of the same angular frequency as that of the input. In this case also, before the steady-state output is reached, the initial samples of the output will be composed of samples of the transient response and the steady-state response.

It is straightforward to develop the expression for the steady-state response of a LTI discrete-time system with a real impulse response $h(n)$ for a sinusoidal input in terms of its frequency response $H(e^{j\omega})$. Let the input to the LTI system be

$$x(n) = A \cos(\omega_0 n), \quad -\infty < n < \infty \quad (6.76)$$

$$x(n) = \frac{A}{2} e^{j\omega_0 n} + \frac{A}{2} e^{-j\omega_0 n}$$

Because of the linearity, the response $y(n)$ to the input $x(n)$ is given by

$$\begin{aligned} y(n) &= \frac{A}{2} H(e^{j\omega_0}) e^{j\omega_0 n} + \frac{A}{2} H(e^{-j\omega_0}) e^{-j\omega_0 n} \\ &= \frac{A}{2} |H(e^{j\omega_0})| e^{j\angle H(e^{j\omega_0})} e^{j\omega_0 n} + \frac{A}{2} |H(e^{-j\omega_0})| e^{j\angle H(e^{-j\omega_0})} e^{-j\omega_0 n} \\ &= \frac{A}{2} |H(e^{j\omega_0})| e^{j\angle H(e^{j\omega_0})} e^{j\omega_0 n} + \frac{A}{2} |H(e^{j\omega_0})| e^{-j\angle H(e^{j\omega_0})} e^{-j\omega_0 n} \end{aligned}$$

where we have used the fact that the magnitude response of the LTI system with a real impulse response is an even function of ω , i.e., $|H(e^{j\omega_0})| = |H(e^{-j\omega_0})|$, and the phase response is an odd function of ω , i.e., $\angle H(e^{j\omega_0}) = -\angle H(e^{-j\omega_0})$. The output of the LTI system thus can be expressed as

$$y(n) = \frac{A}{2} |H(e^{j\omega_0})| e^{j(\omega_0 n + \angle H(e^{j\omega_0}))} + \frac{A}{2} |H(e^{j\omega_0})| e^{-j(\omega_0 n + \angle H(e^{j\omega_0}))}$$

$$y(n) = A |H(e^{j\omega_0})| \cos(\omega_0 n + \angle H(e^{j\omega_0})) \quad (6.77)$$

Thus, the output signal $y(n)$ has the same sinusoidal waveform as the input signal $x(n)$ of Eq. (6.76), with two differences:

1. The amplitude is multiplied by $|H(e^{j\omega_0})|$, the value of the magnitude response of the discrete-time system at $\omega = \omega_0$.
2. The output signal has a phase lag relative to the input by an amount $\angle H(e^{j\omega_0})$, the value of the phase response of the discrete-time system at $\omega = \omega_0$. The expression

developed in Eq. (6.77) is the response of an LTI system to a sinusoidal input signal applied to the system at $n = -\infty$. We usually called such signal eternal sinusoids because they were applied at $n = -\infty$. In such a case, the response $y(n)$ is the steady-state response. There is no transient response in this case.

6.8.3 Response to a Causal Exponential Sequence

Consider a causal sequence applied at the input of a causal LTI discrete-time system at time $n = 0$. The output for such an input when observed at time $n = 0$ will consists of a transient part along with a steady-state component and thus will be different from the one shown in Eq. (6.77). To demonstrate this behaviour, we next develop the expression for the output $y(n)$ of a causal LTI discrete-time system when the input is a causal exponential sequence at $n = 0$, i.e.,

$$x(n) = e^{j\omega_0 n} u(n) \quad (6.78)$$

where $u(n)$ is the unit step sequence. Since $x(n) = 0$, for $n < 0$, we have $y(n) = 0$, for $n < 0$. For $n \geq 0$, the output $y(n)$ is given by

$$\begin{aligned} y(n) &= h(n) * x(n) \\ &= \sum_{m=0}^{\infty} h(m) x(n-m) \\ &= \sum_{m=0}^{\infty} h(m) e^{j\omega_0(n-m)} u(n-m) \\ &= \sum_{m=0}^n h(m) e^{-j\omega_0 m} e^{j\omega_0 n} \\ &= \left(\sum_{m=0}^n h(m) e^{-j\omega_0 m} \right) e^{j\omega_0 n} + \left(\sum_{m=n+1}^{\infty} h(m) e^{-j\omega_0 m} \right) e^{j\omega_0 n} \\ &\quad - \left(\sum_{m=n+1}^{\infty} h(m) e^{-j\omega_0 m} \right) e^{j\omega_0 n} \\ y(n) &= \left(\sum_{m=0}^{\infty} h(m) e^{-j\omega_0 m} \right) e^{j\omega_0 n} - \left(\sum_{m=n+1}^{\infty} h(m) e^{-j\omega_0 m} \right) e^{j\omega_0 n} \\ y(n) &= H(e^{j\omega_0}) e^{j\omega_0 n} - \left(\sum_{m=n+1}^{\infty} h(m) e^{-j\omega_0 m} \right) e^{j\omega_0 n} \quad n \geq 0 \end{aligned} \quad (6.79)$$

The first term in the above equation is the same as that given by Eq. (6.72) and is the *steady-state response*:

$$y_{sr}(n) = H(e^{j\omega_0}) e^{j\omega_0 n} \quad (6.80)$$

The second term in Eq. (6.79) is the *transient response*:

$$y_{tr}(n) = - \left(\sum_{m=n+1}^{\infty} h(m) e^{-j\omega_0 m} \right) e^{j\omega_0 n} \quad (6.81)$$

To determine the effect of the second term on the output response, we observe that

$$|y_{tr}(n)| = \left| \sum_{m=n+1}^{\infty} h(m) e^{-j\omega_0(m-n)} \right| \leq \sum_{m=n+1}^{\infty} |h(m)| \leq \sum_{m=0}^{\infty} |h(m)| \quad (6.82)$$

For a causal and stable LTI discrete-time system, the impulse response is absolutely summable, and as a result the transient response $y_{tr}(n)$ is a bounded sequence. Moreover, as $n \rightarrow \infty$, $\sum_{m=n+1}^{\infty} |h(m)| \rightarrow 0$ and hence, the transient response decays to zero as n gets very large. In most practical cases, the transient response becomes negligibly small after some finite amount of time, and the system can be assumed to be in a steady state. It should be noted that transients will occur whenever an input signal is applied or changed.

Example 6.21 Consider a causal LTI system that is characterized by the difference equation

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = 2x(n)$$

- (a) Find the frequency response $H(e^{j\omega})$ and the impulse response $h(n)$ of the system.
- (b) Find the response $y(n)$ if the input to this system is

$$x(n) = \left(\frac{1}{4} \right)^n u(n)$$

Solution

- (a) Consider the given difference equation

$$y(n) - \frac{3}{4}y(n-1) + \frac{1}{8}y(n-2) = 2x(n)$$

Taking the DTFT of the above equation gives

$$\begin{aligned} Y(e^{j\omega}) - \frac{3}{4}Y(e^{j\omega})e^{-j\omega} + \frac{1}{8}Y(e^{j\omega})e^{-j2\omega} &= 2X(e^{j\omega}) \\ Y(e^{j\omega}) \left(1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega} \right) &= 2X(e^{j\omega}) \end{aligned}$$

From Eq. (6.69), the frequency response of this system is

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{2}{1 - \frac{3}{4}e^{-j\omega} + \frac{1}{8}e^{-j2\omega}}$$

Now, by factorizing the denominator of the above equation, we obtain

$$H(e^{j\omega}) = \frac{2}{\left(1 - \frac{1}{2}e^{-j\omega}\right)\left(1 - \frac{1}{4}e^{-j\omega}\right)}$$

For discrete-time transforms, it is most convenient to substitute p for $e^{-j\omega}$. By making this substitution, we obtain the rational system function

$$H(e^{j\omega}) = \frac{2}{\left(1 - \frac{1}{2}p\right)\left(1 - \frac{1}{4}p\right)}$$

Using the partial fraction expansion, we obtain

$$H(p) = \frac{A_1}{1 - \frac{1}{2}p} + \frac{A_2}{1 - \frac{1}{4}p}$$

$$A_1 = \left[\left(1 - \frac{1}{2}p\right) H(p) \right] \bigg|_{p=2} = \frac{2}{1 - 1/2} = 4$$

$$A_2 = \left[\left(1 - \frac{1}{4}p\right) H(p) \right] \bigg|_{p=4} = \frac{2}{1 - 2} = -2$$

Thus,

$$H(p) = \frac{4}{1 - \left(\frac{1}{2}\right)p} - \frac{2}{1 - \left(\frac{1}{4}\right)p}$$

$$H(e^{j\omega}) = \frac{4}{1 - \left(\frac{1}{2}\right)e^{-j\omega}} - \frac{2}{1 - \left(\frac{1}{4}\right)e^{-j\omega}}$$

and taking the inverse DTFT of the above equation, we obtain the unit impulse response:

$$h(n) = 4 \left(\frac{1}{2}\right)^n u(n) - 2 \left(\frac{1}{4}\right)^n u(n)$$

(b) The DTFT of the given input $x(n) = (1/4)^n u(n)$ is given by

$$X(e^{j\omega}) = \frac{1}{1 - (1/4)e^{-j\omega}}$$

Using Eq. (6.68), we obtain

$$Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega}) = \left[\frac{2}{\left(1 - \left(\frac{1}{2}\right)e^{-j\omega}\right)\left(1 - \left(\frac{1}{4}\right)e^{-j\omega}\right)} \right] \left[\frac{1}{1 - \left(\frac{1}{4}\right)e^{-j\omega}} \right]$$

$$= \frac{2}{\left(1 - \left(\frac{1}{2}\right)e^{-j\omega}\right)\left(1 - \left(\frac{1}{4}\right)e^{-j\omega}\right)^2}$$

Using the partial fraction expansion, we obtain

$$Y(e^{j\omega}) = -\frac{4}{1 - \left(\frac{1}{4}\right)e^{-j\omega}} - \frac{2}{\left(1 - \left(\frac{1}{4}\right)e^{-j\omega}\right)^2} + \frac{8}{1 - \left(\frac{1}{2}\right)e^{-j\omega}}$$

$$Y(e^{j\omega}) = Y_1(e^{j\omega}) + Y_2(e^{j\omega}) + Y_3(e^{j\omega})$$

Taking the inverse DTFT gives

$$y(n) = -y_1(n) - y_2(n) + y_3(n)$$

where $y_1(n) = 4(1/4)^n u(n)$ and $y_3(n) = 8(1/2)^n u(n)$, and to determine $y_2(n)$, substitute $a = 1/4$ in Eq. (6.50) to obtain

$$(n+1) \left(\frac{1}{4}\right)^n u(n) \longleftrightarrow \frac{1}{(1 - \frac{1}{4}e^{-j\omega})^2}$$

$$2(n+1) \left(\frac{1}{4}\right)^n u(n) \longleftrightarrow \frac{2}{(1 - \frac{1}{4}e^{-j\omega})^2}$$

Therefore,

$$y_2(n) = 2(n+1) \left(\frac{1}{4}\right)^n u(n)$$

and

$$y(n) = -4 \left(\frac{1}{4}\right)^n u(n) - 2(n+1) \left(\frac{1}{4}\right)^n u(n) + 8 \left(\frac{1}{2}\right)^n u(n)$$

$$y(n) = \left[-4 \left(\frac{1}{4}\right)^n - 2(n+1) \left(\frac{1}{4}\right)^n + 8 \left(\frac{1}{2}\right)^n \right] u(n)$$

Example 6.22 Suppose that we want to design a discrete-time LTI system with the property that if the input is

$$x(n) = \left(\frac{1}{2}\right)^n u(n) - \frac{1}{4} \left(\frac{1}{2}\right)^{n-1} u(n-1)$$

then the output is

$$y(n) = \left(\frac{1}{3}\right)^n u(n)$$

- Find the impulse response and frequency response of a discrete-time LTI system that has the foregoing property.
- Find a difference equation relating $x(n)$ and $y(n)$ that characterizes the system.

Solution

(a) Taking the DTFT of the given input $x(n)$, we obtain

$$X(e^{j\omega}) = \frac{1}{1 - (\frac{1}{2})e^{-j\omega}} - \frac{1}{4} \frac{1}{1 - (\frac{1}{2})e^{-j\omega}} e^{-j\omega} = \frac{1 - (\frac{1}{4})e^{-j\omega}}{1 - (\frac{1}{2})e^{-j\omega}}$$

Taking the DTFT of the output $y(n)$, we obtain

$$Y(e^{j\omega}) = \frac{1}{1 - (\frac{1}{3})e^{-j\omega}}$$

The frequency response of the system is given by

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1 - \left(\frac{1}{2}\right) e^{-j\omega}}{\left(1 - \left(\frac{1}{3}\right) e^{-j\omega}\right) \left(1 - \left(\frac{1}{4}\right) e^{-j\omega}\right)}$$

(b) From part (a), we have

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1 - \left(\frac{1}{2}\right) e^{-j\omega}}{\left(1 - \left(\frac{1}{3}\right) e^{-j\omega}\right) \left(1 - \left(\frac{1}{4}\right) e^{-j\omega}\right)}$$

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1 - \frac{1}{2} e^{-j\omega}}{1 - \left(\frac{7}{12}\right) e^{-j\omega} + \left(\frac{1}{12}\right) e^{-j2\omega}}$$

$$Y(e^{j\omega}) \left[1 - \frac{7}{12} e^{-j\omega} + \frac{1}{12} e^{-j2\omega}\right] = X(e^{j\omega}) \left[1 - \frac{1}{2} e^{-j\omega}\right]$$

$$Y(e^{j\omega}) - \frac{7}{12} e^{-j\omega} Y(e^{j\omega}) + \frac{1}{12} e^{-j2\omega} Y(e^{j\omega}) = X(e^{j\omega}) - \frac{1}{2} e^{-j\omega} X(e^{j\omega})$$

Taking the inverse DTFT yields

$$y(n) - \frac{7}{12} y(n-1) + \frac{1}{12} y(n-2) = x(n) - \frac{1}{2} x(n-1)$$

Example 6.23 Suppose that a system has the response $(1/4)^n u(n)$ to the input $(n+2)(1/2)^n u(n)$. If the output of this system is $\delta(n) - (-1/2)^n u(n)$, what is the input?

Solution

Given that

$$x(n) = (n+2) \left(\frac{1}{2}\right)^n u(n) = n \left(\frac{1}{2}\right)^n u(n) + 2 \left(\frac{1}{2}\right)^n u(n)$$

Taking the DTFT of the above equation, we obtain

$$X(e^{j\omega}) = \frac{\left(\frac{1}{2}\right) e^{-j\omega}}{\left(1 - \left(\frac{1}{2}\right) e^{-j\omega}\right)^2} + 2 \frac{1}{1 - \left(\frac{1}{2}\right) e^{-j\omega}} = \frac{2 \left(1 - \left(\frac{1}{4}\right) e^{-j\omega}\right)}{\left(1 - \left(\frac{1}{2}\right) e^{-j\omega}\right)^2}$$

The response of the system is given by

$$y(n) = \left(\frac{1}{4}\right)^n u(n)$$

Taking its DTFT, we obtain

$$Y(e^{j\omega}) = \frac{1}{1 - \left(\frac{1}{4}\right) e^{-j\omega}}$$

The frequency response of the system is given by

$$H(e^{j\omega}) = \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{\left(1 - \left(\frac{1}{2}\right) e^{-j\omega}\right)^2}{2 \left(1 - \left(\frac{1}{4}\right) e^{-j\omega}\right)^2}$$

We want to find $x(n)$ when

$$y(n) = \delta(n) - \left(-\frac{1}{2}\right)^n u(n)$$

$$Y(e^{j\omega}) = 1 - \frac{1}{1 + (\frac{1}{2})e^{-j\omega}} = \frac{(\frac{1}{2})e^{-j\omega}}{1 + (\frac{1}{2})e^{-j\omega}}$$

From the expression of the frequency response $H(e^{j\omega})$, we have

$$X(e^{j\omega}) = \frac{2(1 - (\frac{1}{4})e^{-j\omega})^2}{(1 - (\frac{1}{2})e^{-j\omega})^2} Y(e^{j\omega}) = \frac{2(1 - (\frac{1}{4})e^{-j\omega})^2}{(1 - (\frac{1}{2})e^{-j\omega})^2} \frac{(\frac{1}{2})e^{-j\omega}}{(1 + (\frac{1}{2})e^{-j\omega})}$$

$$X(e^{j\omega}) = \frac{e^{-j\omega} (1 - (\frac{1}{4})e^{-j\omega})^2}{(1 - (\frac{1}{2})e^{-j\omega})^2 (1 + (\frac{1}{2})e^{-j\omega})}$$

Using the partial fraction expansion, we obtain

$$X(e^{j\omega}) = \frac{(\frac{3}{8})e^{-j\omega}}{1 + (\frac{1}{2})e^{-j\omega}} + \frac{(\frac{3}{8})e^{-j\omega}}{1 - (\frac{1}{2})e^{-j\omega}} + \frac{(\frac{1}{8})e^{-j\omega}}{(1 - (\frac{1}{2})e^{-j\omega})^2}$$

Taking the inverse DTFT, we obtain

$$x(n) = \frac{3}{8} \left(-\frac{1}{2}\right)^{n-1} u(n-1) + \frac{3}{8} \left(\frac{1}{2}\right)^{n-1} u(n-1) + \frac{1}{4} n \left(\frac{1}{2}\right)^n u(n)$$

$$x(n) = \frac{3}{8} \left(-\frac{1}{2}\right)^{n-1} u(n-1) + \frac{3}{8} \left(\frac{1}{2}\right)^{n-1} u(n-1) + \frac{1}{8} n \left(\frac{1}{2}\right)^{n-1} u(n-1)$$

Example 6.24 Determine the response of the system with impulse response

$$h(n) = \left(\frac{1}{2}\right)^n u(n)$$

when the input is the complex exponential sequence

$$x(n) = A e^{j(\pi/2)n}, \quad -\infty < n < \infty$$

Solution

Given that

$$h(n) = \left(\frac{1}{2}\right)^n u(n)$$

Taking its DTFT, we obtain

$$H(e^{j\omega}) = \frac{1}{1 - (\frac{1}{2})e^{-j\omega}}$$

At $\omega = \pi/2$, the above equation yields

$$H(e^{j\pi/2}) = \frac{1}{1 - (1/2)e^{-j\pi/2}} = \frac{1}{1 + j(1/2)} = \frac{2}{\sqrt{5}} e^{-j26.6^\circ}$$

The response of the system to the complex exponential is given by [Eq. (6.72)]

$$\begin{aligned} y(n) &= AH(e^{j\omega_0}) e^{j\omega_0 n} = AH(e^{j\pi/2}) e^{j\pi/2 n} \\ &= A \left(\frac{2}{\sqrt{5}} e^{-j26.6^\circ} \right) e^{j\pi/2 n} \\ y(n) &= A \frac{2}{\sqrt{5}} e^{j(\pi n/2 - 26.6^\circ)} \end{aligned}$$

Example 6.25 Determine the response of the system with impulse response

$$h(n) = \left(\frac{1}{2} \right)^n u(n)$$

to the input signal

$$x(n) = 5 - 5 \sin\left(\frac{\pi}{2}n\right) + 10 \cos(\pi n), \quad -\infty < n < \infty$$

Solution

Given that

$$h(n) = \left(\frac{1}{2} \right)^n u(n)$$

Taking its DTFT, we obtain

$$H(e^{j\omega}) = \frac{1}{1 - (1/2)e^{-j\omega}}$$

The first term in the input signal $x(n)$ is a constant signal corresponding to $\omega = 0$, thus

$$H(e^{j0}) = \frac{1}{1 - 1/2} = 2$$

The second term in $x(n)$ has a frequency $\pi/2$. At this frequency, the frequency response of the system is

$$H(e^{j\pi/2}) = \frac{1}{1 - \frac{1}{2}e^{-j\pi/2}} = \frac{1}{1 + j1/2} = \frac{2}{\sqrt{5}} e^{-j26.6^\circ}$$

Finally, the third term in $x(n)$ has a frequency $\omega = \pi$. At this frequency, the frequency response of the system is

$$H(e^{j\pi}) = \frac{1}{1 - (1/2)e^{-j\pi}} = \frac{1}{1 + (1/2)} = \frac{1}{3/2} = \frac{2}{3}$$

Hence the response of the system to $x(n)$ is

$$y(n) = 5H(e^{j0}) - H(e^{j\pi/2})5 \sin\left(\frac{\pi}{2}n\right) + H(e^{j\pi})10 \cos(\pi n)$$

$$y(n) = 10 - \frac{10}{\sqrt{5}} \sin\left(\frac{\pi}{2}n - 26.6^\circ\right) + \frac{20}{3} \cos(\pi n)$$

6.8.4 Linear and Nonlinear Phase

Consider a discrete-time LTI system with impulse response $h(n)$ and frequency response $H(e^{j\omega})$. Let a signal $x(n)$ with Fourier transform $X(e^{j\omega})$ be applied to the input of the system. Let a signal $y(n)$ with Fourier transform $Y(e^{j\omega})$ denote the output of the system. In several applications, such as signal amplification or message signal transmission over a communication channel, we require that the output waveform be a replica of the input waveform, i.e., distortionless transmission. As in the continuous-time case, transmission is said to be distortionless if the input and the output have identical waveshapes within a multiplicative constant and a constant time delay. Thus, in distortionless transmission, the input $x(n)$ and the output $y(n)$ satisfy the condition (Fig. 6.18):

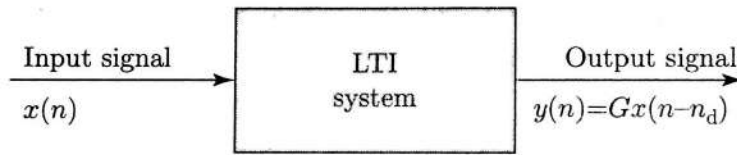


Fig. 6.18

$$y(n) = Gx(n - n_d) \quad (6.83)$$

where the constant G accounts for a change in amplitude and the constant n_d accounts for a delay in transmission. Taking the Fourier transform of the above equation yields

$$Y(e^{j\omega}) = GX(e^{j\omega})e^{-j\omega n_d}$$

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = H(e^{j\omega}) = Ge^{-j\omega n_d}$$

This is the frequency response required of a system for distortionless transmission. From this equation it follows that

1. the magnitude response $|H(e^{j\omega})|$ must be a constant, i.e., we must have

$$|H(e^{j\omega})| = G \quad (6.84)$$

for some constant G .

2. the phase response $\angle H(e^{j\omega})$ must be a linear function of ω with slope $-n_d$ and intercept zero, i.e., we must have

$$\angle H(e^{j\omega}) = -\omega n_d \quad (6.85)$$

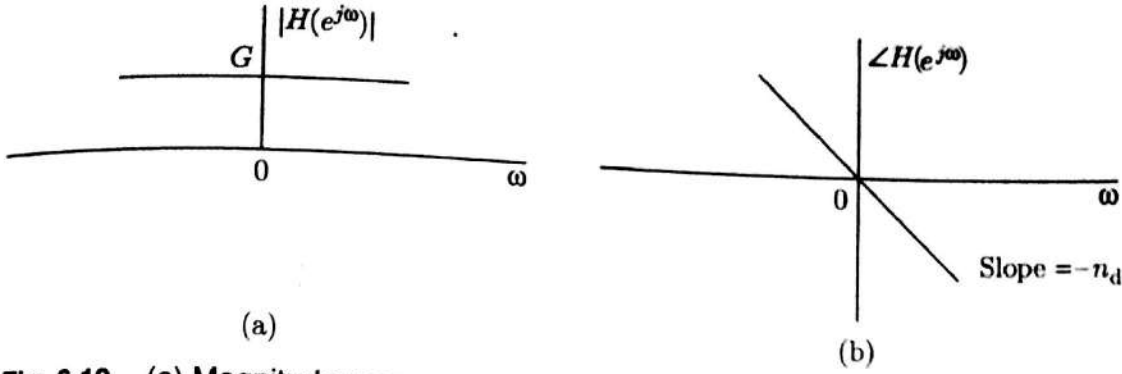


Fig. 6.19 (a) Magnitude response and (b) phase response of a system for distortionless transmission

These two conditions are illustrated in Figs 6.19(a) and (b), respectively. The gain $|H(e^{j\omega})| = G$ means that every spectral component is multiplied by a constant G . A linear phase $\angle H(e^{j\omega}) = -\omega n_d$ means that every spectral component is delayed by n_d seconds. For distortionless transmission, we require a linear-phase characteristic. The phase is not only a linear function of ω but it should also pass through the origin $\omega = 0$. Because each spectral component is attenuated by the same factor (G) and delayed by exactly the same amount (n_d), the output signal is an exact replica of the input (except for attenuating factor G and delay n_d).

6.8.5 Phase Delay and Group Delay

The two important parameters that characterize the form of the response $y(n)$ of an LTI system excited by an input signal $x(n)$ composed of a weighted sum of sinusoidal signals are phase delay and group delay. These two parameters are associated with the frequency response $H(e^{j\omega})$ of the system. For a linear phase system (whose phase varies linearly with frequency), both the phase delay and the group delay are constant.

Phase Delay

The time delay experienced by a single-frequency signal (i.e., a sinusoidal signal) when the signal passes through a system is referred to as the system phase delay. Now, let us assume that the input signal is the single-frequency signal

$$x(n) = A \cos(\omega_0 n + \theta)$$

$$x(n) = \frac{A}{2} e^{j(\omega_0 n + \theta)} + \frac{A}{2} e^{-j(\omega_0 n + \theta)}$$

and the system frequency response is

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})}$$

Then, the system output signal is [Eq. (4.42)]

$$y(n) = \frac{A}{2} |H(e^{j\omega_0})| e^{j(\omega_0 n + \theta + \angle H(e^{j\omega_0}))} + \frac{A}{2} |H(e^{-j\omega_0})| e^{-j(\omega_0 n + \theta - \angle H(e^{-j\omega_0}))}$$

@STC

Since $|H(e^{j\omega_0})| = |H(e^{-j\omega_0})|$ and $\angle H(e^{j\omega_0}) = -\angle H(e^{-j\omega_0})$, we get

$$y(n) = \frac{A}{2}|H(e^{j\omega_0})|e^{j(\omega_0 n + \theta + \angle H(e^{j\omega_0}))} + \frac{A}{2}|H(e^{j\omega_0})|e^{-j(\omega_0 n + \theta + \angle H(e^{j\omega_0}))}$$

$$\begin{aligned} y(t) &= A|H(e^{j\omega_0})|\cos(\omega_0 n + \angle H(e^{j\omega_0}) + \theta) \\ &= A|H(e^{j\omega_0})|\cos\left(\omega_0\left(n + \frac{\angle H(e^{j\omega_0})}{\omega_0}\right) + \theta\right) \\ y(n) &= A|H(e^{j\omega_0})|\cos\left(\omega_0(n - \tau_p(\omega_0)) + \theta\right) \end{aligned}$$

where

$$\tau_p(\omega_0) = -\frac{\angle H(e^{j\omega_0})}{\omega_0} \quad (6.86)$$

is the time delay experienced by the single-frequency signal with frequency ω_0 when it passes through the system. Thus, $\tau_p(\omega_0)$ is the system phase delay for a signal with frequency ω_0 . A negative system phase response at positive frequencies indicates that a signal is delayed in time when it passes through the system, whereas a positive system phase response at positive frequencies indicates that a signal is advanced in time when it passes through the system.

Group Delay

When the input signal contains many sinusoidal components with different frequencies that are not harmonically related, each component will go through different phase delays when processed by a frequency selective LTI system, and the signal delay is determined using a different parameter called the group delay, as defined below:

$$\tau_g(\omega) = -\frac{d\angle H(e^{j\omega})}{d\omega} \quad (6.87)$$

The time delay $\tau_g(\omega)$ is called the group delay or envelope delay. Thus, the group delay at each frequency equals the negative of the slope of the phase at that frequency. If $\tau_g(\omega)$ is constant, all the components are delayed by the same interval.

Example 6.26 Consider a discrete-time LTI system with frequency response $H(e^{j\omega})$ and real impulse response $h(n)$. The group-delay function for such a system is defined as

$$\tau_g(\omega) = -\frac{d\angle H(e^{j\omega})}{d\omega}$$

where $\angle H(e^{j\omega})$ has no discontinuities. Suppose that, for this system,

$$|H(e^{j\pi/2})| = 2, \quad \angle H(e^{j0}) = 0, \quad \tau_g\left(\frac{\pi}{2}\right) = 2$$

Determine the output of the system for each of the following inputs:
 (a) $\cos(\pi/2n)$ (b) $\sin(7\pi/2n + \pi/4)$

Solution

(a) The input signal $x(n) = \cos(\pi/2n)$ has a frequency $\omega_0 = \pi/2$. Hence, the response of the system to $x(n)$ is

$$\begin{aligned} y(n) &= |H(e^{j\omega_0})| \cos(\omega_0(n - \tau_g(\omega_0))) \\ &= |H(e^{j\pi/2})| \cos\left(\frac{\pi}{2}(n - \tau_g(\pi/2))\right) \\ y(n) &= 2 \cos\left(\frac{\pi}{2}(n - 2)\right) \\ y(n) &= 2 \cos\left(\frac{\pi}{2}n - \pi\right) \end{aligned}$$

(b) Consider the given input signal

$$\begin{aligned} x(n) &= \sin\left(\frac{7\pi}{2}n + \frac{\pi}{4}\right) \\ x(n) &= -\sin\left(\frac{\pi}{2}n - \frac{\pi}{4}\right) \end{aligned}$$

The input signal $x(n)$ has a frequency $\omega_0 = \pi/2$. Hence, the response of the system to $x(n)$ is

$$\begin{aligned} y(n) &= |H(e^{j\omega_0})| \sin(\omega_0(n - \tau_g(\omega_0)) + \theta) \\ &= |H(e^{j\pi/2})| \sin\left(\frac{7\pi}{2}(n - \tau_g(\pi/2)) + \frac{\pi}{4}\right) \\ y(n) &= 2 \sin\left(\frac{7\pi}{2}(n - 2) + \frac{\pi}{4}\right) \\ y(n) &= 2 \sin\left(\frac{7\pi}{2}n - 7\pi + \frac{\pi}{4}\right) \\ &= 2 \sin\left(\frac{7\pi}{2}n - \pi + \frac{\pi}{4}\right) \\ y(n) &= 2 \sin\left(\frac{7\pi}{2}n - \frac{3\pi}{4}\right) \end{aligned}$$

Example 6.27 Consider a nonrecursive filter with the impulse response $h(n)$ shown in Fig. 6.20. What is the group delay as a function of frequency for this filter?

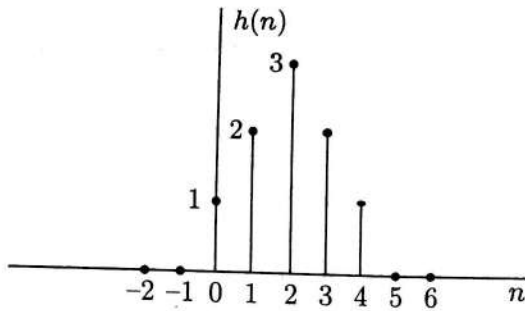


Fig. 6.20

Solution

We can observe that $h_1(n) = h(n+2)$ is real and even. Therefore, (using the properties of the DTFT) the DTFT $H_1(e^{j\omega}) = \mathcal{F}[h_1(n)]$ is real and even. Hence, $H_1(e^{j\omega})$ has zero phase. We also know that

$$\begin{aligned} h_1(n) &= h(n+2) \\ \mathcal{F}[h_1(n)] &= \mathcal{F}[h(n+2)] \\ H_1(e^{j\omega}) &= H(e^{j\omega})e^{j2\omega} \end{aligned}$$

or, equivalently,

$$H(e^{j\omega}) = H_1(e^{j\omega})e^{-j2\omega}$$

Since $H_1(e^{j\omega})$ has zero phase, we have

$$\angle H(e^{j\omega}) = -2\omega$$

Therefore, the group delay is

$$\tau_g(\omega) = -\frac{d\angle H(e^{j\omega})}{d\omega} = 2$$

Example 6.28 Consider a discrete-time low-pass filter whose impulse response $h(n)$ is known to be real and whose frequency response magnitude in the region $-\pi \leq \omega \leq \pi$ is given as

$$|H(e^{j\omega})| = \begin{cases} 1, & |\omega| \leq \pi/4 \\ 0, & \text{otherwise} \end{cases}$$

Determine the real-valued impulse response $h(n)$ for this filter when the corresponding group-delay function is specified as

(a) $\tau_g(\omega) = 5$, (b) $\tau_g(\omega) = 5/2$, (c) $\tau_g(\omega) = -5/2$

Solution

Let $H_0(e^{j\omega}) = |H(e^{j\omega})|$. Then the inverse DTFT of $H_0(e^{j\omega})$ is given by

$$\begin{aligned} h_0(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_0(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\pi/4}^{\pi/4} e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \left. \frac{e^{j\omega n}}{jn} \right|_{-\pi/4}^{\pi/4} \\ &= \frac{1}{n\pi} \frac{e^{j\pi/4n} - e^{-j\pi/4n}}{2j} \\ h_0(n) &= \frac{\sin(\pi/4n)}{n\pi} \end{aligned}$$

If $\tau_g(\omega) = -\frac{d\angle H(e^{j\omega})}{d\omega} = \alpha$ (where α is a constant), then $\angle H(e^{j\omega}) = -\alpha\omega + \beta$, where β is a constant. Given that $h(n)$ is real, therefore $\angle H(e^{j\omega})$ is an odd function, and hence for $\angle H(e^{j\omega})$ to be an odd function, the constant $\beta = 0$. Therefore,

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})} = H_0(e^{j\omega}) e^{-j\alpha\omega}$$

Taking the inverse DTFT, we obtain

$$h(n) = h_0(n - \alpha)$$

$$h(n) = \frac{\sin[\pi/4(n - \alpha)]}{(n - \alpha)\pi}$$

(a) Given that $\tau_g(\omega) = 5 = \alpha$. Substituting $\alpha = 5$ in the expression of $h(n)$ yields

$$h(n) = \frac{\sin[\pi/4(n - 5)]}{(n - 5)\pi}$$

(b) Given that $\tau_g(\omega) = 5/2 = \alpha$. Substituting $\alpha = 5/2$ in the expression of $h(n)$ yields

$$h(n) = \frac{\sin[\pi/4(n - 5/2)]}{(n - 5/2)\pi}$$

(c) Given that $\tau_g(\omega) = -5/2 = \alpha$. Substituting $\alpha = -5/2$ in the expression of $h(n)$ yields

$$h(n) = \frac{\sin[\pi/4(n + 5/2)]}{(n + 5/2)\pi}$$

6.9 IDEAL AND PRACTICAL FILTERS

Filtering is a process by which the essential and useful part of a signal is separated from the extraneous and undesirable components. The idea of filtering using LTI system is based on the convolution property of the DTFT which states that for LTI systems, the DTFT of the output $Y(e^{j\omega})$ is the product of the DTFT of the input $X(e^{j\omega})$ and

the frequency response $H(e^{j\omega})$, i.e., $Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$. In a sense, $H(e^{j\omega})$ acts as a weighting function or spectral shaping function to the different frequency components in the input signal. Ideal filters allow distortionless transmission of a certain band of frequencies and completely suppress the remaining frequencies. The passband of a filter is the band of frequencies that are passed by the system, while the stopband refers to the range of frequencies that are attenuated by the system. In the ideal case, $|H(e^{j\omega})| = 1$ in a passband, while $|H(e^{j\omega})| = 0$ in a stopband. Filters are usually classified according to their frequency-domain characteristics as low-pass, high-pass, bandpass, bandstop or band elimination, and all-pass filters. Following are the characteristics of these filters:

1. Low-pass filters are those characterized by a passband that extends from $\omega = 0$ to $\omega = \omega_c$, where ω_c is called the cutoff frequency of the filter [Fig. 6.21(a)]. A low-pass filter attenuates high-frequency components of the input and passes the low-frequency components.

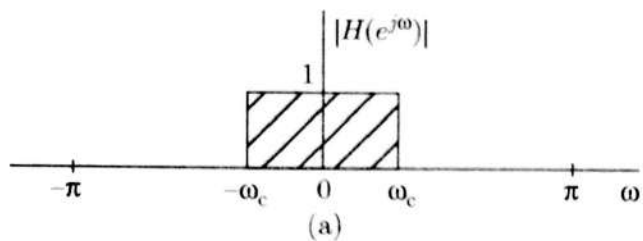


Fig. 6.21 (a) Low-pass filter www.dahiyastech.blogspot.com

2. High-pass filters are characterized by a stopband that extends from $\omega = 0$ to $\omega = \omega_c$ and a passband that extends from $\omega = \omega_c$ to π [Fig. 6.21(b)]. A high-pass filter attenuates low frequencies and passes the high frequencies.

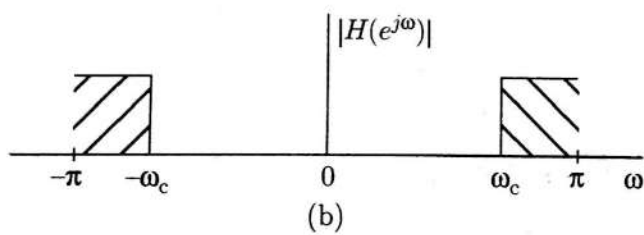


Fig. 6.21 (b) High-pass filter

3. Bandpass filters are characterized by a passband that extends from $\omega = \omega_1$ to $\omega = \omega_2$ and all other frequencies are stopped [Fig. 6.21(c)]. A bandpass filter passes signals within a certain frequency band and attenuates signals outside that band.

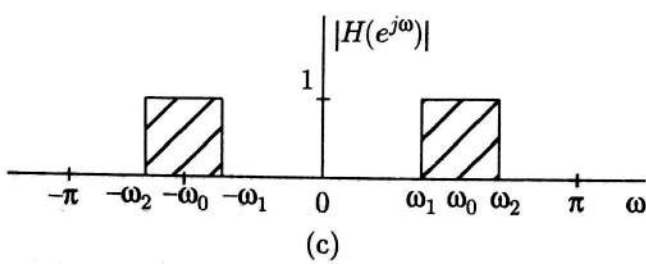


Fig. 6.21 (c) Bandpass filter

4. Bandstop filters stop frequencies extending from $\omega = \omega_1$ to $\omega = \omega_2$ and pass all other frequencies [Fig. 6.21(d)]. A bandstop filter attenuates signals within a certain frequency band and passes signals outside that band.

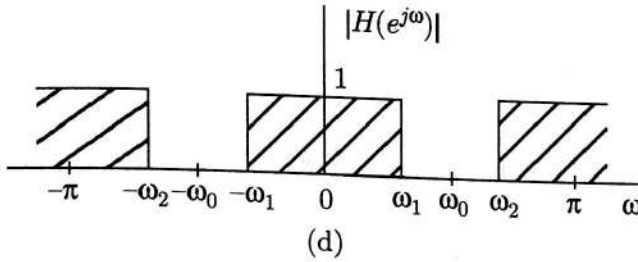


Fig. 6.21 (d) Bandstop filter

5. An all-pass filter is characterized by a magnitude response that is constant for all frequencies [Fig. 6.21(e)]. The characteristics of an all-pass filter are completely determined by its phase-shift characteristics. An all-pass filter has nonconstant group delay and different frequencies in the input are delayed by different amounts.

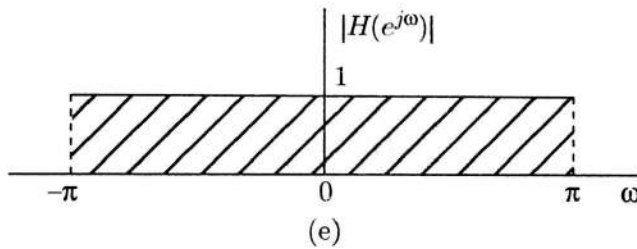


Fig. 6.21 (e) All-pass filter

The filters described so far are referred to as ideal filters because they pass one set of frequencies without any change and completely stop others. Since it is impossible to realize filters with characteristic like those shown in Fig. 6.21, with abrupt changes from passband to stopband and vice versa, most of the filters we deal with in practice have some transition band. Ideal filters are noncausal filters and therefore physically unrealizable. For example, consider an ideal low-pass filter whose magnitude response and phase response are (as shown in Fig. 6.22) given by

$$|H(e^{j\omega})| = \text{rect}\left(\frac{\omega}{2\omega_c}\right) = \begin{cases} 1, & -\omega_c < \omega < \omega_c \\ 0, & \text{otherwise} \end{cases}$$

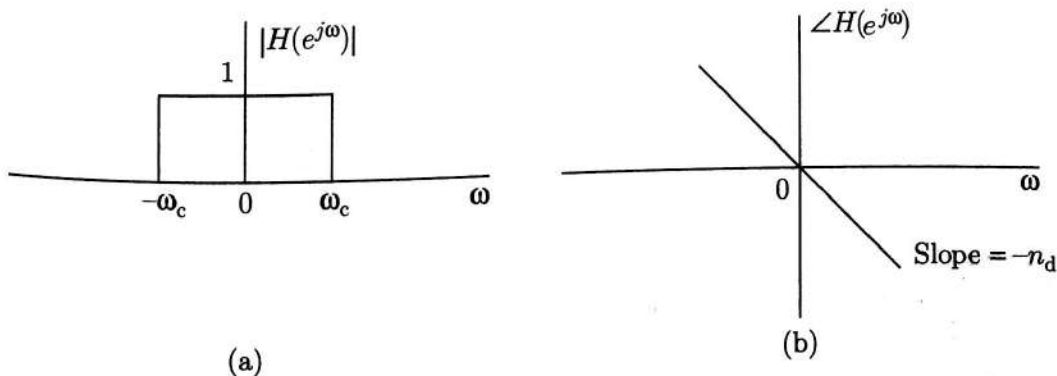


Fig. 6.22 (a) Magnitude response and (b) phase response of an ideal low-pass filter

and

$$\angle H(e^{j\omega}) = -\omega n_d$$

Therefore,

$$H(e^{j\omega}) = |H(e^{j\omega})| e^{j\angle H(e^{j\omega})}$$

$$H(e^{j\omega}) = \text{rect}\left(\frac{\omega}{2\omega_c}\right) e^{-j\omega n_d}$$

Let us define $H_0(e^{j\omega}) = |H(e^{j\omega})|$. Then,

$$|H(e^{j\omega})| = H_0(e^{j\omega}) = \begin{cases} 1, & -\omega_c < \omega < \omega_c \\ 0, & \text{otherwise} \end{cases}$$

Taking the inverse DTFT, we have

$$h_0(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_0(e^{j\omega}) e^{j\omega n} d\omega = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega$$

$$h_0(n) = \frac{\sin(\omega_c n)}{\pi n}$$

Now, consider

$$\begin{aligned} H(e^{j\omega}) &= |H(e^{j\omega})| e^{j\angle H(e^{j\omega})} \\ &= H_0(e^{j\omega}) e^{-j\omega n_d} \end{aligned}$$

Taking the inverse DTFT, we get

$$h(n) = h_0(n - n_d) = \frac{\sin(\omega_c(n - n_d))}{\pi(n - n_d)}$$

Recall that $h(n)$ is the system response to impulse input $\delta(n)$, which is applied at $n = 0$. Figure 6.23 depicts a curious fact: the response $h(n)$ begins even before the input is applied (at $n = 0$). Clearly, the filter is noncausal and therefore physically unrealizable. Similarly, one can show that other ideal filters are also physically unrealizable.

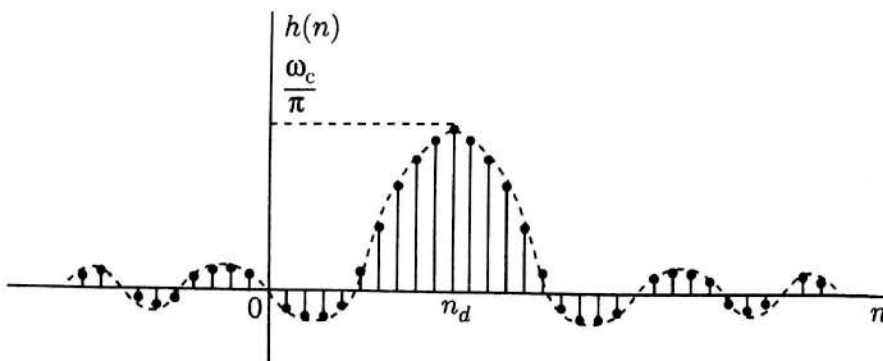


Fig. 6.23 Impulse response

6.9.1 Paley-Wiener Criterion

For a physically realizable system, the impulse response $h(n)$ must be causal and has finite energy, i.e.,

$$h(n) = 0, \quad n < 0$$

In the frequency domain, this condition is equivalent to the well-known Paley-Wiener criterion, which states that the necessary and sufficient condition for the magnitude response $|H(e^{j\omega})|$ to be realizable is

$$\int_{-\pi}^{\pi} |\ln|H(e^{j\omega})|| d\omega < \infty \quad (6.88)$$

If $H(e^{j\omega})$ does not satisfy this condition, it is unrealizable.

Two important conclusion that we draw from the Paley-Wiener theorem are as follows:

1. The magnitude function $|H(e^{j\omega})|$ of a realizable system can be zero at some discrete frequencies, but it cannot be zero over any finite band of frequencies. This means that if $|H(e^{j\omega})| = 0$ over any finite band, then $|\ln|H(e^{j\omega})|| = \infty$ over that band and consequently $H(e^{j\omega})$ is unrealizable.
2. The magnitude function $|H(e^{j\omega})|$ cannot be constant in any finite range of frequencies and the transition from passband to stopband cannot be infinitely sharp [this is a consequence of the Gibbs phenomenon, which results from the truncation of $h(n)$ to achieve causality].

Apparently, causality imposes some tight constraints on a LTI system. In addition to the Paley-Wiener condition, causality also implies a strong relationship between $H_R(e^{j\omega}) = \Re\{H(e^{j\omega})\}$ and $H_I(e^{j\omega}) = \Im\{H(e^{j\omega})\}$, the real and imaginary components of the frequency response $H(e^{j\omega})$. To illustrate this dependence, we decompose $h(n)$ into its even and an odd part, i.e.,

$$h(n) = h_e(n) + h_o(n)$$

where

$$h_e(n) = \frac{1}{2}[h(n) + h(-n)]$$

and

$$h_o(n) = \frac{1}{2}[h(n) - h(-n)]$$

Now, if $h(n)$ is causal, it is possible to recover $h(n)$ from its even part $h_e(n)$ for $0 \leq n \leq \infty$ or from its odd part $h_o(n)$ for $1 \leq n \leq \infty$. Indeed, it can be easily seen that

$$h(n) = 2h_e(n)u(n) - h_e(0)\delta(n), \quad n \geq 0 \quad (6.89)$$

and

$$h(n) = 2h_o(n)u(n) + h(0)\delta(n), \quad n \geq 1 \quad (6.90)$$

Since $h_o(n) = 0$ for $n = 0$, we cannot recover $h(0)$ from $h_o(n)$ and hence we also must know $h(0)$. In any case, it is apparent that $h_o(n) = h_e(n)$ for $n \geq 1$ and so there is a strong relationship between $h_o(n)$ and $h_e(n)$.

If $h(n)$ is absolutely summable (i.e., BIBO stable), the frequency response $H(e^{j\omega})$ exists, and

$$H(e^{j\omega}) = H_R(e^{j\omega}) + jH_I(e^{j\omega})$$

In addition, if $h(n)$ is real valued and causal, the conjugate property of the Fourier transform imply that

$$h_e(n) \longleftrightarrow H_R(e^{j\omega}) \quad \text{and} \quad h_o(n) \longleftrightarrow jH_I(e^{j\omega})$$

Since $h(n)$ is completely specified by $h_e(n)$, it follows that $H(e^{j\omega})$ is completely determined if we know $H_R(e^{j\omega})$. Alternatively, $H(e^{j\omega})$ is completely determined from $H_I(e^{j\omega})$. In short, $H_R(e^{j\omega})$ and $H_I(e^{j\omega})$ are interdependent and cannot be specified independently if the system is causal. Equivalently, the magnitude and phase responses of a causal filter are interdependent and hence cannot be specified independently.

Taking the Fourier transform of Eq. (6.89), we have

$$\begin{aligned} \mathcal{F}[h(n)] &= 2\mathcal{F}[h_e(n)u(n)] - h_e(0)\mathcal{F}[\delta(n)] \\ H(e^{j\omega}) &= H_R(e^{j\omega}) + jH_I(e^{j\omega}) = 2[H_R(e^{j\omega}) * U(e^{j\omega})] - h_e(0) \\ H_R(e^{j\omega}) + jH_I(e^{j\omega}) &= 2 \frac{1}{2\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) U(e^{j(\omega-\theta)}) d\theta - h_e(0) \\ H_R(e^{j\omega}) + jH_I(e^{j\omega}) &= \frac{1}{\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) U(e^{j(\omega-\theta)}) d\theta - h_e(0) \end{aligned} \quad (6.91)$$

where $U(e^{j\omega})$ is the DTFT of the unit step sequence $u(n)$. The DTFT of $u(n)$ is given by

$$\begin{aligned} U(e^{j\omega}) &= \pi\delta(\omega) + \frac{1}{1 - e^{-j\omega}}, \quad -\pi < \omega < \pi \\ U(e^{j\omega}) &= \pi\delta(\omega) + \frac{1}{2} - j\frac{1}{2}\cot\left(\frac{\omega}{2}\right) \end{aligned} \quad (6.92)$$

Substituting Eq. (6.92) into Eq. (6.91) gives

$$\begin{aligned} H_R(e^{j\omega}) + jH_I(e^{j\omega}) &= \frac{1}{\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) \left[\pi\delta(\omega - \theta) + \frac{1}{2} - j\frac{1}{2}\cot\left(\frac{\omega - \theta}{2}\right) \right] d\theta - h_e(0) \\ &= \frac{1}{\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) \pi\delta(\omega - \theta) d\theta + \frac{1}{\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) \frac{1}{2} d\theta - \frac{1}{\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) j\frac{1}{2}\cot\left(\frac{\omega - \theta}{2}\right) d\theta - h_e(0) \end{aligned}$$

$$= \int_{-\pi}^{\pi} H_R(e^{j\theta}) \delta(\omega - \theta) d\theta + \frac{1}{2\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) d\theta - j \frac{1}{2\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) \cot\left(\frac{\omega - \theta}{2}\right) d\theta - h_e(0)$$

$$H_R(e^{j\omega}) + jH_I(e^{j\omega}) = H_R(e^{j\theta}) \Big|_{\theta=\omega} + h_e(0) - j \frac{1}{2\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) \cot\left(\frac{\omega - \theta}{2}\right) d\theta - h_e(0)$$

$$jH_I(e^{j\omega}) = -j \frac{1}{2\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) \cot\left(\frac{\omega - \theta}{2}\right) d\theta$$

$$H_I(e^{j\omega}) = -\frac{1}{2\pi} \int_{-\pi}^{\pi} H_R(e^{j\theta}) \cot\left(\frac{\omega - \theta}{2}\right) d\theta \quad (6.93)$$

Similarly, taking the Fourier transform of Eq. (6.90) yields

$$\mathcal{F}[h(n)] = 2\mathcal{F}[h_o(n)u(n)] + h(0)\mathcal{F}[\delta(n)]$$

$$H(e^{j\omega}) = H_R(e^{j\omega}) + jH_I(e^{j\omega}) = 2[jH_I(e^{j\omega}) * U(e^{j\omega})] + h(0)$$

$$H_R(e^{j\omega}) + jH_I(e^{j\omega}) = 2j \frac{1}{2\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) U(e^{j(\omega-\theta)}) d\theta + h(0)$$

$$H_R(e^{j\omega}) + jH_I(e^{j\omega}) = j \frac{1}{\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) U(e^{j(\omega-\theta)}) d\theta + h(0) \quad (6.94)$$

where $U(e^{j\omega})$ is the DTFT of the unit step sequence $u(n)$. Substituting $U(e^{j\omega})$ from Eq. (6.92) into Eq. (6.94) gives

$$\begin{aligned} H_R(e^{j\omega}) + jH_I(e^{j\omega}) &= j \frac{1}{\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) \left[\pi \delta(\omega - \theta) + \frac{1}{2} - j \frac{1}{2} \cot\left(\frac{\omega - \theta}{2}\right) \right] d\theta + h(0) \\ &= j \frac{1}{\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) \pi \delta(\omega - \theta) d\theta + j \frac{1}{\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) \frac{1}{2} d\theta - j \frac{1}{\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) j \frac{1}{2} \cot\left(\frac{\omega - \theta}{2}\right) d\theta + h(0) \\ &= j \int_{-\pi}^{\pi} H_I(e^{j\theta}) \delta(\omega - \theta) d\theta + j \frac{1}{2\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) d\theta + \frac{1}{2\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) \cot\left(\frac{\omega - \theta}{2}\right) d\theta + h(0) \end{aligned}$$

Since $H_I(e^{j\omega})$ is an odd function, we have $\int_{-\pi}^{\pi} H_I(e^{j\theta}) d\theta = 0$, therefore

$$\begin{aligned}
 H_R(e^{j\omega}) + jH_I(e^{j\omega}) &= jH_I(e^{j\theta}) \Big|_{\theta=\omega} + 0 + \frac{1}{2\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) \cot\left(\frac{\omega - \theta}{2}\right) d\theta + h(0) \\
 H_R(e^{j\omega}) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) \cot\left(\frac{\omega - \theta}{2}\right) d\theta + h(0) \\
 H_R(e^{j\omega}) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_I(e^{j\theta}) \cot\left(\frac{\omega - \theta}{2}\right) d\theta + h(0) \tag{6.95}
 \end{aligned}$$

Equations (6.93) and (6.95) are known as Hilbert transform pair.

To summarize, causality has very important implications in the design of frequency-selective filters. These implications are listed as follows:

1. The frequency response $H(e^{j\omega})$ cannot be zero except at a finite set of points in frequency.
2. The magnitude $|H(e^{j\omega})|$ cannot be constant in any finite range of frequencies and the transition from passband to stopband cannot be infinitely sharp.
3. The real and imaginary parts of $H(e^{j\omega})$ are interdependent and are related by the Hilbert transform. As a consequence, the magnitude $|H(e^{j\omega})|$ and phase $\angle H(e^{j\omega})$ cannot be chosen arbitrarily.

6.10 ENERGY SPECTRAL DENSITY (ESD)

Parseval's theorem relates the total signal energy in a signal $x(n)$ to its Fourier transform through

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega = \int_{-1/2}^{1/2} |X(e^{j\omega})|^2 df$$

Parseval's theorem states that the total energy E_x may be determined either by computing the energy per unit time ($|x(n)|^2$) and summing over all time or by integrating the energy per unit frequency $|X(e^{j\omega})|^2/2\pi$ over a full 2π interval of distinct discrete-time frequencies. For this reason $|X(e^{j\omega})|^2$ represents energy per unit bandwidth and is often referred to as the *energy spectral density* or *energy density spectrum* of the signal $x(n)$ and is denoted by $\Psi_x(e^{j\omega})$. Hence,

$$\Psi_x(e^{j\omega}) = |X(e^{j\omega})|^2 \tag{6.96}$$

For discrete-time signals, the unit of ESD is simply the square of the signal unit (whatever that may be).

6.10.1 Relationship Between Input and Output ESDs of an LTI System

Consider an LTI system with frequency response $H(e^{j\omega})$, input $x(n)$, and output $y(n)$. If $x(n)$ and $y(n)$ are energy signals, then their ESDs are $\Psi_x(e^{j\omega}) = |X(e^{j\omega})|^2$ and $\Psi_y(e^{j\omega}) = |Y(e^{j\omega})|^2$, respectively. Since, we know that

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

it follows that

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$$|Y(e^{j\omega})|^2 = |H(e^{j\omega})X(e^{j\omega})|^2$$

$$|Y(e^{j\omega})|^2 = |H(e^{j\omega})|^2 |X(e^{j\omega})|^2$$

$$\Psi_y(e^{j\omega}) = |H(e^{j\omega})|^2 \Psi_x(e^{j\omega}) \quad (6.97)$$

6.10.2 Relation of ESD to Autocorrelation

The autocorrelation function $R_{xx}(\tau)$ of a real energy signal is defined as [Eq. (2.60)]:

$$R_{xx}(m) = \sum_{n=-\infty}^{\infty} x(n)x(n-m)$$

$$R_{xx}(m) = x(m) * x(-m)$$

taking the Fourier transform of the above equation, we have

$$\mathcal{F}[R_{xx}(m)] = X(e^{j\omega})X(e^{-j\omega})$$

$$= X(e^{j\omega})X^*(e^{j\omega})$$

$$\mathcal{F}[R_{xx}(m)] = |X(e^{j\omega})|^2$$

$$\mathcal{F}[R_{xx}(m)] = \Psi_x(e^{j\omega})$$

$$R_{xx}(m) \longleftrightarrow \Psi_x(e^{j\omega}) \quad (6.98)$$

Thus, the autocorrelation function $R_{xx}(m)$ and ESD makes a Fourier transform pair.

6.11 POWER SPECTRAL DENSITY (PSD)

PSD has the same relation to power signals as ESD has to energy signals. The expression of PSD may be obtained by assuming the power signal as a limiting case of an energy signal.

Consider a power signal of infinite duration. Since the total energy of a power signal cannot be found, let us first find the ESD of a truncated version $x_N(n)$ of the infinite duration signal $x(n)$. The signal $x_N(n)$ may be expressed as

$$x_N(n) = \begin{cases} x(n), & -N \leq n \leq N \\ 0, & \text{otherwise} \end{cases} \quad (6.99)$$

The truncated signal $x_N(n)$ is of finite duration, therefore, it is an energy signal. Now, if $x_N(n) \longleftrightarrow X_N(e^{j\omega})$, then using the Parseval's theorem, we have

$$E_{x_N} = \sum_{n=-\infty}^{\infty} |x_N(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X_N(e^{j\omega})|^2 d\omega$$

Substituting the value of $x_N(n)$ from Eq. (6.99) into the above equation gives

$$\begin{aligned} \sum_{n=-N}^N |x(n)|^2 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} |X_N(e^{j\omega})|^2 d\omega \\ \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \frac{1}{2\pi} \int_{-\pi}^{\pi} |X_N(e^{j\omega})|^2 d\omega \\ \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \lim_{N \rightarrow \infty} \frac{|X_N(e^{j\omega})|^2}{2N+1} d\omega \end{aligned}$$

The LHS of the above equation represents the average power P_x of the signal $x(n)$. Therefore,

$$\begin{aligned} P_x &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x(n)|^2 \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \lim_{N \rightarrow \infty} \frac{|X_N(e^{j\omega})|^2}{2N+1} d\omega \\ P_x &= \frac{1}{2\pi} \int_{-\pi}^{\pi} G_x(e^{j\omega}) d\omega \end{aligned} \tag{6.100}$$

where

$$G_x(e^{j\omega}) = \lim_{N \rightarrow \infty} \frac{|X_N(e^{j\omega})|^2}{2N+1} \tag{6.101}$$

is the PSD. The units of PSD depend on the units of the underlying signal $x(n)$.

6.11.1 Relationship Between Input and Output PSDs of an LTI System

Consider an LTI system with frequency response $H(e^{j\omega})$, input $x_N(n)$, and output $y_N(n)$. If $x(n)$ and $y(n)$ are power signals, then their PSDs are

$$G_x(e^{j\omega}) = \lim_{N \rightarrow \infty} \frac{|X_N(e^{j\omega})|^2}{2N+1}$$

and

$$G_y(e^{j\omega}) = \lim_{N \rightarrow \infty} \frac{|Y_N(e^{j\omega})|^2}{2N+1},$$

respectively. Since, we know that

$$Y_N(e^{j\omega}) = H(e^{j\omega})X_N(e^{j\omega})$$

it follows that

$$\begin{aligned} |Y_N(e^{j\omega})|^2 &= |H(e^{j\omega})X_N(e^{j\omega})|^2 \\ |Y_N(e^{j\omega})|^2 &= |H(e^{j\omega})|^2 |X_N(e^{j\omega})|^2 \\ \lim_{N \rightarrow \infty} \frac{|Y_N(e^{j\omega})|^2}{2N+1} &= |H(e^{j\omega})|^2 \lim_{N \rightarrow \infty} \frac{|X_N(e^{j\omega})|^2}{2N+1} \\ G_y(e^{j\omega}) &= |H(e^{j\omega})|^2 G_x(e^{j\omega}) \end{aligned} \quad (6.102)$$

6.11.2 Relation of PSD to Autocorrelation

The autocorrelation function $R_{xx}(\tau)$ of a power signal is defined as [Eq. (2.61)]:

$$\begin{aligned} R_{xx}(m) &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N x(n)x(n-m) \\ &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-\infty}^{\infty} x_N(n)x_N(n-m) \\ R_{xx}(m) &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} [x_N(m) * x_N(-m)] \end{aligned}$$

taking the Fourier transform of the above equation, we obtain

$$\begin{aligned} \mathcal{F}[R_{xx}(m)] &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} X_N(e^{j\omega})X_N(e^{-j\omega}) \\ &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} X_N(e^{j\omega})X_N^*(e^{j\omega}) \\ \mathcal{F}[R_{xx}(m)] &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} |X_N(e^{j\omega})|^2 \\ \mathcal{F}[R_{xx}(m)] &= G_x(e^{j\omega}) \end{aligned}$$

$$R_{xx}(m) \longleftrightarrow G_x(e^{j\omega}) \quad (6.103)$$

Thus, the autocorrelation function $R_{xx}(m)$ and PSD makes a Fourier transform pair.

SOLVED EXAMPLES

Example 6.29 Determine the Fourier transform for $-\pi \leq \omega < \pi$ in the case of each of the following periodic signals:

(a) $x_1(n) = \sin(\pi/3n + \pi/4)$

(b) $x_2(n) = 2 + \cos(\pi/6n + \pi/8)$

Solution

We know that a periodic signal $x(n)$ with Fourier series representation

$$x(n) = \sum_{k \in \langle N \rangle} X_k e^{jk2\pi/Nn}$$

has the DTFT $X(e^{j\omega})$ given by

$$X(e^{j\omega}) = 2\pi \sum_{k=-\infty}^{\infty} X_k \delta\left(\omega - k \frac{2\pi}{N}\right)$$

(a) Consider the given signal $x_1(n) = \sin(\pi/3n + \pi/4)$. The fundamental period of $x_1(n)$ is $N = 6$.

$$\begin{aligned} x_1(n) &= \sin\left(\frac{\pi}{3}n + \frac{\pi}{4}\right) \\ &= \frac{1}{2j} e^{j(\pi/3n + \pi/4)} - \frac{1}{2j} e^{-j(\pi/3n + \pi/4)} \\ &= \frac{1}{2j} e^{j\pi/4} e^{j\pi/3n} - \frac{1}{2j} e^{-j\pi/4} e^{-j\pi/3n} \\ x_1(n) &= \frac{1}{2j} e^{j\pi/4} e^{j2\pi/6n} - \frac{1}{2j} e^{-j\pi/4} e^{-j2\pi/6n} \\ &= X_1 e^{j2\pi/6n} + X_{-1} e^{-j2\pi/6n} \end{aligned}$$

The nonzero Fourier series coefficients X_k of $x_1(n)$ in the range $-2 \leq k \leq 3$ are

$$X_1 = \frac{1}{2j} e^{j\pi/4}, \quad X_{-1} = -\frac{1}{2j} e^{-j\pi/4}$$

Substituting X_1 and X_{-1} in the expression of the DTFT of the periodic signals gives

$$\begin{aligned} X_1(e^{j\omega}) &= 2\pi X_1 \delta\left(\omega - \frac{2\pi}{6}\right) + 2\pi X_{-1} \delta\left(\omega + \frac{2\pi}{6}\right) \\ &= 2\pi \frac{1}{2j} e^{j\pi/4} \delta\left(\omega - \frac{2\pi}{6}\right) \\ &\quad - 2\pi \frac{1}{2j} e^{-j\pi/4} \delta\left(\omega + \frac{2\pi}{6}\right) \end{aligned}$$

$$\begin{aligned} X_1(e^{j\omega}) &= \frac{\pi}{j} \left[e^{j\pi/4} \delta\left(\omega - \frac{2\pi}{6}\right) \right. \\ &\quad \left. - e^{-j\pi/4} \delta\left(\omega + \frac{2\pi}{6}\right) \right], \quad -\pi \leq \omega < \pi \end{aligned}$$

(b) Consider the given signal $x_2(n) = 2 + \cos(\pi/6n + \pi/8)$. The fundamental period of $x_2(n)$ is $N = 12$.

$$\begin{aligned} x_2(n) &= 2 + \cos\left(\frac{\pi}{6}n + \frac{\pi}{8}\right) \\ &= 2 + \frac{1}{2} e^{j(\pi/6n + \pi/8)} + \frac{1}{2} e^{-j(\pi/6n + \pi/8)} \\ &= 2 + \frac{1}{2} e^{j\pi/8} e^{j\pi/6n} + \frac{1}{2} e^{-j\pi/8} e^{-j\pi/6n} \\ x_2(n) &= 2 + \frac{1}{2} e^{j\pi/8} e^{j2\pi/12n} + \frac{1}{2} e^{-j\pi/8} e^{-j2\pi/12n} \\ &= X_0 + X_1 e^{j2\pi/12n} + X_{-1} e^{-j2\pi/12n} \end{aligned}$$

The nonzero Fourier series coefficients X_k of $x_2(n)$ in the range $-5 \leq k \leq 6$ are

$$X_0 = 2, \quad X_1 = \frac{1}{2} e^{j\pi/8}, \quad X_{-1} = \frac{1}{2} e^{-j\pi/8}$$

Substituting X_0 , X_1 , and X_{-1} in the expression of the DTFT of the periodic signals gives

$$\begin{aligned} X_2(e^{j\omega}) &= 2\pi X_0 \delta(\omega) + 2\pi X_1 \delta\left(\omega - \frac{2\pi}{12}\right) \\ &\quad + 2\pi X_{-1} \delta\left(\omega + \frac{2\pi}{12}\right) \\ &= 4\pi \delta(\omega) + 2\pi \frac{1}{2} e^{j\pi/8} \delta\left(\omega - \frac{2\pi}{12}\right) \\ &\quad + 2\pi \frac{1}{2} e^{-j\pi/8} \delta\left(\omega + \frac{2\pi}{12}\right) \\ X_2(e^{j\omega}) &= 4\pi \delta(\omega) + \pi \left[e^{j\pi/8} \delta\left(\omega - \frac{2\pi}{12}\right) \right. \\ &\quad \left. + e^{-j\pi/8} \delta\left(\omega + \frac{2\pi}{12}\right) \right], \\ &\quad -\pi \leq \omega < \pi \end{aligned}$$

Example 6.30 Determine the inverse Fourier transform of

$$(a) \quad X_1(e^{j\omega}) = \sum_{m=-\infty}^{\infty} \left[2\pi\delta(\omega - 2\pi m) + \pi\delta\left(\omega - \frac{\pi}{2} - 2\pi m\right) + \pi\delta\left(\omega + \frac{\pi}{2} - 2\pi m\right) \right]$$

$$(b) \quad X_2(e^{j\omega}) = \begin{cases} 2j, & 0 < \omega \leq \pi \\ -2j, & -\pi < \omega \leq 0 \end{cases}$$

Solution

(a) The inverse DTFT $x_1(n)$ of $X_1(e^{j\omega})$ is given by

$$\begin{aligned} x_1(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_1(e^{j\omega}) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} \left[2\pi\delta(\omega) + \pi\delta\left(\omega - \frac{\pi}{2}\right) + \pi\delta\left(\omega + \frac{\pi}{2}\right) \right] e^{j\omega n} d\omega \\ &= \int_{-\pi}^{\pi} \delta(\omega) e^{j\omega n} d\omega \\ &\quad + \frac{1}{2} \int_{-\pi}^{\pi} \delta\left(\omega - \frac{\pi}{2}\right) e^{j\omega n} d\omega \\ &\quad + \frac{1}{2} \int_{-\pi}^{\pi} \delta\left(\omega + \frac{\pi}{2}\right) e^{j\omega n} d\omega \\ &= e^{j0} \Big|_{\omega=0} + \frac{1}{2} e^{j\omega n} \Big|_{\omega=\pi/2} + \frac{1}{2} e^{j\omega n} \Big|_{\omega=-\pi/2} \\ &= e^{j0} + \frac{1}{2} e^{j\pi/2n} + \frac{1}{2} e^{-j\pi/2n} \\ x_1(n) &= 1 + \cos\left(\frac{\pi}{2}n\right) \end{aligned}$$

(b) The inverse DTFT $x_2(n)$ of $X_2(e^{j\omega})$ is given by

$$\begin{aligned} x_2(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X_2(e^{j\omega}) e^{j\omega n} d\omega \\ &= -\frac{1}{2\pi} \int_{-\pi}^0 2j e^{j\omega n} d\omega + \frac{1}{2\pi} \int_0^{\pi} 2j e^{j\omega n} d\omega \\ &= -\frac{j}{\pi} \frac{e^{j\omega n}}{jn} \Big|_{-\pi}^0 + \frac{j}{\pi} \frac{e^{j\omega n}}{jn} \Big|_0^{\pi} \\ &= -\frac{1}{n\pi} [1 - e^{-jn\pi}] + \frac{1}{n\pi} [e^{jn\pi} - 1] \\ &= \frac{1}{n\pi} [e^{jn\pi} - 1 - 1 + e^{-jn\pi}] \\ &= \frac{1}{n\pi} [e^{jn\pi} + e^{-jn\pi} - 2] \\ &= \frac{1}{n\pi} [2\cos(n\pi) - 2] \\ &= \frac{2}{n\pi} [\cos(n\pi) - 1] \\ x_2(n) &= \frac{4}{n\pi} \sin^2\left(\frac{n\pi}{2}\right) \end{aligned}$$

Example 6.31 Find the DTFT of the following:

- (a) $x_1(n) = \{1, -1, 2, 2\}$
- (b) $x_2(n) = \delta(n-1) + \delta(n+1)$
- (c) $x_3(n) = (0.5)^n u(n) + 2^n u(-n-1)$

Solution

(a) The DTFT $X_1(e^{j\omega})$ of $x_1(n)$ is given by

$$\begin{aligned} X_1(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_1(n) e^{-j\omega n} \\ &= \sum_{n=0}^3 x_1(n) e^{-j\omega n} \\ &= x(0) + x(1) e^{-j\omega} + x(2) e^{-j2\omega} \\ &\quad + x(3) e^{-j3\omega} \end{aligned}$$

$$\begin{aligned}
X_1(e^{j\omega}) &= 1 - e^{-j\omega} + 2e^{-j2\omega} + 2e^{-j3\omega} \\
&= 2je^{-j\omega/2} \left(\frac{e^{j\omega/2} - e^{-j\omega/2}}{2j} \right) \\
&\quad + 4e^{-j2.5\omega} \left(\frac{e^{j\omega/2} + e^{-j\omega/2}}{2} \right)
\end{aligned}$$

$$X_1(e^{j\omega}) = 2j \sin\left(\frac{\omega}{2}\right) e^{-j\omega/2} + 4 \cos\left(\frac{\omega}{2}\right) e^{-j2.5\omega}$$

(b) Taking the Fourier transform of $x_2(n)$, we obtain

$$X_2(e^{j\omega}) = \mathcal{F}[\delta(n-1)] + \mathcal{F}[\delta(n+1)]$$

Using the time-shifting property of the DTFT, we obtain

$$\begin{aligned}
X_2(e^{j\omega}) &= e^{-j\omega} + e^{j\omega} \\
&= 2\cos(\omega)
\end{aligned}$$

(c) Taking the Fourier transform of $x_3(n)$, we obtain

$$\begin{aligned}
X_3(e^{j\omega}) &= \mathcal{F}[(0.5)^n u(n)] + \mathcal{F}[2^n u(-n-1)] \\
&= \frac{1}{1 - 0.5e^{-j\omega}} + \mathcal{F}\left[\left(\frac{1}{2}\right)^{-n} u(-n-1)\right] \\
&= \frac{1}{1 - 0.5e^{-j\omega}} + \mathcal{F}[(0.5)^{-n} u(-n-1)] \\
&= \frac{1}{1 - 0.5e^{-j\omega}} + \frac{0.5e^{j\omega}}{1 - 0.5e^{j\omega}} \\
&= \frac{1 - (0.5)^2}{1 - \cos(\omega) + (0.5)^2} \\
X_3(e^{j\omega}) &= \frac{0.75}{1 - \cos(\omega) + 0.25}
\end{aligned}$$

Example 6.32 Given that $x(n)$ has Fourier transform $X(e^{j\omega})$, express the Fourier transform of the following signals in terms of $X(e^{j\omega})$:

- (a) $x_1(n) = x(1-n) + x(-1-n)$
- (b) $x_2(n) = \frac{x^*(n) + x(n)}{2}$
- (c) $x_3(n) = (n-1)^2 x(n)$
- (d) $x_4(n) = x(2n+1)$
- (e) $x_5(n) = e^{j\pi/2n} x(n+2)$

Solution

Given that

$$x(n) \longleftrightarrow X(e^{j\omega})$$

(a) Using the time-shifting property [Eq. (6.38)], we have

$$x(n+1) \longleftrightarrow X(e^{j\omega}) e^{j\omega}$$

and

$$x(n-1) \longleftrightarrow X(e^{j\omega}) e^{-j\omega}$$

Using the time-reversal property [Eq. (6.41)] on this, we have

$$x(-n+1) = x(1-n) \longleftrightarrow X(e^{-j\omega}) e^{-j\omega}$$

$$\text{and } x(-n-1) = x(-1-n) \longleftrightarrow X(e^{-j\omega}) e^{j\omega}$$

Therefore,

$$\begin{aligned}
\mathcal{F}[x_1(n)] &= \mathcal{F}[x(1-n) + x(-1-n)] \\
X(e^{-j\omega}) &= X(e^{-j\omega}) e^{-j\omega} + X(e^{-j\omega}) e^{j\omega} \\
X_1(e^{j\omega}) &= 2X(e^{-j\omega}) \cos(\omega)
\end{aligned}$$

(b) Using the conjugation property [Eq. (6.54)], we have

$$x^*(n) \longleftrightarrow X^*(e^{-j\omega})$$

Using the time reversal-property [Eq. (6.41)] on this, we have

$$x^*(-n) \longleftrightarrow X^*(e^{j\omega})$$

Therefore,

$$\begin{aligned}
\mathcal{F}[x_2(n)] &= \mathcal{F}\left[\frac{x^*(n) + x(n)}{2}\right] \\
X_2(e^{j\omega}) &= \frac{X^*(e^{j\omega}) + X(e^{j\omega})}{2} \\
&= \Re\{X(e^{j\omega})\} = X_R(e^{j\omega}) \\
X_2(e^{j\omega}) &= X_R(e^{j\omega})
\end{aligned}$$

(c) Consider the given signal $x_3(n) = (n-1)^2 x(n)$. Using the differentiation in frequency-domain property [Eq. (6.48)], we have

$$nx(n) \longleftrightarrow j \frac{dX(e^{j\omega})}{d\omega}$$

Using the differentiation in frequency-domain property again, we have

$$n[x(n)] \longleftrightarrow j \frac{d}{d\omega} \left(j \frac{dX(e^{j\omega})}{d\omega} \right)$$

$$n^2 x(n) \longleftrightarrow -\frac{d^2 X(e^{j\omega})}{d\omega^2}$$

Therefore,

$$\begin{aligned} \mathcal{F}[x_3(n)] &= \mathcal{F}[(n-1)^2 x(n)] \\ &= \mathcal{F}[n^2 x(n) - 2nx(n) + x(n)] \end{aligned}$$

$$X_3(e^{j\omega}) = -\frac{d^2 X(e^{j\omega})}{d\omega^2} - 2j \frac{dX(e^{j\omega})}{d\omega} + X(e^{j\omega})$$

$$X_3(e^{j\omega}) = -\frac{d^2 X(e^{j\omega})}{d\omega^2} - 2j \frac{dX(e^{j\omega})}{d\omega} + X(e^{j\omega})$$

(d) Using the time-shifting property [Eq. (6.38)], we have

$$x(n+1) \longleftrightarrow X(e^{j\omega}) e^{j\omega}$$

Using the time-expansion property [Eq. (6.45) (choose $m = 1/2$)] on this, we have

$$\mathcal{F}[x_4(n)] = \mathcal{F}[x(2n+1)]$$

$$X_4(e^{j\omega}) = X(e^{j\omega/2}) e^{j\omega/2}$$

$$X_4(e^{j\omega}) = X(e^{j\omega/2}) e^{j\omega/2}$$

(e) Using the time-shifting property [Eq. (6.38)], we have

$$x(n+2) \longleftrightarrow X(e^{j\omega}) e^{j2\omega}$$

Using the frequency-shifting property [Eq. (6.39)] on this, we have

$$x_5(n) = e^{j\pi/2n} x(n+2) \longleftrightarrow X(e^{j(\omega-\pi/2)}) e^{j2(\omega-\pi/2)}$$

$$X_5(e^{j\omega}) = X(e^{j(\omega-\pi/2)}) e^{j2(\omega-\pi/2)}$$

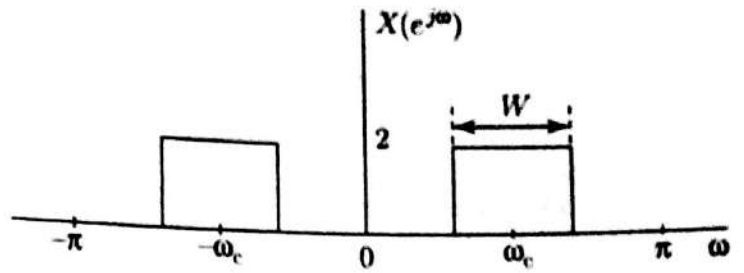


Fig. 6.24

Solution

The given DTFT can be written as

$$X(e^{j\omega}) = \begin{cases} 2, & \omega_c - \frac{W}{2} \leq |\omega| \leq \omega_c + \frac{W}{2} \\ 0, & \omega_c + \frac{W}{2} \leq |\omega| \leq \pi \\ 0, & 0 \leq |\omega| \leq \omega_c - \frac{W}{2} \end{cases}$$

The inverse DTFT of $X(e^{j\omega})$ is given by

$$\begin{aligned} x(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \left[\int_{-\omega_c-(W/2)}^{-\omega_c+(W/2)} 2 e^{j\omega n} d\omega + \int_{\omega_c-(W/2)}^{\omega_c+(W/2)} 2 e^{j\omega n} d\omega \right] \\ &= \frac{1}{\pi} \left[\frac{e^{j\omega n}}{jn} \Big|_{-\omega_c-(W/2)}^{-\omega_c+(W/2)} + \frac{e^{j\omega n}}{jn} \Big|_{\omega_c-(W/2)}^{\omega_c+(W/2)} \right] \\ &= \frac{1}{jn\pi} \left[e^{j(-\omega_c+(W/2))n} - e^{j(-\omega_c-(W/2))n} \right. \\ &\quad \left. + e^{j(\omega_c+(W/2))n} - e^{j(\omega_c-(W/2))n} \right] \\ &= \frac{2}{n\pi} \left[\frac{e^{j(\omega_c+(W/2))n} - e^{-j(\omega_c+(W/2))n}}{2j} \right. \\ &\quad \left. - \frac{e^{j(\omega_c-(W/2))n} - e^{-j(\omega_c-(W/2))n}}{2j} \right] \\ &= \frac{2}{n\pi} \left[\sin \left(\omega_c + \frac{W}{2} \right) n - \sin \left(\omega_c - \frac{W}{2} \right) n \right] \\ &= \frac{4}{n\pi} \cos(\omega_c n) \sin \left(\frac{W}{2} n \right) \end{aligned}$$

Example 6.33 Determine the signal $x(n]$ if its Fourier transform is as given in Fig. 6.24.

Therefore,

$$x(n) = \frac{4}{n\pi} \cos(\omega_c n) \cdot \sin\left(\frac{W}{2}n\right)$$

Example 6.34 Determine the magnitude and phase of $H(e^{j\omega})$ for the three-point moving average system

$$y(n) = \frac{1}{3}[x(n+1) + x(n) + x(n-1)]$$

an plot these two functions for $0 \leq \omega \leq \pi$.

Solution

Consider the given difference equation

$$y(n) = \frac{1}{3}[x(n+1) + x(n) + x(n-1)]$$

Taking the DTFT of the above equation, we obtain

$$Y(e^{j\omega}) = \frac{1}{3}[X(e^{j\omega})e^{j\omega} + X(e^{j\omega}) + X(e^{j\omega})e^{-j\omega}]$$

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = H(e^{j\omega}) = \frac{1}{3}[e^{j\omega} + 1 + e^{-j\omega}]$$

$$H(e^{j\omega}) = \frac{1}{3}[1 + 2\cos(\omega)]$$

The magnitude response is

$$|H(e^{j\omega})| = \frac{1}{3}|[1 + 2\cos(\omega)]|$$

and the phase response is

$$\angle H(e^{j\omega}) = \begin{cases} 0, & 0 \leq \omega \leq \frac{2\pi}{3} \\ \pi, & \frac{2\pi}{3} < \omega < \pi \end{cases}$$

Figure 6.25 illustrates the magnitude and phase of $H(e^{j\omega})$.

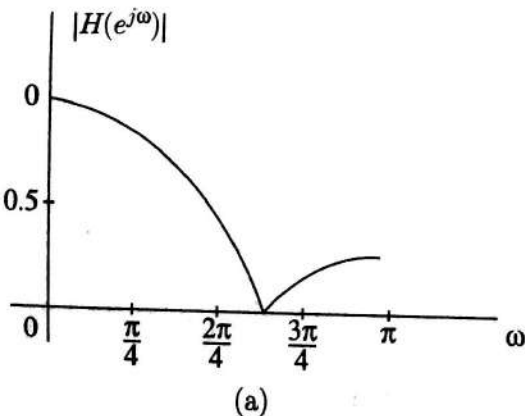


Fig. 6.25 (a)

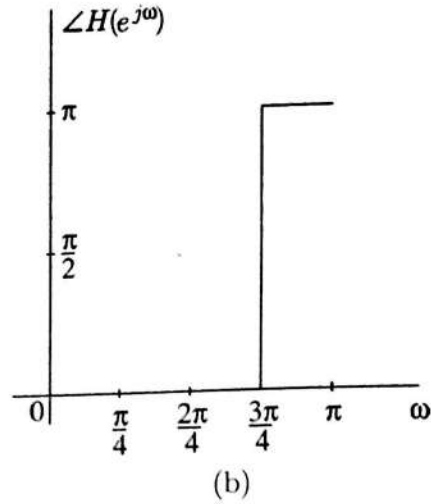


Fig. 6.25 (b)

Example 6.35 Let the sequence $x(n]$ be a real sequence and let

$$x(n) \longleftrightarrow X(e^{j\omega})$$

Prove that

(a)

$$|X(e^{j\omega})| = |X(e^{-j\omega})|$$

and

(b)

$$\angle X(e^{j\omega}) = -\angle X(e^{-j\omega})$$

Solution

Given that

$$x(n) \longleftrightarrow X(e^{j\omega})$$

In general the Fourier transform $X(e^{j\omega})$ is a complex function of the real variable ω and can be written in rectangular form as

$$X(e^{j\omega}) = X_R(e^{j\omega}) + jX_I(e^{j\omega})$$

where $X_R(e^{j\omega})$ and $X_I(e^{j\omega})$ are, respectively, the real and imaginary parts of $X(e^{j\omega})$.

The Fourier transform $X(e^{j\omega})$ can alternatively be expressed in the polar form as

$$X(e^{j\omega}) = |X(e^{j\omega})| e^{j\theta(\omega)}$$

where

$$\theta(\omega) = \angle X(e^{j\omega})$$

The quantity $|X(e^{j\omega})|$ is called the magnitude spectrum and the quantity $\theta(\omega) = \angle X(e^{j\omega})$ is called the phase spectrum.

The relations between the rectangular and polar forms of $X(e^{j\omega})$ is given by

$$X(e^{j\omega}) = X_R(e^{j\omega}) + jX_I(e^{j\omega}) = |X(e^{j\omega})|e^{j\theta(\omega)}$$

$$X_R(e^{j\omega}) + jX_I(e^{j\omega}) = |X(e^{j\omega})| \cos(\theta(\omega)) + j|X(e^{j\omega})| \sin(\theta(\omega))$$

Therefore,

$$X_R(e^{j\omega}) = |X(e^{j\omega})| \cos(\theta(\omega)) = X_R(e^{-j\omega})$$

$$X_I(e^{j\omega}) = |X(e^{j\omega})| \sin(\theta(\omega)) = -X_I(e^{-j\omega})$$

(a) The magnitude spectrum $|X(e^{j\omega})|$ is given by

$$|X(e^{j\omega})| = \sqrt{X_R^2(e^{j\omega}) + X_I^2(e^{j\omega})}$$

Thus, for a real signal,

$$|X(e^{-j\omega})| = \sqrt{X_R^2(e^{-j\omega}) + X_I^2(e^{-j\omega})}$$

$$|X(e^{-j\omega})| = \sqrt{X_R^2(e^{j\omega}) + X_I^2(e^{j\omega})}$$

$$|X(e^{-j\omega})| = |X(e^{j\omega})|$$

The magnitude spectrum $|X(e^{j\omega})|$ is an even function of ω .

(b) We know that the Fourier transform $X(e^{j\omega})$ is a complex function of the real variable ω and can be written in the rectangular form as

$$X(e^{j\omega}) = X_R(e^{j\omega}) + jX_I(e^{j\omega})$$

$$\theta(\omega) = \angle X(e^{j\omega}) = \tan^{-1} \left[\frac{X_I(e^{j\omega})}{X_R(e^{j\omega})} \right]$$

For a real signal,

$$\angle X(e^{-j\omega}) = \tan^{-1} \left[\frac{X_I(e^{-j\omega})}{X_R(e^{-j\omega})} \right]$$

$$= \tan^{-1} \left[-\frac{X_I(e^{j\omega})}{X_R(e^{j\omega})} \right]$$

$$\angle X(e^{-j\omega}) = -\tan^{-1} \left[\frac{X_I(e^{j\omega})}{X_R(e^{j\omega})} \right]$$

$$\angle X(e^{-j\omega}) = -\angle X(e^{j\omega})$$

The above equation implies that the phase spectrum $\angle X(e^{j\omega})$ is an odd function of ω .

Example 6.36 Find the discrete-time Fourier transform of

(a) $x_1(n) = \cos(\omega_0 n)u(n)$

(b) $x_2(n) = \sin(\omega_0 n)u(n)$

Solution

(a) Consider the given signal

$$x_1(n) = \cos(\omega_0 n)u(n)$$

Using the multiplication property [Eq. (6.53)], we have

$$X_1(e^{j\omega}) = \mathcal{F}[\cos(\omega_0 n)] * \mathcal{F}[u(n)], \quad -\pi \leq \omega < \pi$$

$$= \pi[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$* \left[\frac{1}{1 - e^{-j\omega}} + \pi\delta(\omega) \right]$$

$$= \pi \left[\frac{1}{1 - e^{-j\omega_0}} + \frac{1}{1 - e^{j\omega_0}} \right]$$

$$+ \pi^2[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

$$X_1(e^{j\omega}) = \pi + \pi^2[\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$

(b) Consider the given signal

$$x_2(n) = \sin(\omega_0 n)u(n)$$

Using the multiplication property [Eq. (6.53)], we have

$$X_2(e^{j\omega}) = \mathcal{F}[\sin(\omega_0 n)] * \mathcal{F}[u(n)], \quad -\pi \leq \omega < \pi$$

$$= \frac{\pi}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

$$* \left[\frac{1}{1 - e^{-j\omega}} + \pi\delta(\omega) \right]$$

$$= \frac{\pi}{j} \left[\frac{1}{1 - e^{-j\omega_0}} - \frac{1}{1 - e^{j\omega_0}} \right]$$

$$+ \frac{\pi^2}{j}[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

$$= \pi \frac{-\sin(\omega_0)}{1 - \cos(\omega_0)}$$

$$+ \pi^2[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

$$= -\pi \frac{2 \sin\left(\frac{\omega_0}{2}\right) \cos\left(\frac{\omega_0}{2}\right)}{2 \sin^2\left(\frac{\omega_0}{2}\right)}$$

$$+ \pi^2[\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

$$X_2(e^{j\omega}) = -\pi \cot\left(\frac{\omega_0}{2}\right) + \pi^2 [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

Example 6.37 Determine the signal $x(n)$ for the following given discrete-time Fourier transform:

- (a) $X(e^{j\omega}) = e^{-j\alpha\omega}$, for $-\omega_c \leq \omega \leq \omega_c$
 (b) $X(e^{j\omega}) = e^{-j\omega}[1 + \cos(\omega)]$

Solution

(a) The inverse DTFT of $X(e^{j\omega})$ is given by

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\alpha\omega} e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n-\alpha)} d\omega \\ &= \frac{1}{2\pi j(n-\alpha)} e^{j\omega(n-\alpha)} \Big|_{-\omega_c}^{\omega_c} \\ &= \frac{1}{\pi(n-\alpha)} \frac{e^{j\omega_c(n-\alpha)} - e^{-j\omega_c(n-\alpha)}}{2j} \\ x(n) &= \frac{\sin[\omega_c(n-\alpha)]}{\pi(n-\alpha)} \end{aligned}$$

(b) Consider the given DTFT

$$\begin{aligned} X(e^{j\omega}) &= e^{-j\omega}[1 + \cos(\omega)] \\ &= e^{-j\omega} \left[1 + \frac{e^{j\omega} + e^{-j\omega}}{2} \right] \\ &= e^{-j\omega} + 0.5 + 0.5 e^{-j2\omega} \\ X(e^{j\omega}) &= 0.5 + e^{-j\omega} + 0.5 e^{-j2\omega} \end{aligned}$$

Taking the inverse DTFT of the above equation, we obtain

$$\begin{aligned} x(n) &= 0.5\delta(n) + \delta(n-1) + 0.5\delta(n-2) \\ &= \{0.5, 1, 0.5\} \end{aligned}$$

Example 6.38 The following four facts are given about a particular signal $x(n)$ with Fourier transform $X(e^{j\omega})$:

1. $x(n) = 0$, for $n > 0$.
2. $x(0) > 0$.
3. $\Im\{X(e^{j\omega})\} = X_I(e^{j\omega}) = \sin(\omega) - \sin(2\omega)$
4. $\frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega = 3$.

Determine $x(n)$.

Solution

We know that the DTFT of a real and an odd signal is purely imaginary and odd [Eq. (6.59)]. Therefore, we have

$$x_o(n) = \mathcal{O}\{x(n)\} \longleftrightarrow j\Im\{X(e^{j\omega})\} = jX_I(e^{j\omega})$$

$$\begin{aligned} x_o(n) &= \mathcal{F}^{-1}[jX_I(e^{j\omega})] \\ &= \mathcal{F}^{-1}[j\sin(\omega) - j\sin(2\omega)] \\ &= \mathcal{F}^{-1}\left[\frac{1}{2}(e^{j\omega} - e^{-j\omega} - e^{j2\omega} + e^{-j2\omega})\right] \end{aligned}$$

$$\begin{aligned} x_o(n) &= \frac{1}{2} [\delta(n+1) - \delta(n-1) \\ &\quad - \delta(n+2) + \delta(n-2)] \\ &= \left\{ -\frac{1}{2}, \frac{1}{2}, 0, -\frac{1}{2}, \frac{1}{2} \right\} \end{aligned}$$

We also know that

$$x_o(n) = \frac{x(n) - x(-n)}{2}$$

and that $x(n) = 0$, for $n > 0$. Therefore,

$$x(n) = 2x_o(n), \quad \text{for } n < 0$$

$$x(n) = \delta(n+1) - \delta(n+2)$$

Now, we have to find $x(0)$. Using Parseval's relation [Eq. (6.60)], we have

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

From fact number 4, we can write

$$\sum_{n=-\infty}^0 |x(n)|^2 = 3$$

$$x^2(0) + \sum_{n=-\infty}^{-1} |x(n)|^2 = 3$$

$$x^2(0) + 2 = 3$$

$$x^2(0) = 1$$

$$x(0) = \pm 1$$

Since $x(0) > 0$, we conclude that $x(0) = 1$. Therefore,

$$x(n) = \delta(n) + \delta(n+1) - \delta(n+2)$$

$$= \{-1, 1, 1\}$$

Example 6.39 Consider a signal $g(n)$ with Fourier transform $G(e^{j\omega})$. Suppose

$$g(n) = x_{(2)}(n)$$

where the signal $x(n)$ has a Fourier transform $X(e^{j\omega})$. Determine a real number α such that $0 < \alpha < 2\pi$ and $G(e^{j\omega}) = G(e^{j(\omega-\alpha)})$.

Solution

Using the time-expansion property [Eq. (6.45)], we have

$$g(n) = x_{(2)}(n) \longleftrightarrow G(e^{j\omega}) = X(e^{j2\omega})$$

Therefore, $G(e^{j\omega})$ is obtained by compressing $X(e^{j\omega})$ by a factor of 2. Since, we know that $X(e^{j\omega})$ is periodic with period 2π , we may conclude that $G(e^{j\omega})$ is periodic with period $\frac{1}{2}2\pi = \pi$. Hence,

$$G(e^{j\omega}) = G(e^{j(\omega-\pi)}) = G(e^{j(\omega-\alpha)})$$

Therefore, $\alpha = \pi$.

Example 6.40 A linear time-invariant system is described by the following difference equation:

$$y(n) = ay(n-1) + bx(n), \quad 0 < a < 1$$

- Determine the magnitude response $|H(e^{j\omega})|$ and phase response $\angle H(e^{j\omega})$ of the system.
- Choose the parameter b such that the maximum value of $|H(e^{j\omega})|$ is unity.

- Determine the output of the system to the input signal

$$x(n) = 5 + 12 \sin\left(\frac{\pi}{2}n\right) + 20 \cos\left(\pi n + \frac{\pi}{4}\right)$$

for $a = 0.9$.

Solution

- Consider the given difference equation

$$y(n) = ay(n-1) + bx(n), \quad 0 < a < 1$$

Taking the Fourier transform of the above equation, we obtain

$$Y(e^{j\omega}) = aY(e^{j\omega})e^{-j\omega} + bX(e^{j\omega})$$

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = H(e^{j\omega}) = \frac{b}{1 - ae^{-j\omega}}$$

$$H(e^{j\omega}) = \frac{b}{1 - a \cos(\omega) + ja \sin(\omega)}$$

Therefore,

$$|H(e^{j\omega})| = \frac{|b|}{\sqrt{(1 - a \cos(\omega))^2 + (a \sin(\omega))^2}}$$

$$|H(e^{j\omega})| = \frac{|b|}{\sqrt{1 + a^2 - 2a \cos(\omega)}}$$

$$\text{and} \quad \angle H(e^{j\omega}) = \angle b - \tan^{-1} \frac{a \sin(\omega)}{1 - a \cos(\omega)}$$

- Since the parameter a is positive, the denominator of $|H(e^{j\omega})|$ attains a minimum at $\omega = 0$. Therefore, $|H(e^{j\omega})|$ attains its maximum value at $\omega = 0$. At this frequency, we have

$$|H(e^{j0})| = \frac{|b|}{1 - a} = 1$$

$$|b| = 1 - a$$

$$b = \pm(1 - a)$$

- For $a = 0.9$ and $b = (1 - a) = 0.1$, the frequency response of the system is

$$H(e^{j\omega}) = \frac{0.1}{1 - 0.9e^{-j\omega}}$$

The first term in the input signal $x(n)$ is a constant signal corresponding to $\omega = 0$, thus

$$H(e^{j0}) = \frac{0.1}{1 - 0.9} = 1$$

The second term in $x(n)$ has a frequency $\pi/2$. At this frequency the frequency response of the system is

$$\begin{aligned} H(e^{j\pi/2}) &= \frac{0.1}{1 - 0.9e^{-j\pi/2}} \\ &= \frac{0.1}{1 + j0.9} = 0.074e^{-j42^\circ} \end{aligned}$$

Finally, the third term in $x(n)$ has a frequency $\omega = \pi$. At this frequency the frequency response of the system is

$$\begin{aligned} H(e^{j\pi}) &= \frac{0.1}{1 - 0.9e^{-j\pi}} \\ &= \frac{0.1}{1 + 0.9} = \frac{0.1}{1.9} = 0.053 \end{aligned}$$

Hence, the response of the system to $x(n)$ is

$$\begin{aligned} y(n) &= 5H(e^{j0}) + H(e^{j\pi/2})12\sin\left(\frac{\pi}{2}n\right) \\ &\quad - H(e^{j\pi})20\cos\left(\pi n + \frac{\pi}{4}\right) \\ y(n) &= 5 + 0.888\sin\left(\frac{\pi}{2}n - 42^\circ\right) \\ &\quad - 1.06\cos\left(\pi n + \frac{\pi}{4}\right) \end{aligned}$$

Example 6.41 Suppose we are given the following facts about an LTI system with impulse response $h(n)$ and frequency response $H(e^{j\omega})$:

1. $(1/4)^n u(n) \longrightarrow g(n)$, where $g(n) = 0$ for $n \geq 2$ and $n < 0$
2. $H(e^{j\pi/2}) = 1$
3. $H(e^{j\omega}) = H(e^{j(\omega-\pi)})$

Determine $h(n)$.

Solution

Given that $g(n) = 0$ for $n \geq 2$ and $n < 0$. Therefore, the Fourier transform $G(e^{j\omega})$ of $g(n)$ is

$$\begin{aligned} G(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} g(n)e^{-j\omega n} \\ &= \sum_{n=0}^1 g(n)e^{-j\omega n} \\ G(e^{j\omega}) &= g(0) + g(1)e^{-j\omega} \end{aligned}$$

Given that the input to the system is $x(n) = (1/4)^n u(n)$, its Fourier transform is $X(e^{j\omega}) = \frac{1}{1 - \frac{1}{4}e^{-j\omega}}$. Also, when the input to the system is $x(n)$, the output is $g(n)$. Therefore,

$$\begin{aligned} H(e^{j\omega}) &= \frac{G(e^{j\omega})}{X(e^{j\omega})} \\ &= \left(g(0) + g(1)e^{-j\omega}\right) \left(1 - \frac{1}{4}e^{-j\omega}\right) \\ H(e^{j\omega}) &= g(0) + \left[g(1) - \frac{1}{4}g(0)\right]e^{-j\omega} + \frac{1}{4}g(1)e^{-j2\omega} \\ H(e^{j\omega}) &= h(0) + h(1)e^{-j\omega} + h(2)e^{-j2\omega} \end{aligned}$$

Clearly, $h(n)$ is a three-point sequence. From the above equation, we have

$$\begin{aligned} H(e^{j(\omega-\pi)}) &= h(0) + h(1)e^{-j(\omega-\pi)} + h(2)e^{-j2(\omega-\pi)} \\ &= h(0) + h(1)e^{-j\omega}e^{j\pi} + h(2)e^{-j2\omega}e^{j2\pi} \\ H(e^{j(\omega-\pi)}) &= h(0) - h(1)e^{-j\omega} + h(2)e^{-j2\omega} \end{aligned}$$

Given that

$$\begin{aligned} H(e^{j\omega}) &= H(e^{j(\omega-\pi)}) \\ h(0) + h(1)e^{-j\omega} + h(2)e^{-j2\omega} &= h(0) - h(1)e^{-j\omega} \\ &\quad + h(2)e^{-j2\omega} \\ 2h(1)e^{-j\omega} &= 0 \\ h(1) &= 0 \end{aligned}$$

We also have

$$\begin{aligned} H(e^{j\pi/2}) &= 1 \\ h(0) + h(1)e^{-j\pi/2} + h(2)e^{-j2\pi/2} &= 1 \\ h(0) + 0 + h(2)e^{-j\pi} &= 1 \\ h(0) - h(2) &= 1 \end{aligned}$$

Note that

$$\begin{aligned} g(n) &= h(n) * x(n) \\ &= \sum_{k=-\infty}^{\infty} h(k)x(n-k) \\ g(n) &= \sum_{k=0}^2 h(k) \left(\frac{1}{4}\right)^{n-k} u(n-k) \end{aligned}$$

Evaluating this equation at $n = 2$, we have

$$g(2) = \sum_{k=0}^2 h(k) \left(\frac{1}{4}\right)^{2-k} u(2-k) = 0$$

$$\frac{1}{16}h(0) + \frac{1}{4}h(1) = 0$$

Now solving the two equations

$$h(0) - h(2) = 1$$

$$\frac{1}{16}h(0) + \frac{1}{4}h(1) = 0$$

we obtain

$$h(0) = \frac{16}{17} \quad \text{and} \quad h(2) = -\frac{1}{17}$$

Therefore,

$$h(n) = \frac{16}{17}\delta(n) - \frac{1}{17}\delta(n-2)$$

$$= \left\{ \frac{16}{17}, 0, -\frac{1}{17} \right\}$$

Example 6.42 An LTI system with impulse response $h_1(n) = (1/3)^n u(n)$ is connected in parallel with another causal LTI system with impulse response $h_2(n)$. The resulting parallel interconnection has the frequency response

$$H(e^{j\omega}) = \frac{-12 + 5e^{-j\omega}}{12 - 7e^{-j\omega} + e^{-j2\omega}}$$

Determine $h_2(n)$.

Solution

When the two systems are connected in parallel, the impulse response of the overall system is the sum of the impulse responses of the individual systems. Therefore,

$$h(n) = h_1(n) + h_2(n)$$

Taking the Fourier transform of the above equation, we obtain

$$H(e^{j\omega}) = H_1(e^{j\omega}) + H_2(e^{j\omega})$$

Given that $h_1(n) = (1/3)^n u(n)$, we obtain

$$H_1(e^{j\omega}) = \frac{1}{1 - \frac{1}{3}e^{-j\omega}}$$

Therefore,

$$H_2(e^{j\omega}) = H(e^{j\omega}) - H_1(e^{j\omega})$$

$$= \frac{-12 + 5e^{-j\omega}}{12 - 7e^{-j\omega} + e^{-j2\omega}} - \frac{1}{1 - \frac{1}{3}e^{-j\omega}}$$

$$= \frac{-12 + 5e^{-j\omega}}{(4 - e^{-j\omega})(3 - e^{-j\omega})} - \frac{3}{3 - e^{-j\omega}}$$

$$= -\frac{8}{4 - e^{-j\omega}}$$

$$H_2(e^{j\omega}) = -\frac{2}{1 - \frac{1}{4}e^{-j\omega}}$$

Taking the inverse Fourier transform, we obtain

$$h_2(n) = -2 \left(\frac{1}{4}\right)^n u(n)$$

Example 6.43 Consider a discrete-time LTI system with frequency response

$$H(e^{j\omega}) = \left[\frac{1 + e^{-j2\omega} + 4e^{-j4\omega}}{1 + \frac{1}{2}e^{-j2\omega}} \right] e^{-j(\omega - \pi/4)}$$

$$-\pi < \omega \leq \pi$$

Determine the Fourier transform of the output if the input is

$$x(n) = \cos\left(\frac{\pi n}{2}\right)$$

Solution

Consider the given input

$$x(n) = \cos\left(\frac{\pi n}{2}\right)$$

Taking the Fourier transform of the above equation, we obtain

$$X(e^{j\omega}) = \pi \left[\delta\left(\omega - \frac{\pi}{2}\right) + \delta\left(\omega + \frac{\pi}{2}\right) \right]$$

$$-\pi < \omega \leq \pi$$

The Fourier transform of the output is given by

$$\begin{aligned} Y(e^{j\omega}) &= H(e^{j\omega})X(e^{j\omega}) \\ &= H(e^{j\omega})\pi \left[\delta\left(\omega - \frac{\pi}{2}\right) + \delta\left(\omega + \frac{\pi}{2}\right) \right] \\ Y(e^{j\omega}) &= \pi H(e^{j\pi/2})\delta\left(\omega - \frac{\pi}{2}\right) \\ &\quad + \pi H(e^{-j\pi/2})\delta\left(\omega + \frac{\pi}{2}\right) \end{aligned}$$

Given that

$$H(e^{j\omega}) = \left[\frac{1 + e^{-j2\omega} + 4e^{-j4\omega}}{1 + \frac{1}{2}e^{-j2\omega}} \right] e^{-j(\omega - \pi/4)}$$

Therefore,

$$\begin{aligned} H(e^{j\pi/2}) &= \left[\frac{1 + e^{-j2\pi/2} + 4e^{-j4\pi/2}}{1 + 1/2e^{-j2\pi/2}} \right] e^{-j(\pi/2 - \pi/4)} \\ &= \left[\frac{1 + e^{-j\pi} + 4e^{-j2\pi}}{1 + 1/2e^{-j\pi}} \right] e^{-j\pi/4} \\ H(e^{j\pi/2}) &= \left[\frac{1 - 1 + 4}{1 - 1/2} \right] e^{-j\pi/4} \\ H(e^{j\pi/2}) &= 8e^{-j\pi/4} \end{aligned}$$

Similarly, we can obtain

$$H(e^{-j\pi/2}) = 8e^{j3\pi/4}$$

Substituting the values of $H(e^{j\pi/2})$ and $H(e^{-j\pi/2})$ in the expression of $Y(e^{j\omega})$, we obtain

$$Y(e^{j\omega}) = 8\pi \left[e^{-j\pi/4}\delta\left(\omega - \frac{\pi}{2}\right) + e^{j3\pi/4}\delta\left(\omega + \frac{\pi}{2}\right) \right]$$

Example 6.44 A causal system is described by the difference equation

$$y(n) - ay(n-1) = bx(n) + x(n-1)$$

where a is real and less than 1 in magnitude. Find a value of b such that the frequency response of the

system satisfies

$$|H(e^{j\omega})| = 1, \quad \text{for all } \omega$$

This kind of system is called an all-pass system.

Solution

Consider the given difference equation,

$$y(n) - ay(n-1) = bx(n) + x(n-1)$$

Taking the Fourier transform of the above equation, we obtain

$$\begin{aligned} Y(e^{j\omega}) - aY(e^{j\omega})e^{-j\omega} &= bX(e^{j\omega}) + X(e^{j\omega})e^{-j\omega} \\ H(e^{j\omega}) &= \frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{b + e^{-j\omega}}{1 - ae^{-j\omega}} \end{aligned}$$

Taking the magnitude, we obtain

$$\begin{aligned} |H(e^{j\omega})| &= \frac{|b + e^{-j\omega}|}{|1 - ae^{-j\omega}|} \\ &= \frac{|b + \cos(\omega) - j\sin(\omega)|}{|1 - a\cos(\omega) + ja\sin(\omega)|} \\ |H(e^{j\omega})| &= \frac{\sqrt{[b + \cos(\omega)]^2 + \sin^2(\omega)}}{\sqrt{[1 - a\cos(\omega)]^2 + a^2\sin^2(\omega)}} \\ 1 &= \frac{\sqrt{1 + b^2 + 2b\cos(\omega)}}{\sqrt{1 + a^2 - 2a\cos(\omega)}} \end{aligned}$$

$$1 + b^2 + 2b\cos(\omega) = 1 + a^2 - 2a\cos(\omega)$$

From the above equation it is evident that $b = -a$.

Example 6.45 The centre of gravity of a signal $x(n]$ is defined as

$$C = \frac{\sum_{n=-\infty}^{\infty} nx(n)}{\sum_{n=-\infty}^{\infty} x(n)}$$

and provides a measure of 'time delay' of the signal.

(a) Express C in terms of $X(e^{j\omega})$.

(b) Compute C for the signal $x(n]$ whose DTFT is shown in Fig. 6.26.

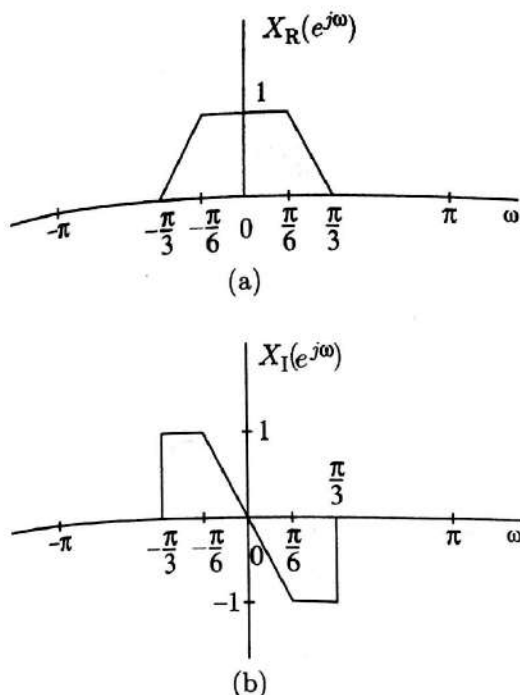


Fig. 6.26

Solution

(a) The DTFT $X(e^{j\omega})$ of a signal $x(n)$ is given by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

Differentiating both sides w.r.t. ω , we obtain

$$\frac{dX(e^{j\omega})}{d\omega} = \sum_{n=-\infty}^{\infty} [-jn x(n)] e^{-j\omega n}$$

$$j \frac{dX(e^{j\omega})}{d\omega} = \sum_{n=-\infty}^{\infty} [n x(n)] e^{-j\omega n}$$

$$j \frac{dX(e^{j\omega})}{d\omega} \Big|_{\omega=0} = \sum_{n=-\infty}^{\infty} n x(n)$$

We also know that [Eq. (6.62)]

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$X(e^{j\omega}) \Big|_{\omega=0} = \sum_{n=-\infty}^{\infty} x(n)$$

Therefore,

$$C = \frac{\sum_{n=-\infty}^{\infty} n x(n)}{\sum_{n=-\infty}^{\infty} x(n)} = \frac{j dX(e^{j\omega}) / (d\omega) \Big|_{\omega=0}}{X(e^{j\omega}) \Big|_{\omega=0}}$$

(b) We know that

$$X(e^{j\omega}) = X_R(e^{j\omega}) + j X_I(e^{j\omega})$$

$$\frac{dX(e^{j\omega})}{d\omega} = \frac{dX_R(e^{j\omega})}{d\omega} + j \frac{dX_I(e^{j\omega})}{d\omega}$$

$$\frac{dX(e^{j\omega})}{d\omega} \Big|_{\omega=0} = \frac{dX_R(e^{j\omega})}{d\omega} \Big|_{\omega=0} + j \frac{dX_I(e^{j\omega})}{d\omega} \Big|_{\omega=0}$$

From Figs 6.26(a) and (b), we have

$$\frac{dX_R(e^{j\omega})}{d\omega} \Big|_{\omega=0} = 0 \quad \text{and} \quad \frac{dX_I(e^{j\omega})}{d\omega} \Big|_{\omega=0} = -\frac{6}{\pi}$$

Therefore,

$$\frac{dX(e^{j\omega})}{d\omega} \Big|_{\omega=0} = 0 - j \frac{6}{\pi}$$

$$j \frac{dX(e^{j\omega})}{d\omega} \Big|_{\omega=0} = \frac{6}{\pi}$$

and

$$\begin{aligned} X(e^{j\omega}) \Big|_{\omega=0} &= X_R(e^{j\omega}) \Big|_{\omega=0} + j X_I(e^{j\omega}) \Big|_{\omega=0} \\ &= 1 + j0 \end{aligned}$$

$$X(e^{j\omega}) \Big|_{\omega=0} = 1$$

Substituting $j dX(e^{j\omega}) / (d\omega) \Big|_{\omega=0} = 6/\pi$ and $X(e^{j\omega}) \Big|_{\omega=0} = 1$ in the expression of C , yields

$$C = \frac{j dX(e^{j\omega}) / (d\omega) \Big|_{\omega=0}}{X(e^{j\omega}) \Big|_{\omega=0}} = \frac{6}{\pi}$$

Example 6.46 An LTI system with unit sample response $h(n)$ and frequency response $H(e^{j\omega})$ is known to have the property that, when $-\pi \leq \omega \leq \pi$,

$$\cos(\omega_0 n) \longrightarrow \omega_0 \cos(\omega_0 n)$$

- Determine $H(e^{j\omega})$.
- Determine $h(n)$.

Solution

(a) Given that

$$\cos(\omega_0 n) \rightarrow \omega_0 \cos(\omega_0 n)$$

$$\left(\frac{e^{j\omega_0 n}}{2} + \frac{e^{-j\omega_0 n}}{2} \right) \rightarrow \omega_0 \left(\frac{e^{j\omega_0 n}}{2} + \frac{e^{-j\omega_0 n}}{2} \right)$$

$$\left(\frac{e^{j\omega_0 n}}{2} + \frac{e^{-j\omega_0 n}}{2} \right) \rightarrow \left(\omega_0 \frac{e^{j\omega_0 n}}{2} + \omega_0 \frac{e^{-j\omega_0 n}}{2} \right)$$

Since the system is linear, we may conclude that

$$\frac{e^{j\omega_0 n}}{2} \rightarrow \omega_0 \frac{e^{j\omega_0 n}}{2}$$

and

$$\frac{e^{-j\omega_0 n}}{2} \rightarrow \omega_0 \frac{e^{-j\omega_0 n}}{2}$$

It is clear that when the input to the system is a complex exponential of frequency ω_0 , the output is a complex exponential of the same frequency but scaled by $|\omega_0|$. Therefore, the frequency response of the system is

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = H(e^{j\omega}) = |\omega| \quad 0 \leq |\omega| \leq \pi$$

$$H(e^{j\omega}) = \begin{cases} \omega, & 0 \leq \omega \leq \pi \\ -\omega, & -\pi \leq \omega \leq 0 \end{cases}$$

(b) Taking the inverse Fourier transform of the frequency response, we obtain

$$\begin{aligned} h(n) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H(e^{j\omega}) e^{j\omega n} d\omega \\ &= \frac{1}{2\pi} \int_{-\pi}^0 -\omega e^{j\omega n} d\omega + \frac{1}{2\pi} \int_0^{\pi} \omega e^{j\omega n} d\omega \\ &= -\frac{1}{2\pi} \left[\frac{\omega e^{j\omega n}}{jn} \Big|_{-\pi}^0 - \frac{e^{j\omega n}}{-n^2} \Big|_{-\pi}^0 \right] \\ &\quad + \frac{1}{2\pi} \left[\frac{\omega e^{j\omega n}}{jn} \Big|_0^{\pi} - \frac{e^{j\omega n}}{-n^2} \Big|_0^{\pi} \right] \\ &= -\frac{1}{2\pi} \left[0 + \frac{\pi e^{-j\pi n}}{jn} + \frac{1}{n^2} (1 - e^{-jn\pi}) \right] \\ &\quad + \frac{1}{2\pi} \left[\frac{\pi e^{j\pi n}}{jn} - 0 + \frac{1}{n^2} (e^{jn\pi} - 1) \right] \end{aligned}$$

@sTc

$$\begin{aligned} &= -\frac{e^{-jn\pi}}{j2n} - \frac{1}{2\pi n^2} (1 - e^{-jn\pi}) \\ &\quad + \frac{e^{jn\pi}}{j2n} + \frac{1}{2\pi n^2} (e^{jn\pi} - 1) \\ h(n) &= -\frac{(-1)^n}{j2n} + \frac{(-1)^n}{j2n} \\ &\quad + \frac{1}{2\pi n^2} [-1 + e^{-jn\pi} + e^{jn\pi} - 1] \\ &= \frac{1}{2\pi n^2} \left[2 \frac{e^{jn\pi} + e^{-jn\pi}}{2} - 2 \right] \\ h(n) &= \frac{1}{\pi n^2} [\cos(n\pi) - 1] \end{aligned}$$

Example 6.47 Let $x(n)$ be a discrete-time signal with DTFT $X(e^{j\omega})$. Determine and sketch the Fourier transform of the following signals:

(a) $y_1(n) = x(2n)$

(b) $y_2(n) = \begin{cases} x(n/2), & n \text{ even} \\ 0, & n \text{ odd} \end{cases}$

Solution

(a) Consider $y_1(n) = x(Mn) = x(2n)$, where $M = 2$. The DTFT of $y_1(n)$ is given by

$$\begin{aligned} Y_1(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} y_1(n) e^{-j\omega n} \\ Y_1(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(Mn) e^{-j\omega n} \end{aligned}$$

Define an intermediate sequence

$$x_1(n) = \begin{cases} x(n), & n = 0, \pm M, \pm 2M, \dots \\ 0, & \text{otherwise} \end{cases}$$

so that $y_1(n) = x(Mn) = x_1(Mn)$. Now

$$\begin{aligned} Y_1(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_1(Mn) e^{-j\omega n} \\ &= \sum_{k=-\infty}^{\infty} x_1(k) e^{-j\omega k/M} \end{aligned}$$

This step is valid because $x_1(k)$ is zero unless k is a multiple of M . So

$$Y_1(e^{j\omega}) = X_1(e^{j\omega/M})$$

It only remains to express $X_1(e^{j\omega})$ in terms of $X(e^{j\omega})$. For this note that

$$x_1(n) = c(n)x(n)$$

where $c(n)$ is a periodic sequence (period $N = M$) defined as

$$c(n) = \begin{cases} 1, & n = 0, \pm M, \pm 2M, \dots \\ 0, & \text{otherwise} \end{cases}$$

Using DTFS, $c(n)$ can be expressed as

$$c(n) = \sum_{k=0}^{M-1} C_k e^{jk\omega_0 n}$$

where $\omega_0 = 2\pi/M$, and

$$C_k = \frac{1}{M} \sum_{m=0}^{M-1} c(n) e^{-jk\omega_0 n}$$

$$= \frac{1}{M} \sum_{m=0}^{M-1} \delta(n) e^{-jk\omega_0 n}$$

$$C_k = \frac{1}{M}$$

Thus, the periodic sequence $c(n)$ can be expressed as

$$c(n) = \frac{1}{M} \sum_{k=0}^{M-1} e^{j2\pi nk/M}$$

We can now obtain

$$\begin{aligned} X_1(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x_1(n) e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} c(n)x(n) e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} \left(\frac{1}{M} \sum_{k=0}^{M-1} e^{j2\pi nk/M} \right) x(n) e^{-j\omega n} \\ &= \frac{1}{M} \sum_{k=0}^{M-1} \left(\sum_{n=-\infty}^{\infty} x(n) e^{-j(\omega - 2\pi k/M)n} \right) \end{aligned}$$

$$X_1(e^{j\omega}) = \frac{1}{M} \sum_{k=0}^{M-1} X(e^{j(\omega - 2\pi k/M)})$$

For $M = 2$, the above equation yields

$$X_1(e^{j\omega}) = \frac{1}{2} \sum_{k=0}^1 X(e^{j(\omega - \pi k)})$$

$$\begin{aligned} X_1(e^{j\omega}) &= \frac{1}{2} [X(e^{j\omega}) + X(e^{j\omega - \pi})] \\ &= \frac{1}{2} [X(e^{j\omega}) + X(e^{j\omega} e^{-j\pi})] \end{aligned}$$

$$X_1(e^{j\omega}) = \frac{1}{2} [X(e^{j\omega}) + X(-e^{j\omega})]$$

Thus, for $M = 2$, we get

$$\begin{aligned} Y_1(e^{j\omega}) &= X_1(e^{j\omega/2}) \\ &= \frac{1}{2} [X(e^{j\omega/2}) + X(-e^{j\omega/2})] \end{aligned}$$

(b) Consider the given signal

$$\begin{aligned} y_2(n) &= \begin{cases} x(n/2), & n \text{ even} \\ 0, & n \text{ odd} \end{cases} \\ &= \begin{cases} x(n/2), & \text{if } n \text{ is a multiple of 2} \\ 0, & \text{if } n \text{ is not a multiple of 2.} \end{cases} \end{aligned}$$

The Fourier transform of $y_2(n)$ is given by

$$\begin{aligned} Y_2(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} y_2(n) e^{-j\omega n} \\ &= \sum_{n=-\infty}^{\infty} x\left(\frac{n}{2}\right) e^{-j\omega n} \end{aligned}$$

A change of variables is performed by letting $r = n/2$, which also yields $r = -\infty$ as $n = -\infty$, and $r = \infty$ as $n = \infty$. Therefore,

$$Y_2(e^{j\omega}) = \sum_{r=-\infty}^{\infty} x(r) e^{-j\omega 2r}$$

$$Y_2(e^{j\omega}) = X(e^{j2\omega})$$

Example 6.48 Consider a discrete-time LTI system with unit sample response

$$h(n) = \left(\frac{1}{2}\right)^n u(n) + \frac{1}{2} \left(\frac{1}{4}\right)^n u(n)$$

Determine a linear constant-coefficient difference equation relating the input and output of the system.

Solution

Consider the given unit sample response

$$h(n) = \left(\frac{1}{2}\right)^n u(n) + \frac{1}{2} \left(\frac{1}{4}\right)^n u(n)$$

Taking the Fourier transform of the unit sample response, we obtain

$$H(e^{j\omega}) = \frac{1}{1 - 1/2 e^{-j\omega}} + \frac{1/2}{1 - 1/4 e^{-j\omega}}$$

$$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{(3/2) - (1/2)e^{-j\omega}}{1 - (3/4)e^{-j\omega} + (1/8)e^{-j2\omega}}$$

$$\begin{aligned} Y(e^{j\omega}) \left[1 - \frac{3}{4} e^{-j\omega} + \frac{1}{8} e^{-j2\omega} \right] &= X(e^{j\omega}) \left[\frac{3}{2} - \frac{1}{2} e^{-j\omega} \right] \\ Y(e^{j\omega}) - \frac{3}{4} e^{-j\omega} Y(e^{j\omega}) + \frac{1}{8} e^{-j2\omega} Y(e^{j\omega}) &= \frac{3}{2} X(e^{j\omega}) - \frac{1}{2} e^{-j\omega} X(e^{j\omega}) \end{aligned}$$

Taking the inverse Fourier transform, we obtain

$$\begin{aligned} y(n) - \frac{3}{4} y(n-1) + \frac{1}{8} y(n-2) &= \frac{3}{2} x(n) - \frac{1}{2} x(n-1) \end{aligned}$$

Example 6.49 Given input $x(n]$ and impulse response $h(n]$, as shown in Fig. 6.27, evaluate $y(n) = x(n) * h(n]$, using the DTFT.

Solution

The DTFT of the given input signal $x(n) = \{1, \frac{1}{2}, 1\}$ is given by

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \\ &= \sum_{n=-1}^1 x(n) e^{-j\omega n} \end{aligned}$$

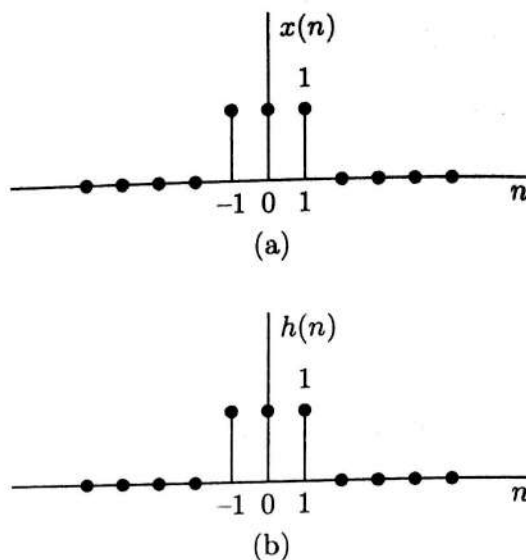


Fig. 6.27

$$= x(-1) e^{j\omega} + x(0) + x(1) e^{-j\omega}$$

$$X(e^{j\omega}) = 1 + e^{j\omega} + e^{-j\omega}$$

Similarly, the DTFT of the impulse response $h(n) = \{1, \frac{1}{2}, 1\}$ is given by

$$H(e^{j\omega}) = 1 + e^{j\omega} + e^{-j\omega}$$

Now using the convolution property of the DTFT, we have

$$y(n) = x(n) * h(n)$$

$$Y(e^{j\omega}) = X(e^{j\omega}) H(e^{j\omega})$$

$$Y(e^{j\omega}) = [1 + e^{j\omega} + e^{-j\omega}][1 + e^{j\omega} + e^{-j\omega}]$$

$$Y(e^{j\omega}) = e^{j2\omega} + 2e^{j\omega} + 3 + 2e^{-j\omega} + e^{-j2\omega}$$

Taking the inverse Fourier transform, we obtain

$$\begin{aligned} y(n) &= \delta(n+2) + 2\delta(n+1) + 3\delta(n) \\ &\quad + 2\delta(n-1) + \delta(n-2) \end{aligned}$$

SUMMARY

1. The Fourier transform of $x(n)$ is defined by

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

2. The inverse Fourier transform of $X(e^{j\omega})$ is defined by

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

3. The magnitude of $X(e^{j\omega})$ plotted against ω is called the magnitude spectrum of $x(n)$ and $|X(e^{j\omega})|^2$ is called the energy spectrum.
 4. The angle of $X(e^{j\omega})$ plotted versus ω is called the phase spectrum.
 5. Parseval's theorem states that

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

6. The power spectral density of $x(n)$ is defined by

$$G_x(e^{j\omega}) = \lim_{N \rightarrow \infty} \frac{|X_N(e^{j\omega})|^2}{2N+1}$$

where

$$x_N(n) \longleftrightarrow X_N(e^{j\omega})$$

and

$$x_N(n) = \begin{cases} x(n), & -N \leq n \leq N \\ 0, & \text{otherwise} \end{cases}$$

7. The DTFT is always periodic with periodic one in the f domain or period 2π in the ω domain.
 8. If a signal is discrete in one domain, it is periodic in the other.
 9. Convolution and multiplication of functions are dual in the time and frequency domains.
 10. The DTFT of a periodic signal consists only of impulses.
 11. The correlation function indicates how correlated two signals are as a function of how much one of them is shifted in time.
 12. Correlation and convolution are closely related mathematical processes.
 13. The correlation of a signal with a shifted version of itself is called autocorrelation.
 14. Autocorrelation is closely related to signal energy or power and contains important information about how rapidly a signal varies in time.
 15. ESD and PSD are the frequency-domain counterparts of autocorrelation and are related through the Fourier transform.
 16. ESD and PSD indicated how the energy or power of a signal varies with frequency.

MULTIPLE-CHOICE QUESTIONS

1. The Fourier transform of a signal $x(n) = a^n u(n)$ is given by

- (a) $\frac{1}{1 - a e^{j\omega}}$
 (b) $\frac{1}{1 + a e^{j\omega}}$
 (c) $\frac{1}{1 - a e^{-j\omega}}$
 (d) $\frac{1}{a + e^{-j\omega}}$

2. The Fourier transform of a unit step function

- (a) does not exist
 (b) is another unit step

- (c) contains impulse functions
 (d) is $1/1 - e^{-j\omega}$

3. If the notation $\circledast$ is used to denote the convolution, and $x(n) \longleftrightarrow X(e^{j\omega})$, $y(n) \longleftrightarrow Y(e^{j\omega})$ makes the Fourier transform pair, then $x(n)y(n) \longleftrightarrow F(e^{j\omega})$ is given by

- (a) $X(e^{j\omega}) \circledast Y(e^{j\omega})$
 (b) $X(e^{j\omega})Y(e^{j\omega})$
 (c) $1/2\pi X(e^{j\omega})Y(e^{j\omega})$
 (d) $1/2\pi[X(e^{j\omega}) \circledast Y(e^{j\omega})]$

4. The DTFT is periodic with period

- (a) π

- (b) 2π
 (c) $\pi/2$
 (d) $\pi/4$.
5. The property of Fourier transform that states that the expansion in time domain is equivalent to compression in the frequency domain is
- (a) duality
 (b) time shifting
 (c) time expansion
 (d) frequency shifting
6. The DTFT of a function $x(n)$ is $X(e^{j\omega})$. The FT of $nx(n)$ will be
- (a) $j \frac{dX(e^{j\omega})}{d\omega}$
 (b) $\frac{dX(e^{j\omega})}{d\omega}$
 (c) $j\omega \frac{dX(e^{j\omega})}{d\omega}$
 (d) $\frac{X(e^{j\omega})}{j\omega}$
7. The Fourier transform of the exponential signal $e^{j\omega_0 n}$, for $0 < \omega \leq \pi$, is
- (a) a constant
 (b) a rectangular pulse
 (c) an impulse
 (d) an another exponential
8. The Fourier transform of a real and even signal $x(n)$ is
- (a) real and even function of frequency
 (b) real and odd function of frequency
 (c) complex, in general
 (d) purely imaginary
9. The Fourier transform of a real and odd signal $x(n)$ is
- (a) imaginary and odd function of frequency
 (b) real and odd function of frequency
 (c) complex, in general
 (d) purely imaginary
10. The frequency response of a system with $h(n) = \delta(n) - \delta(n-1)$ is given by
- (a) $\delta(\omega) - \delta(\omega-1)$
 (b) $1 - e^{j\omega}$
 (c) $u(\omega) - u(\omega-1)$
 (d) $1 - e^{-j\omega}$
11. A transmission is said to be if the response of the system is exact replica of the input signal.
- (a) LTI
 (b) distorted
 (c) distortionless
 (d) undefined
12. The group-delay function $\tau_g(\omega)$ is related to phase function $\phi(\omega)$ as
- (a) $\tau_g(\omega) = -\frac{d\phi(\omega)}{d\omega}$
 (b) $\tau_g(\omega) = \frac{d\phi(\omega)}{d\omega}$
 (c) $\tau_g(\omega) = -\frac{d^2\phi(\omega)}{d\omega^2}$
 (d) $\tau_g(\omega) = -\frac{\phi(\omega)}{\omega}$
13. For a nonperiodic discrete-time signal, the frequency-shift property states that if the DTFT of $x(n)$ is $X(e^{j\omega})$, then the DTFT of $x'(n)$ is $X(e^{j(\omega-\alpha)})$, where $x'(n)$ is given by
- (a) $e^{j\alpha n}x(n)$
 (b) $e^{-j\alpha n}x(n)$
 (c) $e^{jn}x(n-\alpha)$
 (d) $e^{-jn}x(n-\alpha)$

PROBLEMS

6.1 A signal $x(n)$ can be expressed as the sum of even and odd components:

$$x(n) = x_e(n) + x_o(n)$$

- (a) If $x(n) \longleftrightarrow X(e^{j\omega})$, show that for real $x(n)$,
- $$x_e(n) \longleftrightarrow \Re\{X(e^{j\omega})\}$$

and

$$x_o(n) \longleftrightarrow j\Im\{X(e^{j\omega})\}$$

- (b) Verify these results by finding the DTFT of the even and odd components of the signal $(0.8)^n u(n)$.

6.2 An accumulator system has the property that an input $x(n]$ results in the output

$$y(n) = \sum_{k=-\infty}^n x(k)$$

- Find the impulse response $h(n)$ and the frequency response $H(e^{j\omega})$ for the accumulator.
- Use the result in part (a) to find the DTFT of $u(n)$.

6.3 $X(e^{j\omega})$ denotes the DTFT of a length-9 sequence $x(n]$ given by

$$x(n) = \{2, 3, -1, 0, -4, 3, 1, 2, 4\}, \quad -2 \leq n \leq 6$$

Evaluate the following functions of $X(e^{j\omega})$ without computing the transform itself:

- $X(e^{j0})$
- $X(e^{j\pi})$
- $\int_{-\pi}^{\pi} X(e^{j\omega}) d\omega$
- $\int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$
- $\int_{-\pi}^{\pi} \left| \frac{dX(e^{j\omega})}{d\omega} \right|^2 d\omega$

6.4 Let $x(n]$ be a causal and absolutely summable real sequence with a DTFT $X(e^{j\omega})$. If $X_R(e^{j\omega})$ and $X_I(e^{j\omega})$ denote the real and imaginary parts of $X(e^{j\omega})$, show that they are related as

$$X_R(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X_I(e^{j\theta}) \cot\left(\frac{\omega - \theta}{2}\right) d\theta + x(0)$$

$$X_I(e^{j\omega}) = -\frac{1}{2\pi} \int_{-\pi}^{\pi} X_R(e^{j\theta}) \cot\left(\frac{\omega - \theta}{2}\right) d\theta$$

6.5 Consider the filter

$$y(n) = 0.9y(n-1) + bx(n)$$

- Determine b so that $|H(e^{j0})| = 1$.
- Determine the frequency at which $|H(e^{j\omega})| = 1/\sqrt{2}$.
- Is this filter low-pass, high-pass, or bandpass?

(d) Repeat parts (b) and (c) for the filter

$$y(n) = -0.9y(n-1) + 0.1x(n)$$

6.6 The input to a discrete-time system is given by

$$x(n) = \cos\left(\frac{\pi}{4}n\right) + \cos\left(\frac{3\pi}{4}n\right)$$

The impulse response of the system is specified as follows:

$$h(n) \longleftrightarrow H(e^{j\omega}) = \begin{cases} 1, & |\omega| < \pi/4 \\ 0, & \pi/4 \leq |\omega| < \pi \end{cases}$$

Obtain the output $y(n]$ using the convolution property of the DTFT.

6.7 Compute the Fourier transform of each of the following signals:

- $x(n) = u(n-2) - u(n-6)$
- $x(n) = (1/2)^{-n} u(-n-1)$
- $x(n) = (1/2)^{|n|} u(-n-1)$
- $x(n) = (n-1)(1/3)^{|n|}$

6.8 Determine the signals having the following Fourier transforms:

- $X(e^{j\omega}) = \begin{cases} 0, & 0 \leq |\omega| \leq \omega_0 \\ 1, & \omega_0 < |\omega| \leq \pi \end{cases}$
- $X(e^{j\omega}) = \cos^2(\omega)$
- $X(e^{j\omega}) = e^{-j\omega/2}$, for $-\pi \leq \omega \leq \pi$
- $X(e^{j\omega}) = \frac{e^{-j\omega} - 1/5}{1 - 1/5 e^{-j\omega}}$

6.9 A signal $x(n]$ has the following Fourier transform

$$X(e^{j\omega}) = \frac{1}{1 - a e^{-j\omega}}$$

Determine the Fourier transforms of the following signals:

- $x(2n+1)$
- $e^{j(\pi/2)n} x(n+2)$
- $x(-2n)$
- $x(n) \cos(0.3\pi n)$
- $x(n) * x(n-1)$
- $x(n) * x(-n)$

6.10 The following input-output pairs have been observed during the operation of various systems:

- (a) $x(n) = (1/2)^n \rightarrow y(n) = (1/8)^n$
- (b) $x(n) = (1/2)^n u(n) \rightarrow y(n) = (1/8)^n u(n)$
- (c) $x(n) = e^{j\pi/5} \rightarrow y(n) = 3 e^{j\pi/5}$

Determine their frequency response if each of the above system is LTI.

6.11 Determine and sketch the magnitude and phase response of the following systems:

- (a) $y(n) = \frac{1}{2}[x(n) + x(n-1)]$
- (b) $y(n) = \frac{1}{2}[x(n) - x(n-1)]$
- (c) $y(n) = \frac{1}{2}[x(n+1) - x(n-1)]$
- (d) $y(n) = \frac{1}{2}[x(n+1) + x(n-1)]$

ANSWERS TO MULTIPLE-CHOICE QUESTIONS

- | | | | | | | | | |
|---------|---------|---------|---------|--------|--------|--------|--------|--------|
| 1. (a) | 2. (c) | 3. (d) | 4. (b) | 5. (c) | 6. (a) | 7. (c) | 8. (a) | 9. (a) |
| 10. (d) | 11. (c) | 12. (a) | 13. (a) | | | | | |

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